



Android Lab & Studio Projects - Master Documentation

Comprehensive Code Repository: Android Lab & AndroidStudioProjects

27

PROJECTS

195

SOURCE FILES

7,529

LINES OF CODE

Generated: September 23, 2026

Source Folders: F:\Lab Android

Contents: Java Activities/Adapters, Layout XMLs, Menus, Manifests

Table of Contents

Android Lab (3 projects)

MasterApp

23 files • 1189 lines

- [AndroidManifest.xml](#) (51 lines)
- [MainActivity.java](#) (47 lines)
- [MainActivity2.java](#) (13 lines)
- [MainActivity3.java](#) (21 lines)
- [MainActivity4.java](#) (60 lines)
- [MainActivity5.java](#) (72 lines)
- ... and 17 more files

President

12 files • 263 lines

- [AndroidManifest.xml](#) (39 lines)
- [MainActivity.java](#) (31 lines)
- [President.java](#) (20 lines)
- [PresidentActivity.java](#) (13 lines)
- [PresidentDetailFragment.java](#) (26 lines)
- [PresidentListFragment.java](#) (37 lines)
- ... and 6 more files

layout

4 files • 167 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (70 lines)
- [activity_main.xml](#) (62 lines)
- [strings.xml](#) (8 lines)

AndroidStudioProjects (24 projects)

BankApp

9 files • 344 lines

- [AndroidManifest.xml](#) (28 lines)
- [MainActivity.java](#) (65 lines)
- [activity_main.xml](#) (48 lines)
- [activity_canara.xml](#) (38 lines)
- [activity_hdfc.xml](#) (37 lines)
- [activity_icici.xml](#) (36 lines)
- ... and 3 more files

Bankapp2

9 files • 338 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (64 lines)
- [activity_main.xml](#) (48 lines)
- [activity_canara.xml](#) (38 lines)
- [activity_hdfc.xml](#) (37 lines)
- [activity_icici.xml](#) (36 lines)
- ... and 3 more files

Calculator

5 files • 238 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (72 lines)
- [activity_main.xml](#) (60 lines)
- [activity_main2.xml](#) (76 lines)
- [strings.xml](#) (3 lines)

DatabaseProgram

10 files • 626 lines

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (123 lines)
- [CustomAdapter.java](#) (85 lines)
- [DatabaseHelper.java](#) (130 lines)
- [Student.java](#) (38 lines)
- [ViewDataActivity.java](#) (47 lines)
- ... and 4 more files

DateTimePicker

5 files • 221 lines

Demo

6 files • 111 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (108 lines)
- [activity_main.xml](#) (59 lines)
- [main_menu.xml](#) (24 lines)
- [strings.xml](#) (3 lines)

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (20 lines)
- [MainActivity2.java](#) (14 lines)
- [activity_main.xml](#) (35 lines)
- [activity_main2.xml](#) (9 lines)
- [strings.xml](#) (3 lines)

Dialerapp

6 files • 329 lines

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (109 lines)
- [MainActivity2.java](#) (14 lines)
- [activity_main.xml](#) (106 lines)
- [activity_main2.xml](#) (67 lines)
- [strings.xml](#) (3 lines)

Exam1

4 files • 159 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (52 lines)
- [activity_main.xml](#) (77 lines)
- [strings.xml](#) (3 lines)

Intent

6 files • 354 lines

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (101 lines)
- [Intent1.java](#) (34 lines)
- [activity_main.xml](#) (143 lines)
- [activity_intent1.xml](#) (43 lines)
- [strings.xml](#) (3 lines)

LAB1

4 files • 120 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (18 lines)
- [activity_main.xml](#) (70 lines)
- [strings.xml](#) (5 lines)

Lab3activitylifecyle

4 files • 102 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (54 lines)
- [activity_main.xml](#) (18 lines)
- [strings.xml](#) (3 lines)

Lab4

4 files • 136 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (49 lines)
- [activity_main.xml](#) (52 lines)
- [strings.xml](#) (8 lines)

MasterApp

8 files • 247 lines

- [AndroidManifest.xml](#) (33 lines)
- [MainActivity.java](#) (26 lines)
- [MainActivity2.java](#) (13 lines)
- [MainActivity3.java](#) (21 lines)
- [activity_main.xml](#) (101 lines)
- [activity_main2.xml](#) (21 lines)
- ... and 2 more files

Merge

4 files • 141 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (59 lines)
- [activity_main.xml](#) (52 lines)
- [strings.xml](#) (3 lines)

MyApplication

4 files • 62 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (14 lines)
- [activity_main.xml](#) (18 lines)
- [strings.xml](#) (3 lines)

Notification

8 files • 308 lines

- [AndroidManifest.xml](#) (33 lines)
- [MainActivity.java](#) (90 lines)
- [CardActivity.java](#) (26 lines)
- [ScrollingActivity.java](#) (14 lines)
- [activity_main.xml](#) (43 lines)

- [activity_card.xml](#) (41 lines)
- ... and 2 more files

Notificationapp

8 files • 339 lines

- [AndroidManifest.xml](#) (38 lines)
- [MainActivity.java](#) (114 lines)
- [CardActivity.java](#) (29 lines)
- [ScrollingActivity.java](#) (13 lines)
- [activity_main.xml](#) (43 lines)
- [activity_card.xml](#) (41 lines)
- ... and 2 more files

President

12 files • 313 lines

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (46 lines)
- [FragmentDetail.java](#) (36 lines)
- [ListFragment.java](#) (67 lines)
- [President.java](#) (21 lines)
- [PresidentActivity.java](#) (23 lines)
- ... and 6 more files

SharedReference

6 files • 243 lines

- [AndroidManifest.xml](#) (31 lines)
- [MainActivity.java](#) (65 lines)
- [Home.java](#) (45 lines)
- [activity_main.xml](#) (66 lines)
- [activity_home.xml](#) (33 lines)
- [strings.xml](#) (3 lines)

Toast

4 files • 81 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (22 lines)
- [activity_main.xml](#) (29 lines)
- [strings.xml](#) (3 lines)

linearlayoutcalculator

4 files • 178 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (72 lines)
- [activity_main.xml](#) (76 lines)
- [strings.xml](#) (3 lines)

presidentnew - gallery

16 files • 405 lines

- [AndroidManifest.xml](#) (30 lines)
- [MainActivity.java](#) (52 lines)
- [FullImageActivity.java](#) (20 lines)
- [gallery_activity.java](#) (37 lines)
- [President.java](#) (21 lines)
- [PresidentActivity.java](#) (22 lines)
- ... and 10 more files

sum of 2 num

6 files • 253 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (53 lines)
- [activity_main.xml](#) (58 lines)
- [layout.xml](#) (57 lines)
- [layout2.xml](#) (50 lines)
- [strings.xml](#) (8 lines)

userinfo

4 files • 262 lines

- [AndroidManifest.xml](#) (27 lines)
- [MainActivity.java](#) (92 lines)
- [activity_main.xml](#) (140 lines)
- [strings.xml](#) (3 lines)

MasterApp

Android Lab • 23 files

1189 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 51 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.masterapp">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.MasterApp"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity9"
18           android:exported="false" />
19       <activity
20           android:name=".MainActivity8"
21           android:exported="false" />
22       <activity
23           android:name=".MainActivity7"
24           android:exported="false" />
25       <activity
26           android:name=".MainActivity6"
27           android:exported="false" />
28       <activity
29           android:name=".MainActivity5"
30           android:exported="false" />
31       <activity
32           android:name=".MainActivity4"
33           android:exported="false" />
34       <activity
35           android:name=".MainActivity3"
36           android:exported="false" />
37       <activity
38           android:name=".MainActivity2"
39           android:exported="false" />
40       <activity
41           android:name=".MainActivity"
42           android:exported="true">
43           <intent-filter>
44               <action android:name="android.intent.action.MAIN" />
45               <category android:name="android.intent.category.LAUNCHER" />
46           </intent-filter>
47       </activity>
48   </application>
49 </manifest>
50
51
```

app/src/main/java/com/example/masterapp/MainActivity.java

JAVA SOURCE • 47 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;

```

```

4      import android.content.Intent;
5      import android.os.Bundle;
6      import android.view.View;
7
8      public class MainActivity extends AppCompatActivity {
9
10         @Override
11         protected void onCreate(Bundle savedInstanceState) {
12             super.onCreate(savedInstanceState);
13             setContentView(R.layout.activity_main);
14         }
15
16
17         public void hello(View view) {
18             Intent intent = new Intent(MainActivity.this, MainActivity2.class);
19             startActivity(intent);
20         }
21
22         public void toast(View view) {
23             Intent intent = new Intent(MainActivity.this, MainActivity3.class);
24             startActivity(intent);
25         }
26         public void lifeCycle(View view) {
27             Intent intent = new Intent(MainActivity.this, MainActivity4.class);
28             startActivity(intent);
29         }
30         public void sum(View view) {
31
32             Intent intent = new Intent(MainActivity.this, MainActivity5.class);
33             startActivity(intent);
34         }
35         public void userInfo(View view) {
36             Intent intent = new Intent(MainActivity.this, MainActivity6.class);
37             startActivity(intent);
38         }
39         public void userInfoIntent(View view) {
40             Intent intent = new Intent(MainActivity.this, MainActivity7.class);
41             startActivity(intent);
42         }
43         public void calculator(View view) {
44             Intent intent = new Intent(MainActivity.this, MainActivity8.class);
45             startActivity(intent);
46         }
47     }

```

app/src/main/java/com/example/masterapp/MainActivity2.java

JAVA SOURCE • 13 LINES

```

1      package com.example.masterapp;
2
3      import androidx.appcompat.app.AppCompatActivity;
4      import android.os.Bundle;
5
6      public class MainActivity2 extends AppCompatActivity {
7
8          @Override
9          protected void onCreate(Bundle savedInstanceState) {
10              super.onCreate(savedInstanceState);
11              setContentView(R.layout.activity_main2);
12          }
13     }

```

app/src/main/java/com/example/masterapp/MainActivity3.java

JAVA SOURCE • 21 LINES

```

1      package com.example.masterapp;
2
3      import androidx.appcompat.app.AppCompatActivity;

```

```

4
5 import android.os.Bundle;
6 import android.view.View;
7 import android.widget.Toast;
8
9 public class MainActivity3 extends AppCompatActivity {
10
11     @Override
12     protected void onCreate(Bundle savedInstanceState) {
13         super.onCreate(savedInstanceState);
14         setContentView(R.layout.activity_main3);
15
16         Toast.makeText(getApplicationContext(), "Default Toast Message", Toast.LENGTH_LONG).show
17     };
18     public void showtoast(View view){
19         Toast.makeText(getApplicationContext(),"onClickToastMessage",Toast.LENGTH_SHORT).show();
20     }
21 }

```

app/src/main/java/com/example/masterapp/MainActivity4.java

JAVA SOURCE • 60 LINES

```

1 package com.example.masterapp;
2
3 import androidx.appcompat.app.AppCompatActivity;
4
5 import android.os.Bundle;
6
7 import androidx.appcompat.app.AppCompatActivity;
8
9 import android.os.Bundle;
10 import android.util.Log;
11
12 public class MainActivity4 extends AppCompatActivity {
13
14     @Override
15     protected void onCreate(Bundle savedInstanceState) {
16         super.onCreate(savedInstanceState);
17         setContentView(R.layout.activity_main4);
18         Log.d("Activity life cycle","onCreateCalled");
19     }
20     @Override
21     protected void onStart(){
22         super.onStart();
23         Log.d("Activity life cycle","onStartCalled");
24     }
25
26
27     @Override
28     protected void onResume(){
29         super.onResume();
30         Log.d("Activity life cycle","onResumeCalled");
31     }
32
33     @Override
34     protected void onPause(){
35         super.onPause();
36         Log.d("Activity life cycle","onPauseCalled");
37     }
38
39     @Override
40     protected void onRestart(){
41         super.onRestart();
42         Log.d("Activity life cycle","onRestartCalled");
43     }
44
45     @Override
46     protected void onStop(){
47         super.onStop();
48         Log.d("Activity life cycle","onStopCalled");

```

```

49     }
50
51     @Override
52     protected void onDestroy(){
53         super.onDestroy();
54         Log.d("Activity life cycle","onDestroyCalled");
55     }
56
57
58 }

```

app/src/main/java/com/example/masterapp/MainActivity5.java

JAVA SOURCE • 72 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.text.TextUtils;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.EditText;
10 import android.widget.TextView;
11 import android.widget.Toast;
12
13 public class MainActivity5 extends AppCompatActivity {
14
15     EditText et1, et2;
16     TextView tv;
17     Button btn_sum, btn_reset;
18
19     @Override
20     protected void onCreate(Bundle savedInstanceState) {
21         super.onCreate(savedInstanceState);
22         setContentView(R.layout.activity_main5);
23
24         et1 = findViewById(R.id.number1);
25         et2 = findViewById(R.id.number2);
26         tv = findViewById(R.id.textView3);
27
28         btn_sum = findViewById(R.id.calculate_button);
29         btn_reset = findViewById(R.id.reset_button);
30
31
32         btn_sum.setOnClickListener(new View.OnClickListener() {
33             @Override
34             public void onClick(View view) {
35
36                 String val1 = et1.getText().toString().trim();
37                 String val2 = et2.getText().toString().trim();
38
39                 if (!TextUtils.isEmpty(val1) && !TextUtils.isEmpty(val2)) {
40
41                     int sum = Integer.parseInt(val1) + Integer.parseInt(val2);
42
43                     tv.setText("Result = " + sum);
44
45                     Toast.makeText(MainActivity5.this,
46                                 "Sum is " + sum,
47                                 Toast.LENGTH_SHORT).show();
48
49                 } else {
50                     Toast.makeText(MainActivity5.this,
51                                 "Enter both numbers first",
52                                 Toast.LENGTH_SHORT).show();
53                 }
54             }
55         });
56
57

```

```

58         btn_reset.setOnClickListener(new View.OnClickListener() {
59             @Override
60             public void onClick(View view) {
61
62                 et1.setText("");
63                 et2.setText("");
64                 tv.setText("Result = 0");
65
66                 Toast.makeText(MainActivity5.this,
67                     "Inputs cleared",
68                     Toast.LENGTH_SHORT).show();
69             }
70         });
71     }
72 }

```

app/src/main/java/com/example/masterapp/MainActivity6.java

JAVA SOURCE • 92 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.Button;
8  import android.widget.CheckBox;
9  import android.widget.RadioButton;
10 import android.widget.EditText;
11 import android.widget.RadioGroup;
12 import android.widget.TextView;
13 import android.widget.Toast;
14
15 public class MainActivity6 extends AppCompatActivity {
16     EditText editText;
17     RadioGroup radioGender;
18     RadioButton radioMale, radioFemale;
19     CheckBox checkBoxReading, checkBoxMusic, checkBoxSports, checkBoxYoga;
20     Button submit;
21     Button clear;
22     TextView output;
23
24     @Override
25     protected void onCreate(Bundle savedInstanceState) {
26         super.onCreate(savedInstanceState);
27         setContentView(R.layout.activity_main6);
28
29         editText = findViewById(R.id.editText);
30         radioGender = findViewById(R.id.genderGroup);
31         radioMale = findViewById(R.id.radioMale);
32         radioFemale = findViewById(R.id.radioFemale);
33         checkBoxReading = findViewById(R.id.checkBoxReading);
34         checkBoxMusic = findViewById(R.id.checkBoxMusic);
35         checkBoxSports = findViewById(R.id.checkBoxSports);
36         checkBoxYoga = findViewById(R.id.checkBoxYoga);
37         submit = findViewById(R.id.submit);
38         clear = findViewById(R.id.clear);
39         output = findViewById(R.id.output);
40
41         View.OnClickListener radioListener = View → {
42             RadioButton selected = (RadioButton) View;
43             Toast.makeText(MainActivity6.this, selected.getText() + "selected", Toast.LENGTH_SHOR
T).show();
44
45         };
46         radioMale.setOnClickListener(radioListener);
47         radioFemale.setOnClickListener(radioListener);
48
49         View.OnClickListener checkListener = View → {
50             CheckBox box = (CheckBox) View;
51             if (box.isChecked()) {

```

```

52         Toast.makeText(MainActivity6.this, box.getText() + "selected", Toast.LENGTH_SHOR
T).show();
53     }
54 };
55
56     checkBoxReading.setOnClickListener(checkListener);
57     checkBoxMusic.setOnClickListener(checkListener);
58     checkBoxSports.setOnClickListener(checkListener);
59     checkBoxYoga.setOnClickListener(checkListener);
60
61
62     submit.setOnClickListener(v → {
63         String name = editText.getText().toString();
64         int selectedId = radioGender.getCheckedRadioButtonId();
65         String gender = "";
66         if (selectedId ≠ -1) {
67             RadioButton selected = findViewById(selectedId);
68             gender = selected.getText().toString();
69         }
70         StringBuilder hobbies = new StringBuilder();
71         if (checkBoxReading.isChecked()) hobbies.append("Reading");
72         if (checkBoxMusic.isChecked()) hobbies.append("Music");
73         if (checkBoxSports.isChecked()) hobbies.append("Sports");
74         if (checkBoxYoga.isChecked()) hobbies.append("Yoga");
75
76
77         String message = "Name : " + name + "\nGender : " + gender + "\nHobbies : " + hobbies;
78         output.setText(message);
79
80         clear.setOnClickListener(clear → {
81             editText.setText("");
82             radioGender.clearCheck();
83             checkBoxReading.setChecked(false);
84             checkBoxMusic.setChecked(false);
85             checkBoxSports.setChecked(false);
86             checkBoxYoga.setChecked(false);
87             output.setText("Output will appear here");
88         });
89     });
90 }
91 }
92 }

```

app/src/main/java/com/example/masterapp/MainActivity7.java

JAVA SOURCE • 101 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.CheckBox;
10 import android.widget.EditText;
11 import android.widget.RadioButton;
12 import android.widget.RadioGroup;
13 import android.widget.TextView;
14 import android.widget.Toast;
15
16 public class MainActivity7 extends AppCompatActivity {
17     EditText editText;
18     RadioGroup radioGender;
19     RadioButton radioMale,radioFemale;
20     CheckBox checkBoxReading,checkBoxMusic,checkBoxSports,checkBoxYoga;
21     Button submit;
22     Button clear;
23     TextView output;
24
25     @Override

```

```

26     protected void onCreate(Bundle savedInstanceState) {
27         super.onCreate(savedInstanceState);
28         setContentView(R.layout.activity_main7);
29
30         editText = findViewById(R.id.editText);
31         radioGender = findViewById(R.id.genderGroup);
32         radioMale = findViewById(R.id.radioMale);
33         radioFemale = findViewById(R.id.radioFemale);
34         checkBoxReading = findViewById(R.id.checkBoxReading);
35         checkBoxMusic = findViewById(R.id.checkBoxMusic);
36         checkBoxSports = findViewById(R.id.checkBoxSports);
37         checkBoxYoga = findViewById(R.id.checkBoxYoga);
38         submit = findViewById(R.id.submit);
39         clear = findViewById(R.id.clear);
40         output = findViewById(R.id.output);
41
42         View.OnClickListener radiolListener = View → {
43             RadioButton selected = (RadioButton) View;
44             Toast.makeText(MainActivity7.this, selected.getText() + "selected", Toast.LENGTH_SHOR
T).show();
45
46         };
47         radioMale.setOnClickListener(radiolListener);
48         radioFemale.setOnClickListener(radiolListener);
49
50         View.OnClickListener checkListener = View → {
51             CheckBox box = (CheckBox) View;
52             if (box.isChecked()) {
53                 Toast.makeText(MainActivity7.this, box.getText() + "selected", Toast.LENGTH_SHOR
T).show();
54             }
55         };
56
57         checkBoxReading.setOnClickListener(checkListener);
58         checkBoxMusic.setOnClickListener(checkListener);
59         checkBoxSports.setOnClickListener(checkListener);
60         checkBoxYoga.setOnClickListener(checkListener);
61
62
63         submit.setOnClickListener(v → {
64             String name = editText.getText().toString();
65             int selectedId = radioGender.getCheckedRadioButtonId();
66             String gender = "";
67             if (selectedId ≠ -1) {
68                 RadioButton selected = findViewById(selectedId);
69                 gender = selected.getText().toString();
70             }
71
72
73
74
75             StringBuilder hobbies = new StringBuilder();
76             if (checkBoxReading.isChecked()) hobbies.append("Reading");
77             if (checkBoxMusic.isChecked()) hobbies.append("Music");
78             if (checkBoxSports.isChecked()) hobbies.append("Sports");
79             if (checkBoxYoga.isChecked()) hobbies.append("Yoga");
80
81
82             String message = "Name :" + name + "\nGender :" + gender + "\nHobbies :" + hobbies;
83             output.setText(message);
84             Intent intent = new Intent(MainActivity7.this, MainActivity9.class);
85             intent.putExtra("data", message);
86             startActivity(intent);
87
88             clear.setOnClickListener(clear → {
89                 editText.setText("");
90                 radioGender.clearCheck();
91                 checkBoxReading.setChecked(false);
92                 checkBoxMusic.setChecked(false);
93                 checkBoxSports.setChecked(false);
94                 checkBoxYoga.setChecked(false);
95                 output.setText("Output will appear here");
96             });

```

```

97
98
99     });
100 }
101 }

```

app/src/main/java/com/example/masterapp/MainActivity8.java

JAVA SOURCE • 56 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.TextView;
8  public class MainActivity8 extends AppCompatActivity {
9      TextView txtDisplay;
10     String input = "";
11     double num1 = 0;
12     String operator = "";
13     @Override
14     protected void onCreate(Bundle savedInstanceState) {
15         super.onCreate(savedInstanceState);
16         setContentView(R.layout.activity_main8);
17         txtDisplay = findViewById(R.id.txtDisplay);
18     }
19     public void numberClick(View view) {
20         String value = ((android.widget.Button) view).getText().toString();
21         input += value;
22         txtDisplay.setText(input);
23     }
24     public void operatorClick(View view) {
25         num1 = Double.parseDouble(input);
26         operator = ((android.widget.Button) view).getText().toString();
27         input = "";
28     }
29     public void equalClick(View view) {
30         double num2 = Double.parseDouble(input);
31         double result = 0;
32         switch (operator) {
33             case "+":
34                 result = num1 + num2;
35                 break;
36             case "-":
37                 result = num1 - num2;
38                 break;
39
40             case "*":
41                 result = num1 * num2;
42                 break;
43             case "/":
44                 result = num1 / num2;
45                 break;
46         }
47         txtDisplay.setText(String.valueOf(result));
48         input = String.valueOf(result);
49     }
50     public void clearClick(View view) {
51         input = "";
52         num1 = 0;
53         operator = "";
54         txtDisplay.setText("0");
55     }
56 }

```

app/src/main/java/com/example/masterapp/MainActivity9.java

JAVA SOURCE • 34 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.widget.*;
8
9  public class MainActivity9 extends AppCompatActivity {
10     TextView output;
11     Button back;
12
13     @Override
14     protected void onCreate(Bundle savedInstanceState) {
15         super.onCreate(savedInstanceState);
16         setContentView(R.layout.activity_main9);
17
18         output=findViewById(R.id.output);
19         back=findViewById(R.id.back);
20
21         Intent intent=getIntent();
22         String data = intent.getStringExtra("data");
23         output.setText(data);
24
25         back.setOnClickListener(view→ {
26             Intent i = new Intent(MainActivity9.this, MainActivity7.class);
27             startActivity(i);
28             finish();
29         });
30
31     }
32 }
33
34 }
```

app/src/main/java/com/example/masterapp/ui/main/MainFragment.java

JAVA SOURCE • 39 LINES

```

1  package com.example.masterapp.ui.main;
2
3  import androidx.lifecycle.ViewModelProvider;
4
5  import android.os.Bundle;
6
7  import androidx.annotation.NonNull;
8  import androidx.annotation.Nullable;
9  import androidx.fragment.app.Fragment;
10
11 import android.view.LayoutInflater;
12 import android.view.View;
13 import android.view.ViewGroup;
14
15 import com.example.masterapp.R;
16
17 public class MainFragment extends Fragment {
18
19     private MainViewModel mViewModel;
20
21     public static MainFragment newInstance() {
22         return new MainFragment();
23     }
24
25     @Nullable
26     @Override
27     public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup container,
28                             @Nullable Bundle savedInstanceState) {
29         return inflater.inflate(R.layout.fragment_main, container, false);
30     }
31
32 }
```

```

30     }
31
32     @Override
33     public void onActivityCreated(@Nullable Bundle savedInstanceState) {
34         super.onActivityCreated(savedInstanceState);
35         mViewModel = new ViewModelProvider(this).get(MainViewModel.class);
36         // TODO: Use the ViewModel
37     }
38
39 }

```

app/src/main/java/com/example/masterapp/ui/main/MainViewModel.java

JAVA SOURCE • 7 LINES

```

1  package com.example.masterapp.ui.main;
2
3  import androidx.lifecycle.ViewModel;
4
5  public class MainViewModel extends ViewModel {
6      // TODO: Implement the ViewModel
7  }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 58 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="vertical"
6      android:gravity="center_horizontal"
7      android:padding="16dp">
8
9      <TextView
10         android:layout_width="wrap_content"
11         android:layout_height="wrap_content"
12         android:text="Master App"
13         android:textSize="20sp"
14         android:layout_marginBottom="20dp" />
15
16      <Button
17         android:layout_width="250dp"
18         android:layout_height="wrap_content"
19         android:text="Hello World"
20         android:onClick="hello" />
21
22      <Button
23         android:layout_width="250dp"
24         android:layout_height="wrap_content"
25         android:text="Toast"
26         android:onClick="toast" />
27
28      <Button
29         android:layout_width="250dp"
30         android:layout_height="wrap_content"
31         android:text="Life Cycle"
32         android:onClick="lifeCycle" />
33
34      <Button
35         android:layout_width="250dp"
36         android:layout_height="wrap_content"
37         android:text="Sum of 2 nums"
38         android:onClick="sum" />
39
40      <Button
41         android:layout_width="250dp"
42         android:layout_height="wrap_content"
43         android:text="User Info"

```

```

44         android:onClick="userInfo" />
45
46     <Button
47         android:layout_width="250dp"
48         android:layout_height="wrap_content"
49         android:text="User Info Intent"
50         android:onClick="userInfoIntent" />
51
52     <Button
53         android:layout_width="250dp"
54         android:layout_height="wrap_content"
55         android:text="Calculator"
56         android:onClick="calculator" />
57
58 </LinearLayout>

```

app/src/main/res/layout/activity_main2.xml

LAYOUT XML • 21 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      tools:context=".MainActivity2">
9
10     <TextView
11         android:id="@+id/textView"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="Hello World"
15         android:textColor="#AC1B4C"
16         android:textSize="48sp"
17         app:layout_constraintTop_toTopOf="parent"
18         app:layout_constraintBottom_toBottomOf="parent"
19         app:layout_constraintStart_toStartOf="parent"
20         app:layout_constraintEnd_toEndOf="parent" />
21 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main3.xml

LAYOUT XML • 29 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity3">
8
9
10     <Button
11         android:id="@+id/button"
12         android:layout_width="293dp"
13         android:layout_height="159dp"
14         android:insetLeft="50dp"
15         android:insetTop="50dp"
16         android:onClick="showtoast"
17         android:text=" On Click Toast"
18         tools:ignore="MissingConstraints"
19         tools:layout_editor_absoluteX="47dp"
20         tools:layout_editor_absoluteY="243dp" />
21
22     <androidx.constraintlayout.widget.Barrier
23         android:id="@+id/barrier2"

```

```

24         android:layout_width="wrap_content"
25         android:layout_height="wrap_content"
26         app:barrierDirection="top"
27         app:constraint_referenced_ids="button" />
28
29 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main4.xml

LAYOUT XML • 17 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity4">
8
9      <TextView
10         android:id="@+id/textView2"
11         android:layout_width="290dp"
12         android:layout_height="71dp"
13         android:text="Activity life cycle"
14         tools:layout_editor_absoluteX="104dp"
15         tools:layout_editor_absoluteY="203dp"
16         tools:ignore="MissingConstraints" />
17 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main5.xml

LAYOUT XML • 62 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent">
7
8      <EditText
9         android:id="@+id/number1"
10         android:layout_width="250dp"
11         android:layout_height="wrap_content"
12         android:hint="Enter Number 1"
13         android:inputType="number"
14         android:layout_marginTop="80dp"
15         app:layout_constraintTop_toTopOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintEnd_toEndOf="parent" />
18
19      <EditText
20         android:id="@+id/number2"
21         android:layout_width="250dp"
22         android:layout_height="wrap_content"
23         android:hint="Enter Number 2"
24         android:inputType="number"
25         android:layout_marginTop="20dp"
26         app:layout_constraintTop_toBottomOf="@id/number1"
27         app:layout_constraintStart_toStartOf="parent"
28         app:layout_constraintEnd_toEndOf="parent" />
29
30      <Button
31         android:id="@+id/calculate_button"
32         android:layout_width="wrap_content"
33         android:layout_height="wrap_content"
34         android:text="Calculate"
35         android:layout_marginTop="20dp"
36         app:layout_constraintTop_toBottomOf="@id/number2"

```

```

37         app:layout_constraintStart_toStartOf="parent"
38         app:layout_constraintEnd_toEndOf="parent" />
39
40     <Button
41         android:id="@+id/reset_button"
42         android:layout_width="wrap_content"
43         android:layout_height="wrap_content"
44         android:text="Reset"
45         android:layout_marginTop="15dp"
46         app:layout_constraintTop_toBottomOf="@id/calculate_button"
47         app:layout_constraintStart_toStartOf="parent"
48         app:layout_constraintEnd_toEndOf="parent" />
49
50     <TextView
51         android:id="@+id/textView3"
52         android:layout_width="wrap_content"
53         android:layout_height="wrap_content"
54         android:text="Result = 0"
55         android:textSize="24sp"
56         android:textStyle="bold"
57         android:layout_marginTop="30dp"
58         app:layout_constraintTop_toBottomOf="@id/reset_button"
59         app:layout_constraintStart_toStartOf="parent"
60         app:layout_constraintEnd_toEndOf="parent" />
61
62 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main6.xml

LAYOUT XML • 142 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp"
8      tools:context=".MainActivity6">>
9
10
11  <ImageButton
12      android:id="@+id/imageButton2"
13      android:layout_width="185dp"
14      android:layout_height="120dp"
15      android:layout_gravity="center_horizontal"
16      android:layout_marginBottom="16dp"
17      android:scaleType="centerCrop"
18      tools:srcCompat="@tools:sample/avatars[0]" />
19
20
21  <LinearLayout
22      android:layout_width="match_parent"
23      android:layout_height="wrap_content"
24      android:orientation="horizontal"
25      android:gravity="center_vertical"
26      android:layout_marginBottom="12dp">
27
28      <TextView
29          android:id="@+id/textView2"
30          android:layout_width="80dp"
31          android:layout_height="wrap_content"
32          android:text="Name:" />
33
34      <EditText
35          android:id="@+id/editText"
36          android:layout_width="match_parent"
37          android:layout_height="wrap_content"
38          android:ems="10"
39          android:hint="Enter name"
40          android:inputType="textPersonName" />
41  </LinearLayout>

```

```

42
43
44 <LinearLayout
45     android:layout_width="match_parent"
46     android:layout_height="wrap_content"
47     android:orientation="horizontal"
48     android:gravity="center_vertical"
49     android:layout_marginBottom="12dp">
50
51     <TextView
52         android:id="@+id/radioGenderLabel"
53         android:layout_width="80dp"
54         android:layout_height="wrap_content"
55         android:text="Gender: " />
56
57     <RadioGroup
58         android:id="@+id/genderGroup"
59         android:layout_width="match_parent"
60         android:layout_height="wrap_content"
61         android:orientation="horizontal">
62
63         <RadioButton
64             android:id="@+id/radioMale"
65             android:layout_width="wrap_content"
66             android:layout_height="wrap_content"
67             android:text="Male" />
68
69         <RadioButton
70             android:id="@+id/radioFemale"
71             android:layout_width="wrap_content"
72             android:layout_height="wrap_content"
73             android:text="Female" />
74     </RadioGroup>
75 </LinearLayout>
76
77
78 <TextView
79     android:id="@+id/textView5"
80     android:layout_width="wrap_content"
81     android:layout_height="wrap_content"
82     android:text="Hobbies: "
83     android:textStyle="bold"
84     android:layout_marginBottom="8dp" />
85
86 <CheckBox
87     android:id="@+id/checkBoxReading"
88     android:layout_width="wrap_content"
89     android:layout_height="wrap_content"
90     android:text="Reading" />
91
92 <CheckBox
93     android:id="@+id/checkBoxMusic"
94     android:layout_width="wrap_content"
95     android:layout_height="wrap_content"
96     android:text="Music" />
97
98 <CheckBox
99     android:id="@+id/checkBoxSports"
100     android:layout_width="wrap_content"
101     android:layout_height="wrap_content"
102     android:text="Sports" />
103
104 <CheckBox
105     android:id="@+id/checkBoxYoga"
106     android:layout_width="wrap_content"
107     android:layout_height="wrap_content"
108     android:text="Yoga" />
109
110
111 <LinearLayout
112     android:layout_width="match_parent"
113     android:layout_height="wrap_content"
114     android:orientation="horizontal"

```

```

115         android:layout_marginTop="24dp">
116
117         <Button
118             android:id="@+id/submit"
119             android:layout_width="0dp"
120             android:layout_weight="1"
121             android:layout_height="wrap_content"
122             android:layout_marginEnd="8dp"
123             android:text="Submit" />
124
125         <Button
126             android:id="@+id/clear"
127             android:layout_width="0dp"
128             android:layout_weight="1"
129             android:layout_height="wrap_content"
130             android:text="Clear" />
131     </LinearLayout>
132
133
134     <TextView
135         android:id="@+id/output"
136         android:layout_width="match_parent"
137         android:layout_height="53dp"
138         android:layout_marginTop="20dp"
139         android:text="Output will appear here"
140         android:textAlignment="center" />
141
142 </LinearLayout>

```

app/src/main/res/layout/activity_main7.xml

LAYOUT XML • 141 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp">
8
9      <ImageButton
10         android:id="@+id/imageButton2"
11         android:layout_width="185dp"
12         android:layout_height="120dp"
13         android:layout_gravity="center_horizontal"
14         android:layout_marginBottom="16dp"
15         android:scaleType="centerCrop"
16         android:contentDescription="Profile Image"
17         tools:srcCompat="@tools:sample/avatars[0]" />
18
19     <LinearLayout
20         android:layout_width="match_parent"
21         android:layout_height="wrap_content"
22         android:orientation="horizontal"
23         android:gravity="center_vertical"
24         android:layout_marginBottom="12dp">
25
26         <TextView
27             android:id="@+id/textView2"
28             android:layout_width="80dp"
29             android:layout_height="wrap_content"
30             android:text="Name:" />
31
32         <EditText
33             android:id="@+id/editText"
34             android:layout_width="match_parent"
35             android:layout_height="wrap_content"
36             android:ems="10"
37             android:hint="Enter name"
38             android:inputType="textPersonName" />
39     </LinearLayout>

```

```

40
41
42 <LinearLayout
43     android:layout_width="match_parent"
44     android:layout_height="wrap_content"
45     android:orientation="horizontal"
46     android:gravity="center_vertical"
47     android:layout_marginBottom="12dp">
48
49     <TextView
50         android:id="@+id/radioGenderLabel"
51         android:layout_width="80dp"
52         android:layout_height="wrap_content"
53         android:text="Gender: " />
54
55     <RadioGroup
56         android:id="@+id/genderGroup"
57         android:layout_width="match_parent"
58         android:layout_height="wrap_content"
59         android:orientation="horizontal">
60
61         <RadioButton
62             android:id="@+id/radioMale"
63             android:layout_width="wrap_content"
64             android:layout_height="wrap_content"
65             android:text="Male" />
66
67         <RadioButton
68             android:id="@+id/radioFemale"
69             android:layout_width="wrap_content"
70             android:layout_height="wrap_content"
71             android:text="Female" />
72     </RadioGroup>
73 </LinearLayout>
74
75
76 <TextView
77     android:id="@+id/textView5"
78     android:layout_width="wrap_content"
79     android:layout_height="wrap_content"
80     android:layout_marginBottom="8dp"
81     android:text="Hobbies: "
82     android:textStyle="bold" />
83
84 <CheckBox
85     android:id="@+id/checkBoxReading"
86     android:layout_width="wrap_content"
87     android:layout_height="wrap_content"
88     android:text="Reading" />
89
90 <CheckBox
91     android:id="@+id/checkBoxMusic"
92     android:layout_width="wrap_content"
93     android:layout_height="wrap_content"
94     android:text="Music" />
95
96 <CheckBox
97     android:id="@+id/checkBoxSports"
98     android:layout_width="wrap_content"
99     android:layout_height="wrap_content"
100    android:text="Sports" />
101
102 <CheckBox
103     android:id="@+id/checkBoxYoga"
104     android:layout_width="wrap_content"
105     android:layout_height="wrap_content"
106     android:text="Yoga" />
107
108
109 <LinearLayout
110     android:layout_width="match_parent"
111     android:layout_height="wrap_content"
112     android:orientation="horizontal"

```

```

113         android:layout_marginTop="24dp">
114
115         <Button
116             android:id="@+id/submit"
117             android:layout_width="0dp"
118             android:layout_weight="1"
119             android:layout_height="wrap_content"
120             android:layout_marginEnd="8dp"
121             android:text="Submit" />
122
123         <Button
124             android:id="@+id/clear"
125             android:layout_width="0dp"
126             android:layout_weight="1"
127             android:layout_height="wrap_content"
128             android:text="Clear" />
129     </LinearLayout>
130
131
132     <TextView
133         android:id="@+id/output"
134         android:layout_width="match_parent"
135         android:layout_height="53dp"
136         android:layout_marginTop="20dp"
137         android:text="Output will appear here"
138         android:textAlignment="center"
139         android:visibility="invisible" />
140
141 </LinearLayout>

```

app/src/main/res/layout/activity_main8.xml

LAYOUT XML • 61 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="8dp">
8      <TextView
9          android:id="@+id/txtDisplay"
10         android:layout_width="match_parent"
11         android:layout_height="120dp"
12         android:gravity="end|center_vertical"
13         android:text="0"
14         android:textSize="36sp"
15         android:padding="16dp" />
16      <GridLayout
17          android:layout_width="match_parent"
18          android:layout_height="wrap_content"
19          android:columnCount="4">
20          <Button android:id="@+id/btnClear" android:text="C"
21              android:onClick="clearClick" />
22          <Button android:id="@+id/btnDivide" android:text="/"
23              android:onClick="operatorClick" />
24          <Button android:id="@+id/btnMultiply" android:text="*"
25              android:onClick="operatorClick" />
26          <Button android:id="@+id/btnMinus" android:text="-"
27              android:onClick="operatorClick" />
28
29          <Button android:id="@+id/btn7" android:text="7"
30              android:onClick="numberClick" />
31          <Button android:id="@+id/btn8" android:text="8"
32              android:onClick="numberClick" />
33          <Button android:id="@+id/btn9" android:text="9"
34              android:onClick="numberClick" />
35          <Button android:id="@+id/btnPlus" android:text="+"
36              android:onClick="operatorClick" />
37
38          <Button android:id="@+id/btn4" android:text="4"

```

```

39         android:onClick="numberClick" />
40     <Button android:id="@+id/btn5" android:text="5"
41         android:onClick="numberClick" />
42     <Button android:id="@+id/btn6" android:text="6"
43         android:onClick="numberClick" />
44     <Button android:id="@+id/btnEqual" android:text="="
45         android:onClick="equalClick" />
46
47     <Button android:id="@+id/btn1" android:text="1"
48         android:onClick="numberClick" />
49     <Button android:id="@+id/btn2" android:text="2"
50         android:onClick="numberClick" />
51     <Button android:id="@+id/btn3" android:text="3"
52         android:onClick="numberClick" />
53
54     <Button android:text="" android:enabled="false" />
55     <Button android:id="@+id/btn0"
56         android:layout_columnSpan="2"
57         android:text="0" android:onClick="numberClick" />
58
59     <Button android:id="@+id/btnDot" android:text="." />
60 </GridLayout>
61 </LinearLayout>

```

app/src/main/res/layout/activity_main9.xml

LAYOUT XML • 42 LINES

```

1  <androidx.constraintlayout.widget.ConstraintLayout
2      xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent">
7
8      <RelativeLayout
9          android:id="@+id/container"
10         android:layout_width="match_parent"
11         android:layout_height="match_parent">
12
13         <ImageView
14             android:id="@+id/imageView"
15             android:layout_width="100dp"
16             android:layout_height="100dp"
17             android:layout_centerHorizontal="true"
18             android:layout_marginTop="40dp"
19             android:contentDescription="Profile image"
20             tools:srcCompat="@tools:sample/avatars" />
21
22         <TextView
23             android:id="@+id/output"
24             android:layout_width="wrap_content"
25             android:layout_height="wrap_content"
26             android:text="TextView"
27             android:layout_below="@id/imageView"
28             android:layout_centerHorizontal="true"
29             android:layout_marginTop="20dp" />
30
31         <Button
32             android:id="@+id/back"
33             android:layout_width="wrap_content"
34             android:layout_height="wrap_content"
35             android:text="Back"
36             android:layout_below="@id/output"
37             android:layout_centerHorizontal="true"
38             android:layout_marginTop="30dp" />
39
40     </RelativeLayout>
41
42 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/fragment_main.xml

LAYOUT XML • 20 LINES

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:id="@+id/main"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      tools:context=".ui.main.MainFragment">
9
10     <TextView
11         android:id="@+id/message"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="MainFragment"
15         app:layout_constraintBottom_toBottomOf="parent"
16         app:layout_constraintEnd_toEndOf="parent"
17         app:layout_constraintStart_toStartOf="parent"
18         app:layout_constraintTop_toTopOf="parent" />
19
20 </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1  <resources>
2      <string name="app_name">Master App</string>
3  </resources>
```

President

Android Lab • 12 files

263 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 39 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.president">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.President"
15       tools:targetApi="31">
16       <activity
17           android:name=".President"
18           android:exported="false" />
19       <activity
20           android:name=".PresidentListFragment"
21           android:exported="false" />
22       <activity
23           android:name=".PresidentDetailFragment"
24           android:exported="false" />
25       <activity
26           android:name=".PresidentActivity"
27           android:exported="false" />
28       <activity
29           android:name=".MainActivity"
30           android:exported="true">
31           <intent-filter>
32               <action android:name="android.intent.action.MAIN" />
33
34               <category android:name="android.intent.category.LAUNCHER" />
35           </intent-filter>
36       </activity>
37   </application>
38
39 </manifest>

```

app/src/main/java/com/example/president/MainActivity.java

JAVA SOURCE • 31 LINES

```

1  package com.example.president;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.content.Intent;
5  import android.os.Bundle;
6  import android.view.Menu;
7  import android.view.MenuItem;
8  import android.widget.Toast;
9  import com.example.president.R;
10 public class MainActivity extends AppCompatActivity {
11
12     @Override
13     protected void onCreate(Bundle savedInstanceState) {
14         super.onCreate(savedInstanceState);
15         setContentView(R.layout.activity_main);}

```

```

16     @Override
17     public boolean onCreateOptionsMenu(Menu menu) {
18         getMenuInflater().inflate(R.menu.menu, menu);
19         return true;
20     }
21     @Override
22     public boolean onOptionsItemSelected(MenuItem item) {
23         switch (item.getItemId()) {
24             case R.id.action_toast: Toast.makeText(this, "This is a toast message", Toast.LENGTH_SH
ORT).show();
25             return true;
26             case R.id.action_exit: finish();
27             return true;
28             case R.id.action_president: startActivity(new Intent(this, PresidentActivity.class));
29             return true;
30             default: return super.onOptionsItemSelected(item);
31         }
32     }

```

app/src/main/java/com/example/president/President.java

JAVA SOURCE • 20 LINES

```

1  package com.example.president;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  public class President extends AppCompatActivity {
5      String name;
6      String period;
7      String qualification;
8      String achievements;
9      String experience;
10     int imageResId;
11     public President(String name, String period, String qualification, String achievements, String exp
erience, int imageResId)
12     {
13         this.name = name;
14         this.period = period;
15         this.qualification = qualification;
16         this.achievements = achievements;
17         this.experience = experience;
18         this.imageResId = imageResId;
19     }
20 }

```

app/src/main/java/com/example/president/PresidentActivity.java

JAVA SOURCE • 13 LINES

```

1  package com.example.president;
2  import androidx.appcompat.app.AppCompatActivity;
3  import android.os.Bundle;
4  import com.example.president.R;
5  public class PresidentActivity extends AppCompatActivity {
6      @Override
7      protected void onCreate(Bundle savedInstanceState) {
8          super.onCreate(savedInstanceState);
9          setContentView(R.layout.activity_president);
10         public void showPresidentDetails(President president) {
11             PresidentDetailFragment detailFragment = (PresidentDetailFragment) getFragmentManager().f
indFragmentById(R.id.fragment_detail);
12             detailFragment.updateDetails(president);
13         }
14     }

```

app/src/main/java/com/example/president/PresidentDetailFragment.java

JAVA SOURCE • 26 LINES

```

1  package com.example.president;
2

```

```

3  import android.app.Fragment;
4  import android.os.Bundle;
5  import android.view.LayoutInflater;
6  import android.view.View;
7  import android.view.ViewGroup;
8  import android.widget.ImageView;
9  import android.widget.TextView;
10
11 import com.example.president.R;
12 public class PresidentDetailFragment extends Fragment {
13     private TextView detailsView;
14     private ImageView imageView;
15     @Override
16     public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle savedInstanceState) {
17         View view=inflater.inflate(R.layout.fragment_president_detail,container,false);
18         detailsView=view.findViewById(R.id.presidentDetails);
19         imageView=view.findViewById(R.id.presidentImage);
20         return view;}
21     public void updateDetails(President p) {
22         String detail="Name: " +p.name + "\n" +"Period: "+p.period+"\n"+"Qualification: "+p.qualification+
23             "\n"+"Achievements: "+p.achievements+"\n"+"Experience: "+p.experience;
24         detailsView.setText(detail);
25         imageView.setImageResource(p.imageResId);}
26 }

```

app/src/main/java/com/example/president/PresidentListFragment.java

JAVA SOURCE • 37 LINES

```

1  package com.example.president;
2  import android.app.Fragment;
3  import android.os.Bundle;
4  import android.view.LayoutInflater;
5  import android.view.View;
6  import android.view.ViewGroup;
7  import android.widget.AdapterView;
8  import android.widget.AdapterView;
9  import android.widget.AdapterView;
10 import com.example.president.R;
11 public class PresidentListFragment extends Fragment {
12     String[] names = {"Dr.Rajendra Prasad","Dr.S.Radhakrishnan","Dr.A.P.J.Abdul Kalam","Shri Ramnath Kovind",
13         "Smt.Droupadi Murmu","Giani Zail Singh","Pranab Mukherjee","Pratibha Patil",
14         "Ramaswamy Venkataraman","Shankar Dayal Sharma"};
15     President[] presidents;
16     @Override
17     public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle savedInstanceState) {
18         View view=inflater.inflate(R.layout.fragment_president_list,container,false);
19         ListView listView=view.findViewById(R.id.presidentListView);
20         presidents=new President[] {
21             new President("Dr. Rajendra Prasad", "1950-1962", "Lawyer", "Indian Independence Movement", "Lifetime service to the nation", R.drawable.rajendraprasad),
22             new President("Dr. S. Radhakrishnan", "1962-1967", "Philosopher", "Education & Culture", "Promoted Indian philosophy", R.drawable.radhakrishnan),
23             new President("Dr. A.P.J. Abdul Kalam", "2002-2007", "Scientist", "Missile Man of India", "ISRO, DRDO, and education reform", R.drawable.kalam),
24             new President("Shri Ramnath Kovind", "2017-2022", "Lawyer", "Dalit rights advocacy", "Legal and social justice", R.drawable.ramnathkovind),
25             new President("Smt. Droupadi Murmu", "2022-present", "Teacher", "Tribal upliftment", "Governance and tribal education", R.drawable.draupadimurmu),
26             new President("Giani Zail Singh", "1982-1987", "Political Leader", "Punjab & National Unity", "Ministerial roles and unity efforts", R.drawable.gianizailsingh),
27             new President("Pranab Mukherjee", "2012-2017", "Economist", "Finance & Governance", "Decades of political leadership", R.drawable.pranabmukherjee),
28             new President("Pratibha Patil", "2007-2012", "Lawyer", "Women's Rights", "First woman president", R.drawable.pratibhapatil),
29             new President("Ramaswamy Venkataraman", "1987-1992", "Lawyer", "Indian law & independence", "Indian Constituent Assembly", R.drawable.ramaswamyvenkataraman),
30             new President("Shankar Dayal Sharma", "1992-1997", "Lawyer", "Governance", "Former

```

```

r VP and Governor", R.drawable.shrishankardayalsharma)
31     };
32     ArrayAdapter<String> adapter = new ArrayAdapter<>(getActivity(), android.R.layout.simple_
list_item_1, names);
33     listView.setAdapter(adapter);
34     listView.setOnItemClickListener((AdapterView<?> parent, View v, int position, long id) →
{((PresidentActivity) getActivity()).showPresidentDetails(presidents[position]);});
35     return view;
36 }
37 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 14 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <RelativeLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent">
6
7      <TextView
8          android:id="@+id/mainText"
9          android:layout_width="wrap_content"
10         android:layout_height="wrap_content"
11         android:text="Welcome to the President App"
12         android:layout_centerInParent="true"
13         android:textSize="18sp" />
14 </RelativeLayout>

```

app/src/main/res/layout/activity_president.xml

LAYOUT XML • 28 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent">
6      <!--xmlns:app="http://schemas.android.com/apk/res-auto"
7      xmlns:tools="http://schemas.android.com/tools"tools:context=".PresidentActivity"-->
8
9      <LinearLayout
10         xmlns:android="http://schemas.android.com/apk/res/android"
11         android:layout_width="match_parent"
12         android:layout_height="match_parent"
13         android:orientation="horizontal">
14
15         <fragment
16             android:id="@+id/fragment_list"
17             android:name="com.example.president.PresidentListFragment"
18             android:layout_width="0dp"
19             android:layout_weight="1"
20             android:layout_height="match_parent" />
21         <fragment
22             android:id="@+id/fragment_detail"
23             android:name="com.example.president.PresidentDetailFragment"
24             android:layout_width="0dp"
25             android:layout_weight="2"
26             android:layout_height="match_parent" />
27     </LinearLayout>
28 </LinearLayout>

```

app/src/main/res/layout/fragment_president_detail.xml

LAYOUT XML • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"

```

```

3     android:layout_width="match_parent"
4     android:layout_height="match_parent">
5
6     <LinearLayout
7         android:orientation="vertical"
8         android:padding="16dp"
9         android:layout_width="match_parent"
10        android:layout_height="wrap_content">
11
12        <ImageView
13            android:id="@+id/presidentImage"
14            android:layout_width="200dp"
15            android:layout_height="200dp"
16            android:layout_gravity="center"
17            android:scaleType="centerCrop"
18            android:src="@drawable/ic_launcher_background" />
19
20        <TextView
21            android:id="@+id/presidentDetails"
22            android:layout_width="wrap_content"
23            android:layout_height="wrap_content"
24            android:textSize="16sp"
25            android:layout_marginTop="16dp" />
26    </LinearLayout>
27</ScrollView>

```

app/src/main/res/layout/fragment_president_list.xml

LAYOUT XML • 6 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <ListView
3     xmlns:android="http://schemas.android.com/apk/res/android"
4     android:id="@+id/presidentListView"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent" />

```

app/src/main/res/menu/menu.xml

MENU XML • 17 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <menu xmlns:android="http://schemas.android.com/apk/res/android"
3     xmlns:app="http://schemas.android.com/apk/res-auto">
4     <item
5         android:id="@+id/action_toast"
6         android:title="Show Toast"
7         app:showAsAction="never" />
8     <item
9         android:id="@+id/action_exit"
10        android:title="Exit"
11        app:showAsAction="never" />
12    <item
13        android:id="@+id/action_president"
14        android:title="President"
15        app:showAsAction="never" />
16
17 </menu>

```

app/src/main/res/values/strings.xml

STRINGS XML • 5 LINES

```

1 <resources>
2     <string name="app_name">President</string>
3     <!-- TODO: Remove or change this placeholder text -->
4     <string name="hello_blank_fragment">Hello blank fragment</string>
5 </resources>

```

layout

Android Lab • 4 files

167 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.layout">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@android:drawable/ic_menu_compass"
11         android:label="@string/Ashok"
12         android:roundIcon="@android:drawable/ic_menu_compass"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Layout"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/layout/MainActivity.java

JAVA SOURCE • 70 LINES

```

1  package com.example.layout;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.text.TextUtils;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.EditText;
10 import android.widget.TextView;
11 import android.widget.Toast;
12
13 public class MainActivity extends AppCompatActivity {
14
15     EditText et1, et2;
16     TextView tv;
17     Button btn_sum, btn_reset;
18
19     @Override
20     protected void onCreate(Bundle savedInstanceState) {
21         super.onCreate(savedInstanceState);
22         setContentView(R.layout.activity_main);
23
24         et1 = findViewById(R.id.number1);
25         et2 = findViewById(R.id.number2);
26         tv = findViewById(R.id.textView3);
27

```

```

28     btn_sum = findViewById(R.id.calculate_button);
29     btn_reset = findViewById(R.id.reset_button);
30
31     btn_sum.setOnClickListener(new View.OnClickListener() {
32         @Override
33         public void onClick(View view) {
34
35             String val1 = et1.getText().toString().trim();
36             String val2 = et2.getText().toString().trim();
37
38             if (!TextUtils.isEmpty(val1) && !TextUtils.isEmpty(val2)) {
39
40                 int sum = Integer.parseInt(val1) + Integer.parseInt(val2);
41
42                 tv.setText("Result = " + sum);
43
44                 Toast.makeText(MainActivity.this,
45                     "Sum is " + sum,
46                     Toast.LENGTH_SHORT).show();
47
48             } else {
49                 Toast.makeText(MainActivity.this,
50                     "Enter both numbers first",
51                     Toast.LENGTH_SHORT).show();
52             }
53         }
54     });
55
56     btn_reset.setOnClickListener(new View.OnClickListener() {
57         @Override
58         public void onClick(View view) {
59
60             et1.setText("");
61             et2.setText("");
62             tv.setText("Result = 0");
63
64             Toast.makeText(MainActivity.this,
65                 "Inputs cleared",
66                 Toast.LENGTH_SHORT).show();
67         }
68     });
69 }
70 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 62 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent">
7
8      <EditText
9          android:id="@+id/number1"
10         android:layout_width="250dp"
11         android:layout_height="wrap_content"
12         android:hint="Enter Number 1"
13         android:inputType="number"
14         android:layout_marginTop="80dp"
15         app:layout_constraintTop_toTopOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintEnd_toEndOf="parent" />
18
19     <EditText
20         android:id="@+id/number2"
21         android:layout_width="250dp"
22         android:layout_height="wrap_content"
23         android:hint="Enter Number 2"
24         android:inputType="number"

```

```

25         android:layout_marginTop="20dp"
26         app:layout_constraintTop_toBottomOf="@id/number1"
27         app:layout_constraintStart_toStartOf="parent"
28         app:layout_constraintEnd_toEndOf="parent" />
29
30     <Button
31         android:id="@+id/calculate_button"
32         android:layout_width="wrap_content"
33         android:layout_height="wrap_content"
34         android:text="Calculate"
35         android:layout_marginTop="20dp"
36         app:layout_constraintTop_toBottomOf="@id/number2"
37         app:layout_constraintStart_toStartOf="parent"
38         app:layout_constraintEnd_toEndOf="parent" />
39
40     <Button
41         android:id="@+id/reset_button"
42         android:layout_width="wrap_content"
43         android:layout_height="wrap_content"
44         android:text="Reset"
45         android:layout_marginTop="15dp"
46         app:layout_constraintTop_toBottomOf="@id/calculate_button"
47         app:layout_constraintStart_toStartOf="parent"
48         app:layout_constraintEnd_toEndOf="parent" />
49
50     <TextView
51         android:id="@+id/textView3"
52         android:layout_width="wrap_content"
53         android:layout_height="wrap_content"
54         android:text="Result = 0"
55         android:textSize="24sp"
56         android:textStyle="bold"
57         android:layout_marginTop="30dp"
58         app:layout_constraintTop_toBottomOf="@id/reset_button"
59         app:layout_constraintStart_toStartOf="parent"
60         app:layout_constraintEnd_toEndOf="parent" />
61
62 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 8 LINES

```

1 <resources>
2     <string name="Ashok">layout</string>
3     <string name="input_hint1">Enter first number</string>
4     <string name="input_hint2">Enter second number</string>
5     <string name="Result_text">Result will appear here</string>
6     <string name="sum_button">Calculate sum</string>
7     <string name="reset_button">Reset</string>
8 </resources>

```

BankApp

AndroidStudioProjects • 9 files

344 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 28 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.bankapp">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.BankApp"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true"
19           tools:ignore="Instantiatable">
20           <intent-filter>
21               <action android:name="android.intent.action.MAIN" />
22
23               <category android:name="android.intent.category.LAUNCHER" />
24           </intent-filter>
25       </activity>
26   </application>
27
28 </manifest>

```

app/src/main/java/com/example/bankapp/MainActivity.java

JAVA SOURCE • 65 LINES

```

1  package com.example.bankapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5  import android.view.View;
6  import android.widget.AdapterView;
7  import android.widget.AdapterView.OnItemClickListener;
8  import android.widget.ArrayAdapter;
9  import android.widget.Spinner;
10 import android.widget.Toast;
11 import android.view.LayoutInflater;
12
13 public class MainActivity extends AppCompatActivity implements AdapterView.OnItemClickListener {
14
15     String[] Bank = {"--Select--", "SBI", "Union", "Canara", "HDFC", "ICICI"};
16
17     @Override
18     protected void onCreate(Bundle savedInstanceState) {
19         super.onCreate(savedInstanceState);
20         setContentView(R.layout.activity_main);
21
22         Spinner spin = findViewById(R.id.spinner);
23         ArrayAdapter<String> adapter = new ArrayAdapter<>(this, android.R.layout.simple_spinner_item, Bank);
24         adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
25         spin.setAdapter(adapter);
26     }
27 }

```

```

25         spin.setOnItemSelectedListener(this);
26     }
27
28     @Override
29     public void onItemSelected(AdapterView<?> adapterView, View view, int pos, long l) {
30         switch (pos) {
31             case 1:
32                 showBankToast(R.layout.activity_sbi);
33                 break;
34             case 2:
35                 showBankToast(R.layout.activity_union);
36                 break;
37             case 3:
38                 showBankToast(R.layout.activity_canara);
39                 break;
40             case 4:
41                 showBankToast(R.layout.activity_hdfc);
42                 break;
43             case 5:
44                 showBankToast(R.layout.activity_icici);
45                 break;
46             default:
47                 break;
48         }
49     }
50
51     private void showBankToast(int layoutResId) {
52         LayoutInflater inflater = getLayoutInflater();
53         View layout = inflater.inflate(layoutResId, null);
54         Toast toast = new Toast(getApplicationContext());
55         toast.setDuration(Toast.LENGTH_LONG);
56         toast.setView(layout);
57         toast.show();
58     }
59
60     @Override
61     public void onNothingSelected(AdapterView<?> adapterView)
62     {
63
64     }
65 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 48 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      tools:context=".MainActivity">
9
10     <TextView
11         android:id="@+id/tv1"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="Bank App"
15         android:textStyle="bold"
16         android:textColor="@color/black"
17         android:textSize="40sp"
18         app:layout_constraintBottom_toBottomOf="parent"
19         app:layout_constraintEnd_toEndOf="parent"
20         app:layout_constraintHorizontal_bias="0.538"
21         app:layout_constraintStart_toStartOf="parent"
22         app:layout_constraintTop_toTopOf="parent"
23         app:layout_constraintVertical_bias="0.096" />
24
25     <Spinner
26         android:id="@+id/spinner"

```

```

27         android:layout_width="407dp"
28         android:layout_height="43dp"
29         app:layout_constraintBottom_toBottomOf="parent"
30         app:layout_constraintEnd_toEndOf="parent"
31         app:layout_constraintStart_toStartOf="parent"
32         app:layout_constraintTop_toTopOf="parent"
33         app:layout_constraintVertical_bias="0.305" />
34
35     <TextView
36         android:id="@+id/tv2"
37         android:layout_width="135dp"
38         android:layout_height="28dp"
39         android:text="@string/bank"
40         android:textSize="20sp"
41         app:layout_constraintBottom_toBottomOf="parent"
42         app:layout_constraintEnd_toEndOf="parent"
43         app:layout_constraintHorizontal_bias="0.006"
44         app:layout_constraintStart_toStartOf="parent"
45         app:layout_constraintTop_toTopOf="parent"
46         app:layout_constraintVertical_bias="0.247" />
47
48 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_canara.xml

LAYOUT XML • 38 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv1"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="314dp"
14          android:layout_height="229dp"
15          android:layout_marginEnd="72dp"
16          android:background="@color/white"
17          android:src="@drawable/canara"
18          app:layout_constraintBottom_toBottomOf="parent"
19          app:layout_constraintEnd_toEndOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.82" />
22
23      <TextView
24          android:id="@+id/txt1"
25          android:layout_width="221dp"
26          android:layout_height="42dp"
27          android:text="@string/IFSCCanara"
28          android:textAlignment="center"
29          android:textSize="20dp"
30          android:background="@color/white"
31          app:layout_constraintBottom_toBottomOf="parent"
32          app:layout_constraintEnd_toEndOf="parent"
33          app:layout_constraintHorizontal_bias="0.547"
34          app:layout_constraintStart_toStartOf="parent"
35          app:layout_constraintTop_toTopOf="parent"
36          app:layout_constraintVertical_bias="0.436" />
37
38 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_hdfc.xml

LAYOUT XML • 37 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv2"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="223dp"
14          android:layout_height="198dp"
15          android:src="@drawable/hdfc"
16          app:layout_constraintBottom_toBottomOf="parent"
17          app:layout_constraintEnd_toEndOf="parent"
18          app:layout_constraintHorizontal_bias="0.42"
19          app:layout_constraintStart_toStartOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.774" />
22
23      <TextView
24          android:id="@+id/txt2"
25          android:layout_width="221dp"
26          android:layout_height="42dp"
27          android:text="@string/IFSCHDFC"
28          android:textAlignment="center"
29          android:textSize="20dp"
30          app:layout_constraintBottom_toBottomOf="parent"
31          app:layout_constraintEnd_toEndOf="parent"
32          app:layout_constraintHorizontal_bias="0.415"
33          app:layout_constraintStart_toStartOf="parent"
34          app:layout_constraintTop_toTopOf="parent"
35          app:layout_constraintVertical_bias="0.358" />
36
37  </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_icici.xml

LAYOUT XML • 36 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv3"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="154dp"
14          android:layout_height="163dp"
15          android:src="@drawable/icici"
16          app:layout_constraintBottom_toBottomOf="parent"
17          app:layout_constraintEnd_toEndOf="parent"
18          app:layout_constraintHorizontal_bias="0.497"
19          app:layout_constraintStart_toStartOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.591" />
22
23      <TextView
24          android:id="@+id/txt3"
25          android:layout_width="221dp"
26          android:layout_height="42dp"

```

```

27         android:text="@string/IFSCICICI"
28         android:textAlignment="center"
29         android:textSize="20dp"
30         app:layout_constraintBottom_toBottomOf="parent"
31         app:layout_constraintEnd_toEndOf="parent"
32         app:layout_constraintStart_toStartOf="parent"
33         app:layout_constraintTop_toTopOf="parent"
34         app:layout_constraintVertical_bias="0.275" />
35
36     </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_sbi.xml

LAYOUT XML • 37 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv4"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11
12      <ImageView
13          android:id="@+id/imageView"
14          android:layout_width="186dp"
15          android:layout_height="183dp"
16          android:src="@drawable/sbi"
17          app:layout_constraintBottom_toBottomOf="parent"
18          app:layout_constraintEnd_toEndOf="parent"
19          app:layout_constraintHorizontal_bias="0.496"
20          app:layout_constraintStart_toStartOf="parent"
21          app:layout_constraintTop_toTopOf="parent"
22          app:layout_constraintVertical_bias="0.749" />
23
24      <TextView
25          android:id="@+id/txt4"
26          android:layout_width="221dp"
27          android:layout_height="42dp"
28          android:text="@string/IFSCSBI"
29          android:textAlignment="center"
30          android:textSize="20dp"
31          app:layout_constraintBottom_toBottomOf="parent"
32          app:layout_constraintEnd_toEndOf="parent"
33          app:layout_constraintHorizontal_bias="0.521"
34          app:layout_constraintStart_toStartOf="parent"
35          app:layout_constraintTop_toTopOf="parent"
36          app:layout_constraintVertical_bias="0.368" />
37  </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_union.xml

LAYOUT XML • 36 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv5"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11
12      <ImageView
13          android:id="@+id/imageView"

```

```

14         android:layout_width="164dp"
15         android:layout_height="172dp"
16         android:src="@drawable/union"
17         app:layout_constraintBottom_toBottomOf="parent"
18         app:layout_constraintEnd_toEndOf="parent"
19         app:layout_constraintHorizontal_bias="0.496"
20         app:layout_constraintStart_toStartOf="parent"
21         app:layout_constraintTop_toTopOf="parent"
22         app:layout_constraintVertical_bias="0.649" />
23
24     <TextView
25         android:id="@+id/txt5"
26         android:layout_width="221dp"
27         android:layout_height="42dp"
28         android:text="@string/IFSCUnion"
29         android:textAlignment="center"
30         android:textSize="20dp"
31         app:layout_constraintBottom_toBottomOf="parent"
32         app:layout_constraintEnd_toEndOf="parent"
33         app:layout_constraintStart_toStartOf="parent"
34         app:layout_constraintTop_toTopOf="parent"
35         app:layout_constraintVertical_bias="0.307" />
36 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 19 LINES

```

1  <resources>
2      <string name="app_name">Bank App</string>
3
4
5
6      <string name="bank">Select Bank</string>
7
8      <string name="IFSCSBI">IFSC : SBIN0000001</string>
9
10     <string name="IFSCUnion">IFSC : UBIN0000001</string>
11
12     <string name="IFSCCanara">IFSC : CNRB0000001</string>
13
14     <string name="IFSCHDFC">IFSC : HDFC0000001</string>
15
16     <string name="IFSCICICI">IFSC : ICIC0000001</string>
17
18
19 </resources>

```

Bankapp2

AndroidStudioProjects • 9 files

338 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.bankapp">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.BankApp"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/bankapp/MainActivity.java

JAVA SOURCE • 64 LINES

```

1  package com.example.bankapp;
2  import androidx.appcompat.app.AppCompatActivity;
3  import android.os.Bundle;
4  import android.view.View;
5  import android.widget.AdapterView;
6  import android.widget.AdapterView.OnItemClickListener;
7  import android.widget.AdapterView.OnItemSelectedListener;
8  import android.widget.Toast;
9  import android.view.LayoutInflater;
10
11 public class MainActivity extends AppCompatActivity implements AdapterView.OnItemClickListener
12 {
13     String[] Bank = {"--Select--", "SBI", "Union", "Canara", "HDFC", "ICICI"};
14
15     @Override
16     protected void onCreate(Bundle savedInstanceState) {
17         super.onCreate(savedInstanceState);
18         setContentView(R.layout.activity_main);
19
20         Spinner spin = findViewById(R.id.spinner);
21         ArrayAdapter<String> adapter = new ArrayAdapter<(this, android.R.layout.simple_spinner_i
22 tem, Bank);
23         adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
24         spin.setAdapter(adapter);
25         spin.setOnItemSelectedListener(this);
26     }
27 }

```

```

26
27 @Override
28 public void onItemClick(AdapterView<?> adapterView, View view, int pos, long l) {
29     switch (pos) {
30         case 1:
31             showBankToast(R.layout.activity_sbi);
32             break;
33         case 2:
34             showBankToast(R.layout.activity_union);
35             break;
36         case 3:
37             showBankToast(R.layout.activity_canara);
38             break;
39         case 4:
40             showBankToast(R.layout.activity_hdfc);
41             break;
42         case 5:
43             showBankToast(R.layout.activity_icici);
44             break;
45         default:
46             break;
47     }
48 }
49
50 private void showBankToast(int layoutResId) {
51     LayoutInflater inflater = getLayoutInflater();
52     View layout = inflater.inflate(layoutResId, null);
53     Toast toast = new Toast(getApplicationContext());
54     toast.setDuration(Toast.LENGTH_LONG);
55     toast.setView(layout);
56     toast.show();
57 }
58
59 @Override
60 public void onNothingSelected(AdapterView<?> adapterView)
61 {
62     //todo
63 }
64 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 48 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      tools:context=".MainActivity">
9
10     <TextView
11         android:id="@+id/tv1"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="Bank App"
15         android:textStyle="bold"
16         android:textColor="@color/black"
17         android:textSize="40dp"
18         app:layout_constraintBottom_toBottomOf="parent"
19         app:layout_constraintEnd_toEndOf="parent"
20         app:layout_constraintHorizontal_bias="0.538"
21         app:layout_constraintStart_toStartOf="parent"
22         app:layout_constraintTop_toTopOf="parent"
23         app:layout_constraintVertical_bias="0.096" />
24
25     <Spinner
26         android:id="@+id/spinner"
27         android:layout_width="407dp"
28         android:layout_height="43dp"

```

```

29         app:layout_constraintBottom_toBottomOf="parent"
30         app:layout_constraintEnd_toEndOf="parent"
31         app:layout_constraintStart_toStartOf="parent"
32         app:layout_constraintTop_toTopOf="parent"
33         app:layout_constraintVertical_bias="0.305" />
34
35     <TextView
36         android:id="@+id/tv2"
37         android:layout_width="135dp"
38         android:layout_height="28dp"
39         android:text="@string/bank"
40         android:textSize="20dp"
41         app:layout_constraintBottom_toBottomOf="parent"
42         app:layout_constraintEnd_toEndOf="parent"
43         app:layout_constraintHorizontal_bias="0.006"
44         app:layout_constraintStart_toStartOf="parent"
45         app:layout_constraintTop_toTopOf="parent"
46         app:layout_constraintVertical_bias="0.247" />
47
48 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_canara.xml

LAYOUT XML • 38 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv1"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="265dp"
14          android:layout_height="212dp"
15          android:layout_marginEnd="72dp"
16          android:src="@drawable/canara"
17          android:background="@color/white"
18          app:layout_constraintBottom_toBottomOf="parent"
19          app:layout_constraintEnd_toEndOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.82" />
22
23      <TextView
24          android:id="@+id/txt1"
25          android:layout_width="221dp"
26          android:layout_height="42dp"
27          android:text="@string/IFSCCanara"
28          android:textAlignment="center"
29          android:textSize="20dp"
30          android:background="@color/white"
31          app:layout_constraintBottom_toBottomOf="parent"
32          app:layout_constraintEnd_toEndOf="parent"
33          app:layout_constraintHorizontal_bias="0.547"
34          app:layout_constraintStart_toStartOf="parent"
35          app:layout_constraintTop_toTopOf="parent"
36          app:layout_constraintVertical_bias="0.436" />
37
38 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_hdfc.xml

LAYOUT XML • 37 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout

```

```

3      android:id="@+id/tv2"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="223dp"
14          android:layout_height="198dp"
15          android:src="@drawable/hdfe"
16          app:layout_constraintBottom_toBottomOf="parent"
17          app:layout_constraintEnd_toEndOf="parent"
18          app:layout_constraintHorizontal_bias="0.42"
19          app:layout_constraintStart_toStartOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.774" />
22
23      <TextView
24          android:id="@+id/txt2"
25          android:layout_width="221dp"
26          android:layout_height="42dp"
27          android:text="@string/IFSCHDFC"
28          android:textAlignment="center"
29          android:textSize="20dp"
30          app:layout_constraintBottom_toBottomOf="parent"
31          app:layout_constraintEnd_toEndOf="parent"
32          app:layout_constraintHorizontal_bias="0.415"
33          app:layout_constraintStart_toStartOf="parent"
34          app:layout_constraintTop_toTopOf="parent"
35          app:layout_constraintVertical_bias="0.358" />
36
37  </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_icici.xml

LAYOUT XML • 36 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv3"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".ICICI">
10
11      <ImageView
12          android:id="@+id/imageView"
13          android:layout_width="154dp"
14          android:layout_height="163dp"
15          android:src="@drawable/icici"
16          app:layout_constraintBottom_toBottomOf="parent"
17          app:layout_constraintEnd_toEndOf="parent"
18          app:layout_constraintHorizontal_bias="0.497"
19          app:layout_constraintStart_toStartOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          app:layout_constraintVertical_bias="0.591" />
22
23      <TextView
24          android:id="@+id/txt3"
25          android:layout_width="221dp"
26          android:layout_height="42dp"
27          android:text="@string/IFSCICICI"
28          android:textAlignment="center"
29          android:textSize="20dp"
30          app:layout_constraintBottom_toBottomOf="parent"
31          app:layout_constraintEnd_toEndOf="parent"
32          app:layout_constraintStart_toStartOf="parent"

```

```

33         app:layout_constraintTop_toTopOf="parent"
34         app:layout_constraintVertical_bias="0.275" />
35
36     </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_sbi.xml

LAYOUT XML • 37 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv4"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11
12      <ImageView
13          android:id="@+id/imageView"
14          android:layout_width="186dp"
15          android:layout_height="183dp"
16          android:src="@drawable/sbi"
17          app:layout_constraintBottom_toBottomOf="parent"
18          app:layout_constraintEnd_toEndOf="parent"
19          app:layout_constraintHorizontal_bias="0.496"
20          app:layout_constraintStart_toStartOf="parent"
21          app:layout_constraintTop_toTopOf="parent"
22          app:layout_constraintVertical_bias="0.749" />
23
24      <TextView
25          android:id="@+id/txt4"
26          android:layout_width="221dp"
27          android:layout_height="42dp"
28          android:text="@string/IFSCSBI"
29          android:textAlignment="center"
30          android:textSize="20dp"
31          app:layout_constraintBottom_toBottomOf="parent"
32          app:layout_constraintEnd_toEndOf="parent"
33          app:layout_constraintHorizontal_bias="0.521"
34          app:layout_constraintStart_toStartOf="parent"
35          app:layout_constraintTop_toTopOf="parent"
36          app:layout_constraintVertical_bias="0.368" />
37  </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_union.xml

LAYOUT XML • 36 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      android:id="@+id/tv5"
4      xmlns:android="http://schemas.android.com/apk/res/android"
5      xmlns:app="http://schemas.android.com/apk/res-auto"
6      xmlns:tools="http://schemas.android.com/tools"
7      android:layout_width="match_parent"
8      android:layout_height="match_parent"
9      tools:context=".MainActivity">
10
11
12      <ImageView
13          android:id="@+id/imageView"
14          android:layout_width="164dp"
15          android:layout_height="172dp"
16          android:src="@drawable/union"
17          app:layout_constraintBottom_toBottomOf="parent"
18          app:layout_constraintEnd_toEndOf="parent"
19          app:layout_constraintHorizontal_bias="0.496"

```

```

20     app:layout_constraintStart_toStartOf="parent"
21     app:layout_constraintTop_toTopOf="parent"
22     app:layout_constraintVertical_bias="0.649" />
23
24     <TextView
25         android:id="@+id/txt5"
26         android:layout_width="221dp"
27         android:layout_height="42dp"
28         android:text="@string/IFSCUnion"
29         android:textAlignment="center"
30         android:textSize="20dp"
31         app:layout_constraintBottom_toBottomOf="parent"
32         app:layout_constraintEnd_toEndOf="parent"
33         app:layout_constraintStart_toStartOf="parent"
34         app:layout_constraintTop_toTopOf="parent"
35         app:layout_constraintVertical_bias="0.307" />
36 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 15 LINES

```

1 <resources>
2     <string name="app_name">Bank app</string>
3     <string name="bank">Select Bank</string>
4
5     <string name="IFSCSBI">IFSC : SBIN0000001</string>
6
7     <string name="IFSCUnion">IFSC : UBIN0000001</string>
8
9     <string name="IFSCCanara">IFSC : CNRB0000001</string>
10
11    <string name="IFSCHDFC">IFSC : HDFC0000001</string>
12
13    <string name="IFSCICICI">IFSC : ICIC0000001</string>
14
15 </resources>

```

Calculator

AndroidStudioProjects • 5 files

238 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.calculator">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Calculator"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/calculator/MainActivity.java

JAVA SOURCE • 72 LINES

```

1  package com.example.calculator;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.TextView;
9
10
11 public class MainActivity extends AppCompatActivity {
12
13     TextView txtDisplay;
14     String input="";
15     double num1=0;
16     String operator="";
17
18     @Override
19     protected void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_main);
22         txtDisplay=findViewById(R.id.txtDisplay);
23     }
24
25     public void numberClick(View view){
26         String value=((android.widget.Button)view).getText().toString();
27         input+=value;

```

```

28     txtDisplay.setText(input);
29 }
30
31 public void operatorClick(View view){
32     num1=Double.parseDouble(input);
33     operator=((android.widget.Button)view).getText().toString();
34     input="";
35 }
36
37 public void equalClick(View view){
38     double num2=Double.parseDouble(input);
39     double result=0;
40     switch (operator){
41         case "+":
42             result=num1+num2;
43             break;
44
45         case "-":
46             result=num1-num2;
47             break;
48
49         case "*":
50             result=num1*num2;
51             break;
52
53
54         case "/":
55             result=num1/num2;
56             break;
57
58     }
59     txtDisplay.setText(String.valueOf(result));
60     input=String.valueOf(result);
61 }
62
63 public void clearClick(View view){
64     input="";
65     num1=0;
66     operator="";
67
68     txtDisplay.setText("0");
69 }
70
71
72 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 60 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="vertical"
6      android:padding="16dp">
7
8      <TextView
9          android:id="@+id/txtDisplay"
10         android:layout_width="match_parent"
11         android:layout_height="wrap_content"
12         android:gravity="end"
13         android:padding="16dp"
14         android:text="0"
15         android:textSize="32sp" />
16
17     <GridLayout
18         android:layout_width="match_parent"
19         android:layout_height="wrap_content"
20         android:columnCount="4"
21         android:rowCount="5"
22         android:layout_marginTop="16dp">

```

```

23
24
25         <Button android:text="7" android:layout_row="0" android:layout_column="0" android:onClick
="numberClick" />
26
27         <Button
28             android:layout_row="0"
29             android:layout_column="1"
30             android:onClick="numberClick"
31             android:text="8" />
32
33         <Button android:text="9" android:layout_row="0" android:layout_column="2" android:onClick
="numberClick" />
34         <Button android:text="/" android:layout_row="0" android:layout_column="3" android:onClick
="operatorClick" />
35
36
37         <Button android:text="4" android:layout_row="1" android:layout_column="0" android:onClick
="numberClick" />
38         <Button android:text="5" android:layout_row="1" android:layout_column="1" android:onClick
="numberClick" />
39         <Button android:text="6" android:layout_row="1" android:layout_column="2" android:onClick
="numberClick" />
40         <Button android:text="*" android:layout_row="1" android:layout_column="3" android:onClick
="operatorClick" />
41
42
43         <Button android:text="1" android:layout_row="2" android:layout_column="0" android:onClick
="numberClick" />
44         <Button android:text="2" android:layout_row="2" android:layout_column="1" android:onClick
="numberClick" />
45         <Button android:text="3" android:layout_row="2" android:layout_column="2" android:onClick
="numberClick" />
46         <Button android:text="-" android:layout_row="2" android:layout_column="3" android:onClick
="operatorClick" />
47
48
49         <Button android:text="0" android:layout_row="3" android:layout_column="0" android:onClick
="numberClick" />
50         <Button android:text="." android:layout_row="3" android:layout_column="1" android:onClick
="numberClick" />
51         <Button android:text="=" android:layout_row="3" android:layout_column="2" android:onClick
="equalClick" />
52         <Button android:text="+" android:layout_row="3" android:layout_column="3" android:onClick
="operatorClick" />
53
54         <Button android:text="C" android:layout_row="4" android:layout_column="0"
55             android:layout_columnSpan="4"
56             android:onClick="clearClick"
57         />
58     </GridLayout>
59
60 </LinearLayout>

```

app/src/main/res/layout/activity_main2.xml

LAYOUT XML • 76 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp">
8
9
10     <TextView
11         android:id="@+id/tvDisplay"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"
14         android:text="0"

```

```

15         android:textSize="32sp"
16         android:gravity="end"
17         android:padding="12dp"
18         android:background="#EEEEEE" />
19
20
21     <LinearLayout
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
24         android:orientation="horizontal">
25
26         <Button android:text="7" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
27         <Button android:text="8" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
28         <Button android:text="9" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
29         <Button android:text="/" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
30     </LinearLayout>
31
32
33     <LinearLayout
34         android:layout_width="match_parent"
35         android:layout_height="wrap_content"
36         android:orientation="horizontal">
37
38         <Button android:text="4" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
39         <Button android:text="5" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
40         <Button android:text="6" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
41         <Button android:text="*" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
42     </LinearLayout>
43
44
45     <LinearLayout
46         android:layout_width="match_parent"
47         android:layout_height="wrap_content"
48         android:orientation="horizontal">
49
50         <Button android:text="1" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
51         <Button android:text="2" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
52         <Button android:text="3" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
53         <Button android:text="-" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
54     </LinearLayout>
55
56     <LinearLayout
57         android:layout_width="match_parent"
58         android:layout_height="wrap_content"
59         android:orientation="horizontal">
60
61         <Button android:text="0" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="2" />
62         <Button android:text="." android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
63         <Button android:text="+" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" />
64     </LinearLayout>
65
66
67     <LinearLayout
68         android:layout_width="match_parent"
69         android:layout_height="wrap_content"
70         android:orientation="horizontal">
71
72         <Button android:text="C" android:layout_width="0dp" android:layout_height="wrap_content"

```

```
android:layout_weight="1"/>
73     <Button android:text="=" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="3"/>
74     </LinearLayout>
75
76 </LinearLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">Calculator</string>
3 </resources>
```

DatabaseProgram

AndroidStudioProjects • 10 files

626 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.databaseprogram">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.DatabaseProgram"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25
26       <activity android:name=".ViewDataActivity" android:exported="false" />
27
28   </application>
29
30 </manifest>

```

app/src/main/java/com/example/databaseprogram/MainActivity.java

JAVA SOURCE • 123 LINES

```

1  package com.example.databaseprogram;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.app.DatePickerDialog;
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.ArrayAdapter;
9  import android.widget.Button;
10 import android.widget.DatePicker;
11 import android.widget.EditText;
12 import android.widget.Spinner;
13 import android.widget.Toast;
14 import java.util.Calendar;
15
16 public class MainActivity extends AppCompatActivity {
17
18     EditText etName, etPassword, etDob, etEmail, etPhone;
19     Spinner spinnerState;
20     Button btnSave, btnView;
21     DatabaseHelper db;
22
23     boolean isEdit = false;
24     int editId = -1;

```

```

25
26 @Override
27 protected void onCreate(Bundle savedInstanceState) {
28     super.onCreate(savedInstanceState);
29     setContentView(R.layout.activity_main);
30
31     etName = findViewById(R.id.etName);
32     etPassword = findViewById(R.id.etPassword);
33     etDob = findViewById(R.id.etDob);
34     etEmail = findViewById(R.id.etEmail);
35     etPhone = findViewById(R.id.etPhone);
36     spinnerState = findViewById(R.id.spState);
37     btnSave = findViewById(R.id.btnSave);
38     btnView = findViewById(R.id.btnView);
39
40     db = new DatabaseHelper(this);
41
42     // Load states from strings.xml
43     ArrayAdapter<CharSequence> adapter = ArrayAdapter.createFromResource(
44         this,
45         R.array.states_array,
46         android.R.layout.simple_spinner_item
47     );
48     adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
49     spinnerState.setAdapter(adapter);
50
51     // Date Picker
52     etDob.setOnClickListener(v → showDatePicker());
53
54     // Check if edit mode
55     Intent intent = getIntent();
56     if (intent.hasExtra("isEdit") && intent.getBooleanExtra("isEdit", false)) {
57         isEdit = true;
58         editId = intent.getIntExtra("id", -1);
59         etName.setText(intent.getStringExtra("name"));
60         etPassword.setText(intent.getStringExtra("password"));
61         etDob.setText(intent.getStringExtra("dob"));
62         etEmail.setText(intent.getStringExtra("email"));
63         etPhone.setText(intent.getStringExtra("phone"));
64
65         // Set spinner selection
66         String state = intent.getStringExtra("state");
67         int pos = adapter.getPosition(state);
68         spinnerState.setSelection(pos);
69     }
70
71     // Save / Update button
72     btnSave.setOnClickListener(v → {
73         String name = etName.getText().toString();
74         String password = etPassword.getText().toString();
75         String dob = etDob.getText().toString();
76         String email = etEmail.getText().toString();
77         String phone = etPhone.getText().toString();
78         String state = spinnerState.getSelectedItem().toString();
79
80         if (name.isEmpty() || password.isEmpty() || dob.isEmpty() || email.isEmpty() || phone.isEmpty() || state.equals("Select State")) {
81             Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show();
82             return;
83         }
84
85         if (isEdit) {
86             Student student = new Student(editId, name, password, dob, email, phone, state);
87             db.updateStudent(student);
88             Toast.makeText(this, "Updated Successfully", Toast.LENGTH_SHORT).show();
89         } else {
90             Student student = new Student(0, name, password, dob, email, phone, state);
91             db.insertStudent(student);
92             Toast.makeText(this, "Saved Successfully", Toast.LENGTH_SHORT).show();
93         }
94         clearFields();
95     });
96

```

```

97      // View all students
98      btnView.setOnClickListener(v → {
99          startActivity(new Intent(MainActivity.this, ViewDataActivity.class));
100      });
101  }
102
103  private void showDatePicker() {
104      Calendar calendar = Calendar.getInstance();
105      int year = calendar.get(Calendar.YEAR);
106      int month = calendar.get(Calendar.MONTH);
107      int day = calendar.get(Calendar.DAY_OF_MONTH);
108
109      DatePickerDialog dialog = new DatePickerDialog(this, (view, year1, month1, dayOfMonth) →
{
110          etDob.setText(dayOfMonth + "/" + (month1 + 1) + "/" + year1);
111      }, year, month, day);
112      dialog.show();
113  }
114
115  private void clearFields() {
116      etName.setText("");
117      etPassword.setText("");
118      etDob.setText("");
119      etEmail.setText("");
120      etPhone.setText("");
121      spinnerState.setSelection(0);
122  }
123  }

```

app/src/main/java/com/example/databaseprogram/CustomAdapter.java

JAVA SOURCE • 85 LINES

```

1  package com.example.databaseprogram;
2
3  import android.content.Context;
4  import android.content.Intent;
5  import android.view.LayoutInflater;
6  import android.view.View;
7  import android.view.ViewGroup;
8  import android.widget.BaseAdapter;
9  import android.widget.Button;
10 import android.widget.TextView;
11 import android.widget.Toast;
12 import java.util.List;
13
14 public class CustomAdapter extends BaseAdapter {
15
16     private Context context;
17     private List<Student> studentList;
18     private DatabaseHelper db;
19
20     public CustomAdapter(Context context, List<Student> studentList, DatabaseHelper db) {
21         this.context = context;
22         this.studentList = studentList;
23         this.db = db;
24     }
25
26     @Override
27     public int getCount() {
28         return studentList.size();
29     }
30
31     @Override
32     public Object getItem(int position) {
33         return studentList.get(position);
34     }
35
36     @Override
37     public long getItemId(int position) {
38         return studentList.get(position).getId();
39     }

```

```

40
41 @Override
42 public View getView(int position, View convertView, ViewGroup parent) {
43     if (convertView == null) {
44         convertView = LayoutInflater.from(context).inflate(R.layout.list_item, parent, false);
45     }
46
47     TextView tvDetails = convertView.findViewById(R.id.tvDetails);
48     Button btnEdit = convertView.findViewById(R.id.btnEdit);
49     Button btnDelete = convertView.findViewById(R.id.btnDelete);
50
51     Student student = studentList.get(position);
52
53     String details = "ID: " + student.getId() +
54         "\nName: " + student.getName() +
55         "\nDOB: " + student.getDob() +
56         "\nEmail: " + student.getEmail() +
57         "\nPhone: " + student.getPhone() +
58         "\nState: " + student.getState();
59     tvDetails.setText(details);
60
61     // Edit Button
62     btnEdit.setOnClickListener(v → {
63         Intent intent = new Intent(context, MainActivity.class);
64         intent.putExtra("isEdit", true);
65         intent.putExtra("id", student.getId());
66         intent.putExtra("name", student.getName());
67         intent.putExtra("password", student.getPassword());
68         intent.putExtra("dob", student.getDob());
69         intent.putExtra("email", student.getEmail());
70         intent.putExtra("phone", student.getPhone());
71         intent.putExtra("state", student.getState());
72         context.startActivity(intent);
73     });
74
75     // Delete Button
76     btnDelete.setOnClickListener(v → {
77         db.deleteStudent(student.getId());
78         studentList.remove(position);
79         notifyDataSetChanged();
80         Toast.makeText(context, "Deleted Successfully", Toast.LENGTH_SHORT).show();
81     });
82
83     return convertView;
84 }
85 }

```

app/src/main/java/com/example/databaseprogram/DatabaseHelper.java

JAVA SOURCE • 130 LINES

```

1 package com.example.databaseprogram;
2
3 import android.content.ContentValues;
4 import android.content.Context;
5 import android.database.Cursor;
6 import android.database.sqlite.SQLiteDatabase;
7 import android.database.sqlite.SQLiteOpenHelper;
8
9 import java.util.ArrayList;
10 import java.util.List;
11
12 public class DatabaseHelper extends SQLiteOpenHelper {
13
14     private static final String DATABASE_NAME = "myDB";
15     private static final int DATABASE_VERSION = 1;
16
17     private static final String TABLE_NAME = "registration";
18     private static final String COL_ID = "id";
19     private static final String COL_NAME = "name";
20     private static final String COL_PASSWORD = "password";

```

```

21 private static final String COL_DOB = "dob";
22 private static final String COL_EMAIL = "email";
23 private static final String COL_PHONE = "phone";
24 private static final String COL_STATE = "state";
25
26 public DatabaseHelper(Context context) {
27     super(context, DATABASE_NAME, null, DATABASE_VERSION);
28 }
29
30 @Override
31 public void onCreate(SQLiteDatabase db) {
32     String createTable = "CREATE TABLE " + TABLE_NAME + " ("
33         + COL_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
34         + COL_NAME + " TEXT, "
35         + COL_PASSWORD + " TEXT, "
36         + COL_DOB + " TEXT, "
37         + COL_EMAIL + " TEXT, "
38         + COL_PHONE + " TEXT, "
39         + COL_STATE + " TEXT)";
40     db.execSQL(createTable);
41 }
42
43 @Override
44 public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
45     db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
46     onCreate(db);
47 }
48
49 // Insert new student
50 public void insertStudent(Student student) {
51     SQLiteDatabase db = this.getWritableDatabase();
52     ContentValues values = new ContentValues();
53     values.put(COL_NAME, student.getName());
54     values.put(COL_PASSWORD, student.getPassword());
55     values.put(COL_DOB, student.getDob());
56     values.put(COL_EMAIL, student.getEmail());
57     values.put(COL_PHONE, student.getPhone());
58     values.put(COL_STATE, student.getState());
59     db.insert(TABLE_NAME, null, values);
60     db.close();
61 }
62
63 // Update existing student
64 public void updateStudent(Student student) {
65     SQLiteDatabase db = this.getWritableDatabase();
66     ContentValues values = new ContentValues();
67     values.put(COL_NAME, student.getName());
68     values.put(COL_PASSWORD, student.getPassword());
69     values.put(COL_DOB, student.getDob());
70     values.put(COL_EMAIL, student.getEmail());
71     values.put(COL_PHONE, student.getPhone());
72     values.put(COL_STATE, student.getState());
73
74     db.update(TABLE_NAME, values, COL_ID + " = ?", new String[]{String.valueOf(student.getId
75 ())});
76     db.close();
77 }
78
79 // Delete student by ID
80 public void deleteStudent(int id) {
81     SQLiteDatabase db = this.getWritableDatabase();
82     db.delete(TABLE_NAME, COL_ID + " = ?", new String[]{String.valueOf(id)});
83     db.close();
84 }
85
86 // Get all students (no filter)
87 public List<Student> getAllStudents() {
88     return getAllStudents(null);
89 }
90
91 // Get all students with optional search
92 public List<Student> getAllStudents(String searchQuery) {
93     List<Student> studentList = new ArrayList<>();

```

```

93     SQLiteDatabase db = this.getReadableDatabase();
94
95     String sql;
96     String[] args = null;
97
98     if (searchQuery != null && !searchQuery.isEmpty()) {
99         sql = "SELECT * FROM " + TABLE_NAME +
100             " WHERE " + COL_NAME + " LIKE ? OR " +
101             COL_STATE + " LIKE ? OR " +
102             COL_ID + " LIKE ?";
103         String filter = "%" + searchQuery + "%";
104         args = new String[]{filter, filter, filter};
105     } else {
106         sql = "SELECT * FROM " + TABLE_NAME;
107     }
108
109     Cursor cursor = db.rawQuery(sql, args);
110
111     if (cursor.moveToFirst()) {
112         do {
113             int id = cursor.getInt(cursor.getColumnIndexOrThrow(COL_ID));
114             String name = cursor.getString(cursor.getColumnIndexOrThrow(COL_NAME));
115             String password = cursor.getString(cursor.getColumnIndexOrThrow(COL_PASSWORD));
116             String dob = cursor.getString(cursor.getColumnIndexOrThrow(COL_DOB));
117             String email = cursor.getString(cursor.getColumnIndexOrThrow(COL_EMAIL));
118             String phone = cursor.getString(cursor.getColumnIndexOrThrow(COL_PHONE));
119             String state = cursor.getString(cursor.getColumnIndexOrThrow(COL_STATE));
120
121             Student student = new Student(id, name, password, dob, email, phone, state);
122             studentList.add(student);
123         } while (cursor.moveToNext());
124     }
125
126     cursor.close();
127     db.close();
128     return studentList;
129 }
130 }

```

app/src/main/java/com/example/databaseprogram/Student.java

JAVA SOURCE • 38 LINES

```

1  package com.example.databaseprogram;
2
3  public class Student {
4      private int id;
5      private String name;
6      private String password;
7      private String dob;
8      private String email;
9      private String phone;
10     private String state;
11
12     public Student(int id, String name, String password, String dob, String email, String phone,
String state) {
13         this.id = id;
14         this.name = name;
15         this.password = password;
16         this.dob = dob;
17         this.email = email;
18         this.phone = phone;
19         this.state = state;
20     }
21
22     // Getters
23     public int getId() { return id; }
24     public String getName() { return name; }
25     public String getPassword() { return password; }
26     public String getDob() { return dob; }
27     public String getEmail() { return email; }
28     public String getPhone() { return phone; }

```

```

29     public String getState() { return state; }
30
31     // Setters
32     public void setName(String name) { this.name = name; }
33     public void setPassword(String password) { this.password = password; }
34     public void setDob(String dob) { this.dob = dob; }
35     public void setEmail(String email) { this.email = email; }
36     public void setPhone(String phone) { this.phone = phone; }
37     public void setState(String state) { this.state = state; }
38 }

```

app/src/main/java/com/example/databaseprogram/ViewDataActivity.java

JAVA SOURCE • 47 LINES

```

1  package com.example.databaseprogram;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5  import android.text.Editable;
6  import android.text.TextWatcher;
7  import android.widget.EditText;
8  import android.widget.ListView;
9  import java.util.List;
10
11  public class ViewDataActivity extends AppCompatActivity {
12
13      EditText etSearch;
14      ListView listView;
15      DatabaseHelper db;
16      List<Student> students;
17      CustomAdapter adapter;
18
19      @Override
20      protected void onCreate(Bundle savedInstanceState) {
21          super.onCreate(savedInstanceState);
22          setContentView(R.layout.activity_view_data);
23
24          etSearch = findViewById(R.id.etSearch);
25          listView = findViewById(R.id.listView);
26          db = new DatabaseHelper(this);
27
28          loadStudents("");
29
30          etSearch.addTextChangedListener(new TextWatcher() {
31              @Override
32              public void beforeTextChanged(CharSequence s, int start, int count, int after) {}
33              @Override
34              public void onTextChanged(CharSequence s, int start, int before, int count) {
35                  loadStudents(s.toString());
36              }
37              @Override
38              public void afterTextChanged(Editable s) {}
39          });
40      }
41
42      private void loadStudents(String query) {
43          students = db.getAllStudents(query);
44          adapter = new CustomAdapter(this, students, db);
45          listView.setAdapter(adapter);
46      }
47  }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 73 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"

```

```

4      android:layout_height="match_parent"
5      android:padding="16dp">
6
7      <LinearLayout
8          android:orientation="vertical"
9          android:layout_width="match_parent"
10         android:layout_height="wrap_content">
11
12         <EditText
13             android:id="@+id/etName"
14             android:layout_width="match_parent"
15             android:layout_height="wrap_content"
16             android:hint="Student Name"
17             android:inputType="textPersonName" />
18
19         <EditText
20             android:id="@+id/etPassword"
21             android:layout_width="match_parent"
22             android:layout_height="wrap_content"
23             android:hint="Password"
24             android:inputType="textPassword" />
25
26         <!-- Date of Birth -->
27         <EditText
28             android:id="@+id/etDob"
29             android:layout_width="match_parent"
30             android:layout_height="wrap_content"
31             android:hint="Date of Birth"
32             android:focusable="false"
33             android:clickable="true"
34             android:inputType="none" />
35
36         <EditText
37             android:id="@+id/etEmail"
38             android:layout_width="match_parent"
39             android:layout_height="wrap_content"
40             android:hint="Email"
41             android:inputType="textEmailAddress" />
42
43         <EditText
44             android:id="@+id/etPhone"
45             android:layout_width="match_parent"
46             android:layout_height="wrap_content"
47             android:hint="Phone Number"
48             android:inputType="phone" />
49
50         <!-- State Spinner -->
51         <Spinner
52             android:id="@+id/spState"
53             android:layout_width="match_parent"
54             android:layout_height="wrap_content" />
55
56         <!-- Save Button -->
57         <Button
58             android:id="@+id/btnSave"
59             android:layout_width="match_parent"
60             android:layout_height="wrap_content"
61             android:text="Save"
62             android:layout_marginTop="16dp" />
63
64         <!-- View Data Button -->
65         <Button
66             android:id="@+id/btnView"
67             android:layout_width="match_parent"
68             android:layout_height="wrap_content"
69             android:text="View All Data"
70             android:layout_marginTop="8dp" />
71
72     </LinearLayout>
73 </ScrollView>

```

app/src/main/res/layout/activity_view_data.xml

LAYOUT XML • 28 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="vertical"
6      android:padding="16dp">
7
8      <!-- Search Bar -->
9      <EditText
10         android:id="@+id/etSearch"
11         android:layout_width="match_parent"
12         android:layout_height="wrap_content"
13         android:hint="Search by name, ID, or state"
14         android:padding="10dp"
15         android:drawableStart="@android:drawable/ic_menu_search"
16         android:inputType="text"
17         android:background="@android:drawable/edit_text"
18         android:layout_marginBottom="10dp" />
19
20     <!-- List View -->
21     <ListView
22         android:id="@+id/listView"
23         android:layout_width="match_parent"
24         android:layout_height="match_parent"
25         android:dividerHeight="1dp"
26         android:divider="@android:color/darker_gray" />
27
28 </LinearLayout>

```

app/src/main/res/layout/list_item.xml

LAYOUT XML • 37 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:orientation="vertical"
4      android:padding="10dp"
5      android:background="@android:color/white"
6      android:layout_width="match_parent"
7      android:layout_height="wrap_content">
8
9      <TextView
10         android:id="@+id/tvDetails"
11         android:textSize="16sp"
12         android:textColor="@android:color/black"
13         android:layout_width="match_parent"
14         android:layout_height="wrap_content" />
15
16     <LinearLayout
17         android:orientation="horizontal"
18         android:layout_width="match_parent"
19         android:layout_height="wrap_content"
20         android:gravity="end"
21         android:layout_marginTop="5dp">
22
23         <Button
24             android:id="@+id/btnEdit"
25             android:text="Edit"
26             android:layout_width="wrap_content"
27             android:layout_height="wrap_content" />
28
29         <Button
30             android:id="@+id/btnDelete"
31             android:text="Delete"
32             android:layout_width="wrap_content"
33             android:layout_height="wrap_content"
34             android:layout_marginLeft="10dp" />
35     </LinearLayout>

```

```
36  
37 </LinearLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 35 LINES

```
1 <resources>  
2   <string name="app_name">Database Program</string>  
3  
4   <string-array name="states_array">  
5     <item>Select State</item>  
6     <item>Andhra Pradesh</item>  
7     <item>Arunachal Pradesh</item>  
8     <item>Assam</item>  
9     <item>Bihar</item>  
10    <item>Chhattisgarh</item>  
11    <item>Goa</item>  
12    <item>Gujarat</item>  
13    <item>Haryana</item>  
14    <item>Himachal Pradesh</item>  
15    <item>Jharkhand</item>  
16    <item>Karnataka</item>  
17    <item>Kerala</item>  
18    <item>Madhya Pradesh</item>  
19    <item>Maharashtra</item>  
20    <item>Manipur</item>  
21    <item>Meghalaya</item>  
22    <item>Mizoram</item>  
23    <item>Nagaland</item>  
24    <item>Odisha</item>  
25    <item>Punjab</item>  
26    <item>Rajasthan</item>  
27    <item>Sikkim</item>  
28    <item>Tamil Nadu</item>  
29    <item>Telangana</item>  
30    <item>Tripura</item>  
31    <item>Uttar Pradesh</item>  
32    <item>Uttarakhand</item>  
33    <item>West Bengal</item>  
34   </string-array>  
35 </resources>
```

DateTimePicker

AndroidStudioProjects • 5 files

221 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.datetimepicker">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.DateTimePicker"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/datetimepicker/MainActivity.java

JAVA SOURCE • 108 LINES

```

1  package com.example.datetimepicker;
2
3  import android.app.DatePickerDialog;
4  import android.app.TimePickerDialog;
5  import android.os.Bundle;
6  import android.os.Handler;
7  import android.view.Menu;
8  import android.view.MenuItem;
9  import android.widget.Button;
10 import android.widget.DatePicker;
11 import android.widget.ProgressBar;
12 import android.widget.TimePicker;
13 import android.widget.Toast;
14
15 import androidx.appcompat.app.AppCompatActivity;
16
17 import com.google.android.material.floatingactionbutton.FloatingActionButton;
18
19 import java.util.Calendar;
20
21 public class MainActivity extends AppCompatActivity {
22
23     private Button btnDate, btnTime, btnStartProgress;
24     private ProgressBar progressBar;
25     private FloatingActionButton fab;
26     private int progressStatus = 0;
27     private Handler handler = new Handler();

```

```

28
29 @Override
30 protected void onCreate(Bundle savedInstanceState) {
31     super.onCreate(savedInstanceState);
32     setContentView(R.layout.activity_main);
33
34     btnDate = findViewById(R.id.btnDate);
35     btnTime = findViewById(R.id.btnTime);
36     btnStartProgress = findViewById(R.id.btnStartProgress);
37     progressBar = findViewById(R.id.progressBar);
38     fab = findViewById(R.id.fab);
39
40     btnDate.setOnClickListener(v → showDatePicker());
41     btnTime.setOnClickListener(v → showTimePicker());
42     btnStartProgress.setOnClickListener(v → startProgress());
43     fab.setOnClickListener(v → Toast.makeText(this, "FAB Clicked!", Toast.LENGTH_SHORT).show
44 );
45 }
46
47 private void showDatePicker() {
48     Calendar calendar = Calendar.getInstance();
49     new DatePickerDialog(this, (DatePicker view, int year, int month, int dayOfMonth) →
50         Toast.makeText(this, "Date: " + dayOfMonth + "/" + (month + 1) + "/" + year, Toas
51 t.LENGTH_SHORT).show(),
52         calendar.get(Calendar.YEAR),
53         calendar.get(Calendar.MONTH),
54         calendar.get(Calendar.DAY_OF_MONTH)).show();
55 }
56
57 private void showTimePicker() {
58     Calendar calendar = Calendar.getInstance();
59     new TimePickerDialog(this, (TimePicker view, int hour, int minute) →
60         Toast.makeText(this, "Time: " + hour + ":" + minute, Toast.LENGTH_SHORT).show(),
61         calendar.get(Calendar.HOUR_OF_DAY),
62         calendar.get(Calendar.MINUTE),
63         true).show();
64 }
65
66 private void startProgress() {
67     progressStatus = 0;
68     progressBar.setVisibility(ProgressBar.VISIBLE);
69     progressBar.setProgress(0);
70
71     new Thread() → {
72         while (progressStatus < 100) {
73             progressStatus += 10;
74             handler.post() → progressBar.setProgress(progressStatus));
75             try {
76                 Thread.sleep(300);
77             } catch (InterruptedException e) {
78                 e.printStackTrace();
79             }
80             handler.post() → Toast.makeText(this, "Progress Done!", Toast.LENGTH_SHORT).show
81 ();
82         }.start();
83     }
84
85 @Override
86 public boolean onCreateOptionsMenu(Menu menu) {
87     getMenuInflater().inflate(R.menu.main_menu, menu);
88     return true;
89 }
90
91 @Override
92 public boolean onOptionsItemSelected(MenuItem item) {
93     switch (item.getItemId()) {
94         case R.id.option1:
95             Toast.makeText(this, "Option 1 selected", Toast.LENGTH_SHORT).show();
96             return true;
97         case R.id.option2:
98             Toast.makeText(this, "Option 2 selected", Toast.LENGTH_SHORT).show();
99             return true;
100     }

```

```

98         case R.id.sub_option1:
99             Toast.makeText(this, "Sub Option 1 selected", Toast.LENGTH_SHORT).show();
100             return true;
101         case R.id.sub_option2:
102             Toast.makeText(this, "Sub Option 2 selected", Toast.LENGTH_SHORT).show();
103             return true;
104         default:
105             return super.onOptionsItemSelected(item);
106     }
107 }
108 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 59 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2
3  <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      android:padding="16dp">
8
9      <!-- Date Picker Button -->
10     <Button
11         android:id="@+id/btnDate"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="Pick Date" />
15
16     <!-- Time Picker Button -->
17     <Button
18         android:id="@+id/btnTime"
19         android:layout_width="wrap_content"
20         android:layout_height="wrap_content"
21         android:layout_below="@id/btnDate"
22         android:layout_marginTop="12dp"
23         android:text="Pick Time" />
24
25     <!-- Progress Bar and Start Button -->
26     <Button
27         android:id="@+id/btnStartProgress"
28         android:layout_width="wrap_content"
29         android:layout_height="wrap_content"
30         android:layout_below="@id/btnTime"
31         android:layout_marginTop="16dp"
32         android:text="Start Progress" />
33
34     <ProgressBar
35         android:id="@+id/progressBar"
36         style="@android:style/Widget.DeviceDefault.Light.ProgressBar.Horizontal"
37         android:layout_width="match_parent"
38         android:layout_height="match_parent"
39         android:max="100"
40         android:progress="0"
41         android:layout_below="@id/btnStartProgress"
42         android:layout_marginTop="10dp"
43         android:visibility="gone" />
44
45     <!-- Floating Action Button -->
46     <com.google.android.material.floatingactionbutton.FloatingActionButton
47         android:id="@+id/fab"
48         android:layout_width="wrap_content"
49         android:layout_height="wrap_content"
50         android:layout_alignParentEnd="true"
51         android:layout_alignParentBottom="true"
52         android:layout_marginStart="16dp"
53         android:layout_marginTop="16dp"
54         android:layout_marginEnd="16dp"
55         android:layout_marginBottom="16dp"
56         android:contentDescription="Add"

```

```
57         app:srcCompat="@android:drawable/ic_input_add" />
58
59     </RelativeLayout>
```

app/src/main/res/menu/main_menu.xml

MENU XML • 24 LINES

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <menu xmlns:android="http://schemas.android.com/apk/res/android">
3
4      <item
5          android:id="@+id/option1"
6          android:title="Option 1" />
7
8      <item
9          android:id="@+id/option2"
10         android:title="Option 2" />
11
12     <item
13         android:title="More Options">
14         <menu>
15             <item
16                 android:id="@+id/sub_option1"
17                 android:title="Sub Option 1" />
18             <item
19                 android:id="@+id/sub_option2"
20                 android:title="Sub Option 2" />
21         </menu>
22     </item>
23
24 </menu>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1  <resources>
2      <string name="app_name">DateTime Picker</string>
3  </resources>
```

Demo

AndroidStudioProjects • 6 files

111 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.demo">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.Demo"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity2"
18           android:exported="false" />
19       <activity
20           android:name=".MainActivity"
21           android:exported="true">
22           <intent-filter>
23               <action android:name="android.intent.action.MAIN" />
24
25               <category android:name="android.intent.category.LAUNCHER" />
26           </intent-filter>
27       </activity>
28   </application>
29
30 </manifest>

```

app/src/main/java/com/example/demo/MainActivity.java

JAVA SOURCE • 20 LINES

```

1  package com.example.demo;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8
9  public class MainActivity extends AppCompatActivity {
10
11      @Override
12      protected void onCreate(Bundle savedInstanceState) {
13          super.onCreate(savedInstanceState);
14          setContentView(R.layout.activity_main);
15      }
16      public void toast(View view) {
17          Intent intent = new Intent();
18          startActivity(intent);
19      }
20  }

```

app/src/main/java/com/example/demo/MainActivity2.java

JAVA SOURCE • 14 LINES

```

1  package com.example.demo;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6
7  public class MainActivity2 extends AppCompatActivity {
8
9      @Override
10     protected void onCreate(Bundle savedInstanceState) {
11         super.onCreate(savedInstanceState);
12         setContentView(R.layout.activity_main2);
13     }
14 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 35 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <TextView
10         android:layout_width="327dp"
11         android:layout_height="102dp"
12         android:text="Hello World!"
13         android:textSize="24sp"
14         app:layout_constraintBottom_toBottomOf="parent"
15         app:layout_constraintEnd_toEndOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintTop_toTopOf="parent"
18         app:layout_constraintVertical_bias="0.12" />
19
20
21
22         <Button
23             android:id="@+id/button5"
24             android:layout_width="wrap_content"
25             android:layout_height="wrap_content"
26             android:text="Button"
27             android:onClick="toast"
28             tools:layout_editor_absoluteX="48dp"
29             tools:layout_editor_absoluteY="200dp" />
30
31
32
33
34
35 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main2.xml

LAYOUT XML • 9 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"

```

```
7     tools:context=".MainActivity2">  
8  
9 </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>  
2     <string name="app_name">Demo</string>  
3 </resources>
```

Dialerapp

AndroidStudioProjects • 6 files

329 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.dialerapp">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@android:drawable/sym_action_call"
11       android:label="@string/app_name"
12       android:roundIcon="@android:drawable/ic_menu_call"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.DialerApp"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity2"
18           android:exported="false" />
19       <activity
20           android:name=".MainActivity"
21           android:exported="true">
22           <intent-filter>
23               <action android:name="android.intent.action.MAIN" />
24
25               <category android:name="android.intent.category.LAUNCHER" />
26           </intent-filter>
27       </activity>
28   </application>
29
30 </manifest>

```

app/src/main/java/com/example/dialerapp/MainActivity.java

JAVA SOURCE • 109 LINES

```

1  package com.example.dialerapp;
2
3  import android.content.Intent;
4  import android.net.Uri;
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.Button;
8  import android.widget.EditText;
9  import android.widget.ImageButton;
10 import android.widget.Toast;
11
12 import androidx.appcompat.app.AppCompatActivity;
13
14 public class MainActivity extends AppCompatActivity {
15
16     EditText edtPhone;
17
18     Button btn0, btn1, btn2, btn3, btn4, btn5, btn6, btn7, btn8, btn9;
19     Button btnPlus, btnClear;
20
21     ImageButton btnCall;
22
23     @Override
24     protected void onCreate(Bundle savedInstanceState) {

```

```

25     super.onCreate(savedInstanceState);
26     setContentView(R.layout.activity_main);
27
28     edtPhone = findViewById(R.id.etNumber);
29
30     btn0 = findViewById(R.id.btn0);
31     btn1 = findViewById(R.id.btn1);
32     btn2 = findViewById(R.id.btn2);
33     btn3 = findViewById(R.id.btn3);
34     btn4 = findViewById(R.id.btn4);
35     btn5 = findViewById(R.id.btn5);
36     btn6 = findViewById(R.id.btn6);
37     btn7 = findViewById(R.id.btn7);
38     btn8 = findViewById(R.id.btn8);
39     btn9 = findViewById(R.id.btn9);
40
41     btnPlus = findViewById(R.id.btnPlus);
42     btnClear = findViewById(R.id.btnClear);
43     btnCall = findViewById(R.id.btnCall);
44
45
46     View.OnClickListener numClick = v → {
47         Button b = (Button) v;
48         edtPhone.append(b.getText().toString());
49     };
50
51     btn0.setOnClickListener(numClick);
52     btn1.setOnClickListener(numClick);
53     btn2.setOnClickListener(numClick);
54     btn3.setOnClickListener(numClick);
55     btn4.setOnClickListener(numClick);
56     btn5.setOnClickListener(numClick);
57     btn6.setOnClickListener(numClick);
58     btn7.setOnClickListener(numClick);
59     btn8.setOnClickListener(numClick);
60     btn9.setOnClickListener(numClick);
61     btnPlus.setOnClickListener(numClick);
62
63
64     btnClear.setOnClickListener(v → edtPhone.setText(""));
65
66
67     btnCall.setOnClickListener(v → {
68
69         String number = edtPhone.getText().toString().trim();
70
71         if (number.isEmpty()) {
72             edtPhone.setError("Enter number");
73             return;
74         }
75
76
77         String cleaned = number.replaceAll("[^0-9+]", "");
78
79         if (cleaned.length() < 10) {
80             edtPhone.setError("Enter valid number");
81             return;
82         }
83
84
85         String country;
86
87         if (cleaned.startsWith("+91")) {
88             country = "India IN";
89         } else if (cleaned.startsWith("+1")) {
90             country = "USA us";
91         } else if (cleaned.startsWith("+44")) {
92             country = "UK GB";
93         } else {
94             country = "Unknown 🌐";
95         }
96
97

```

```

98         Toast.makeText(MainActivity.this,
99             "Calling " + country,
100             Toast.LENGTH_SHORT).show();
101
102
103         Intent intent = new Intent(Intent.ACTION_DIAL);
104         intent.setData(Uri.parse("tel:" + number));
105         startActivity(intent);
106     });
107 }
108 }

```

app/src/main/java/com/example/dialerapp/MainActivity2.java

JAVA SOURCE • 14 LINES

```

1  package com.example.dialerapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6
7  public class MainActivity2 extends AppCompatActivity {
8
9      @Override
10     protected void onCreate(Bundle savedInstanceState) {
11         super.onCreate(savedInstanceState);
12         setContentView(R.layout.activity_main2);
13     }
14 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 106 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:orientation="vertical"
7      android:padding="20dp"
8      android:background="#F2F2F2"
9      android:gravity="center_horizontal"
10     tools:context=".MainActivity">
11
12     <TextView
13         android:layout_width="wrap_content"
14         android:layout_height="wrap_content"
15         android:text="PHONE DIALER"
16         android:textSize="24sp"
17         android:textStyle="bold"
18         android:textColor="#000000"
19         android:layout_marginBottom="20dp" />
20
21     <EditText
22         android:id="@+id/etNumber"
23         android:layout_width="match_parent"
24         android:layout_height="60dp"
25         android:hint="Enter Number"
26         android:textSize="22sp"
27         android:gravity="center"
28         android:inputType="phone"
29         android:background="#FFFFFF"
30         android:layout_marginBottom="25dp"
31         android:padding="10dp" />
32
33
34     <LinearLayout
35         android:layout_width="match_parent"

```

```

36         android:layout_height="wrap_content"
37         android:orientation="horizontal">
38
39         <Button android:id="@+id/btn1" android:layout_width="0dp"
40             android:layout_height="70dp" android:layout_weight="1" android:text="1" />
41
42         <Button android:id="@+id/btn2" android:layout_width="0dp"
43             android:layout_height="70dp" android:layout_weight="1" android:text="2" />
44
45         <Button android:id="@+id/btn3" android:layout_width="0dp"
46             android:layout_height="70dp" android:layout_weight="1" android:text="3" />
47     </LinearLayout>
48
49
50     <LinearLayout
51         android:layout_width="match_parent"
52         android:layout_height="wrap_content"
53         android:orientation="horizontal">
54
55         <Button android:id="@+id/btn4" android:layout_width="0dp"
56             android:layout_height="70dp" android:layout_weight="1" android:text="4" />
57
58         <Button android:id="@+id/btn5" android:layout_width="0dp"
59             android:layout_height="70dp" android:layout_weight="1" android:text="5" />
60
61         <Button android:id="@+id/btn6" android:layout_width="0dp"
62             android:layout_height="70dp" android:layout_weight="1" android:text="6" />
63     </LinearLayout>
64
65
66     <LinearLayout
67         android:layout_width="match_parent"
68         android:layout_height="wrap_content"
69         android:orientation="horizontal">
70
71         <Button android:id="@+id/btn7" android:layout_width="0dp"
72             android:layout_height="70dp" android:layout_weight="1" android:text="7" />
73
74         <Button android:id="@+id/btn8" android:layout_width="0dp"
75             android:layout_height="70dp" android:layout_weight="1" android:text="8" />
76
77         <Button android:id="@+id/btn9" android:layout_width="0dp"
78             android:layout_height="70dp" android:layout_weight="1" android:text="9" />
79     </LinearLayout>
80
81
82     <LinearLayout
83         android:layout_width="match_parent"
84         android:layout_height="wrap_content"
85         android:orientation="horizontal">
86
87         <Button android:id="@+id/btnPlus" android:layout_width="0dp"
88             android:layout_height="70dp" android:layout_weight="1" android:text="+" />
89
90         <Button android:id="@+id/btn0" android:layout_width="0dp"
91             android:layout_height="70dp" android:layout_weight="1" android:text="0" />
92
93         <Button android:id="@+id/btnClear" android:layout_width="0dp"
94             android:layout_height="70dp" android:layout_weight="1" android:text="C" />
95     </LinearLayout>
96
97
98     <ImageButton
99         android:id="@+id/btnCall"
100         android:layout_width="90dp"
101         android:layout_height="90dp"
102         android:layout_marginTop="20dp"
103         android:background="#4CAF50"
104         android:src="@android:drawable/ic_menu_call"
105         android:contentDescription="Call" />
106 </LinearLayout>

```

app/src/main/res/layout/activity_main2.xml

LAYOUT XML • 67 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="20dp"
8      android:background="#F2F2F2"
9      android:gravity="center_horizontal"
10     tools:context=".MainActivity">
11
12     <TextView
13         android:layout_width="wrap_content"
14         android:layout_height="wrap_content"
15         android:text="PHONE DIALER"
16         android:textSize="24sp"
17         android:textStyle="bold"
18         android:textColor="#000000"
19         android:layout_marginBottom="20dp" />
20
21     <EditText
22         android:id="@+id/etNumber"
23         android:layout_width="match_parent"
24         android:layout_height="60dp"
25         android:hint="Enter Number"
26         android:textSize="22sp"
27         android:gravity="center"
28         android:inputType="phone"
29         android:background="FFFFFF"
30         android:layout_marginBottom="25dp"
31         android:padding="10dp" />
32
33     <!-- Dial pad using GridLayout -->
34     <GridLayout
35         android:layout_width="match_parent"
36         android:layout_height="wrap_content"
37         android:columnCount="3"
38         android:rowCount="4"
39         android:alignmentMode="alignMargins"
40         android:rowOrderPreserved="false">
41
42         <Button android:id="@+id/btn1" android:layout_width="0dp" android:layout_height="70dp" an
43             droid:layout_columnWeight="1" android:text="1" />
44         <Button android:id="@+id/btn2" android:layout_width="0dp" android:layout_height="70dp" an
45             droid:layout_columnWeight="1" android:text="2" />
46         <Button android:id="@+id/btn3" android:layout_width="0dp" android:layout_height="70dp" an
47             droid:layout_columnWeight="1" android:text="3" />
48         <Button android:id="@+id/btn4" android:layout_width="0dp" android:layout_height="70dp" an
49             droid:layout_columnWeight="1" android:text="4" />
50         <Button android:id="@+id/btn5" android:layout_width="0dp" android:layout_height="70dp" an
51             droid:layout_columnWeight="1" android:text="5" />
52         <Button android:id="@+id/btn6" android:layout_width="0dp" android:layout_height="70dp" an
53             droid:layout_columnWeight="1" android:text="6" />
54         <Button android:id="@+id/btn7" android:layout_width="0dp" android:layout_height="70dp" an
55             droid:layout_columnWeight="1" android:text="7" />
56         <Button android:id="@+id/btn8" android:layout_width="0dp" android:layout_height="70dp" an
57             droid:layout_columnWeight="1" android:text="8" />
58         <Button android:id="@+id/btn9" android:layout_width="0dp" android:layout_height="70dp" an
59             droid:layout_columnWeight="1" android:text="9" />
60         <Button android:id="@+id/btnPlus" android:layout_width="0dp" android:layout_height="70dp"
61             android:layout_columnWeight="1" android:text="+" />
62         <Button android:id="@+id/btn0" android:layout_width="0dp" android:layout_height="70dp" an
63             droid:layout_columnWeight="1" android:text="0" />
64         <Button android:id="@+id/btnClear" android:layout_width="0dp" android:layout_height="70d
65             p" android:layout_columnWeight="1" android:text="C" />
66     </GridLayout>

```

```
58
59     <ImageButton
60         android:id="@+id/btnCall"
61         android:layout_width="90dp"
62         android:layout_height="90dp"
63         android:layout_marginTop="20dp"
64         android:background="#4CAF50"
65         android:src="@android:drawable/ic_menu_call"
66         android:contentDescription="Call" />
67 </LinearLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">Dialer app</string>
3 </resources>
```

Exam1

AndroidStudioProjects • 4 files

159 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.exam1">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Exam1"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/exam1/MainActivity.java

JAVA SOURCE • 52 LINES

```

1  package com.example.exam1;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.Button;
8  import android.widget.RadioButton;
9  import android.widget.EditText;
10 import android.widget.RadioGroup;
11 import android.widget.Toast;
12 import android.widget.TextView;
13
14 public class MainActivity extends AppCompatActivity {
15     Button submit, Reset;
16     EditText sname, Regno, uno;
17     RadioButton male, female;
18     TextView Toast;
19
20
21     @Override
22     protected void onCreate(Bundle savedInstanceState) {
23         super.onCreate(savedInstanceState);
24         setContentView(R.layout.activity_main);
25         sname = findViewById(R.id.sname);
26         Regno = findViewById(R.id.Regno);
27         uno = findViewById(R.id.uno);

```

```

28
29     male = findViewById(R.id.male);
30     female = findViewById(R.id.female);
31     submit = findViewById(R.id.submit);
32     Reset = findViewById(R.id.Reset);
33     view onclickListner.Onclear(View)→{
34         sname.setText("");
35         Regno.setText("");
36         uno.setText("");
37
38     }
39     view onclickListner.Onsubmit(View)→{
40         if (sname.isChecked() || Regno.isChecked() || uno.isChecked()){
41             Toast.makeText("")
42         }
43     }
44
45 }
46
47 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 77 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <EditText
10         android:id="@+id/sname"
11         android:layout_width="wrap_content"
12         android:layout_height="wrap_content"
13         android:ems="10"
14         android:inputType="textPersonName"
15         android:text="Enter student name"
16         tools:layout_editor_absoluteX="4dp"
17         tools:layout_editor_absoluteY="57dp" />
18
19      <EditText
20         android:id="@+id/Regno"
21         android:layout_width="wrap_content"
22         android:layout_height="wrap_content"
23         android:ems="10"
24         android:inputType="textPersonName"
25         android:text="Enter reg no"
26         tools:layout_editor_absoluteX="4dp"
27         tools:layout_editor_absoluteY="110dp" />
28
29      <EditText
30         android:id="@+id/uno"
31         android:layout_width="wrap_content"
32         android:layout_height="wrap_content"
33         android:ems="10"
34         android:inputType="textPersonName"
35         android:text="Enter university name"
36         tools:layout_editor_absoluteX="4dp"
37         tools:layout_editor_absoluteY="199dp" />
38
39      <RadioGroup
40         android:layout_width="wrap_content"
41         android:layout_height="wrap_content"
42         tools:layout_editor_absoluteX="149dp"
43         tools:layout_editor_absoluteY="338dp" />
44
45      <RadioButton
46         android:id="@+id/male"

```

```

47     android:layout_width="89dp"
48     android:layout_height="45dp"
49     android:text="Male"
50     tools:layout_editor_absoluteX="16dp"
51     tools:layout_editor_absoluteY="293dp" />
52
53     <RadioButton
54         android:id="@+id/female"
55         android:layout_width="88dp"
56         android:layout_height="52dp"
57         android:text="Female"
58         tools:layout_editor_absoluteX="149dp"
59         tools:layout_editor_absoluteY="286dp" />
60
61     <Button
62         android:id="@+id/submit"
63         android:layout_width="wrap_content"
64         android:layout_height="wrap_content"
65         android:text="submit"
66         tools:layout_editor_absoluteX="14dp"
67         tools:layout_editor_absoluteY="391dp" />
68
69     <Button
70         android:id="@+id/Reset"
71         android:layout_width="wrap_content"
72         android:layout_height="wrap_content"
73         android:text="reset"
74         tools:layout_editor_absoluteX="149dp"
75         tools:layout_editor_absoluteY="391dp" />
76
77 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```

1 <resources>
2     <string name="app_name">Exam1</string>
3 </resources>

```

Intent

AndroidStudioProjects • 6 files

354 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      package="com.example.intent">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Intent"
15         tools:targetApi="31">
16         <activity
17             android:name=".Intent1"
18             android:exported="false" />
19         <activity
20             android:name=".MainActivity"
21             android:exported="true">
22             <intent-filter>
23                 <action android:name="android.intent.action.MAIN" />
24
25                 <category android:name="android.intent.category.LAUNCHER" />
26             </intent-filter>
27         </activity>
28     </application>
29
30 </manifest>

```

app/src/main/java/com/example/intent/MainActivity.java

JAVA SOURCE • 101 LINES

```

1  package com.example.intent;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.CheckBox;
10 import android.widget.EditText;
11 import android.widget.RadioButton;
12 import android.widget.RadioGroup;
13 import android.widget.TextView;
14 import android.widget.Toast;
15
16 public class MainActivity extends AppCompatActivity {
17     EditText editText;
18     RadioGroup radioGender;
19     RadioButton radioMale,radioFemale;
20     CheckBox checkBoxReading,checkBoxMusic,checkBoxSports,checkBoxYoga;
21     Button submit;
22     Button clear;
23     TextView output;
24

```

```

25     @Override
26     protected void onCreate(Bundle savedInstanceState) {
27         super.onCreate(savedInstanceState);
28         setContentView(R.layout.activity_main);
29
30         editText = findViewById(R.id.editText);
31         radioGender = findViewById(R.id.genderGroup);
32         radioMale = findViewById(R.id.radioMale);
33         radioFemale = findViewById(R.id.radioFemale);
34         checkBoxReading = findViewById(R.id.checkBoxReading);
35         checkBoxMusic = findViewById(R.id.checkBoxMusic);
36         checkBoxSports = findViewById(R.id.checkBoxSports);
37         checkBoxYoga = findViewById(R.id.checkBoxYoga);
38         submit = findViewById(R.id.submit);
39         clear = findViewById(R.id.clear);
40         output = findViewById(R.id.output);
41
42         View.OnClickListener radioListener = View → {
43             RadioButton selected = (RadioButton) View;
44             Toast.makeText(MainActivity.this, selected.getText() + "selected", Toast.LENGTH_SHOR
T).show();
45
46         };
47         radioMale.setOnClickListener(radioListener);
48         radioFemale.setOnClickListener(radioListener);
49
50         View.OnClickListener checkListener = View → {
51             CheckBox box = (CheckBox) View;
52             if (box.isChecked()) {
53                 Toast.makeText(MainActivity.this, box.getText() + "selected", Toast.LENGTH_SHOR
T).show();
54             }
55         };
56
57         checkBoxReading.setOnClickListener(checkListener);
58         checkBoxMusic.setOnClickListener(checkListener);
59         checkBoxSports.setOnClickListener(checkListener);
60         checkBoxYoga.setOnClickListener(checkListener);
61
62
63         submit.setOnClickListener(v → {
64             String name = editText.getText().toString();
65             int selectedId = radioGender.getCheckedRadioButtonId();
66             String gender = "";
67             if (selectedId ≠ -1) {
68                 RadioButton selected = findViewById(selectedId);
69                 gender = selected.getText().toString();
70             }
71
72
73
74
75             StringBuilder hobbies = new StringBuilder();
76             if (checkBoxReading.isChecked()) hobbies.append("Reading");
77             if (checkBoxMusic.isChecked()) hobbies.append("Music");
78             if (checkBoxSports.isChecked()) hobbies.append("Sports");
79             if (checkBoxYoga.isChecked()) hobbies.append("Yoga");
80
81
82             String message = "Name :" + name + "\nGender :" + gender + "\nHobbies :" + hobbies;
83             output.setText(message);
84             Intent intent = new Intent(MainActivity.this, Intent1.class);
85             intent.putExtra("data", message);
86             startActivity(intent);
87
88             clear.setOnClickListener(clear → {
89                 editText.setText("");
90                 radioGender.clearCheck();
91                 checkBoxReading.setChecked(false);
92                 checkBoxMusic.setChecked(false);
93                 checkBoxSports.setChecked(false);
94                 checkBoxYoga.setChecked(false);
95                 output.setText("Output will appear here");

```

```

96         });
97
98
99     });
100 }
101 }

```

app/src/main/java/com/example/intent/Intent1.java

JAVA SOURCE • 34 LINES

```

1  package com.example.intent;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.widget.*;
8
9  public class Intent1 extends AppCompatActivity {
10     TextView output;
11     Button back;
12
13     @Override
14     protected void onCreate(Bundle savedInstanceState) {
15         super.onCreate(savedInstanceState);
16         setContentView(R.layout.activity_intent1);
17
18         output=findViewById(R.id.output);
19         back=findViewById(R.id.back);
20
21         Intent intent=getIntent();
22         String data = intent.getStringExtra("data");
23         output.setText(data);
24
25         back.setOnClickListener(view→ {
26             Intent i = new Intent(Intent1.this, MainActivity.class);
27             startActivity(i);
28             finish();
29         });
30
31     }
32
33 }
34 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 143 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp">
8
9
10     <ImageButton
11         android:id="@+id/imageButton2"
12         android:layout_width="185dp"
13         android:layout_height="120dp"
14         android:layout_gravity="center_horizontal"
15         android:layout_marginBottom="16dp"
16         android:scaleType="centerCrop"
17         android:src="@drawable/window11"
18         android:contentDescription="Profile Image" />
19
20 />

```

```

21 <LinearLayout
22     android:layout_width="match_parent"
23     android:layout_height="wrap_content"
24     android:orientation="horizontal"
25     android:gravity="center_vertical"
26     android:layout_marginBottom="12dp">
27
28     <TextView
29         android:id="@+id/textView2"
30         android:layout_width="80dp"
31         android:layout_height="wrap_content"
32         android:text="Name:" />
33
34     <EditText
35         android:id="@+id/editText"
36         android:layout_width="match_parent"
37         android:layout_height="wrap_content"
38         android:ems="10"
39         android:hint="Enter name"
40         android:inputType="textPersonName" />
41 </LinearLayout>
42
43
44 <LinearLayout
45     android:layout_width="match_parent"
46     android:layout_height="wrap_content"
47     android:orientation="horizontal"
48     android:gravity="center_vertical"
49     android:layout_marginBottom="12dp">
50
51     <TextView
52         android:id="@+id/radioGenderLabel"
53         android:layout_width="80dp"
54         android:layout_height="wrap_content"
55         android:text="Gender:" />
56
57     <RadioGroup
58         android:id="@+id/genderGroup"
59         android:layout_width="match_parent"
60         android:layout_height="wrap_content"
61         android:orientation="horizontal">
62
63         <RadioButton
64             android:id="@+id/radioMale"
65             android:layout_width="wrap_content"
66             android:layout_height="wrap_content"
67             android:text="Male" />
68
69         <RadioButton
70             android:id="@+id/radioFemale"
71             android:layout_width="wrap_content"
72             android:layout_height="wrap_content"
73             android:text="Female" />
74     </RadioGroup>
75 </LinearLayout>
76
77
78 <TextView
79     android:id="@+id/textView5"
80     android:layout_width="wrap_content"
81     android:layout_height="wrap_content"
82     android:text="Hobbies:"
83     android:textStyle="bold"
84     android:layout_marginBottom="8dp" />
85
86 <CheckBox
87     android:id="@+id/checkBoxReading"
88     android:layout_width="wrap_content"
89     android:layout_height="wrap_content"
90     android:text="Reading" />
91
92 <CheckBox
93     android:id="@+id/checkBoxMusic"

```

```

94         android:layout_width="wrap_content"
95         android:layout_height="wrap_content"
96         android:text="Music" />
97
98     <CheckBox
99         android:id="@+id/checkBoxSports"
100         android:layout_width="wrap_content"
101         android:layout_height="wrap_content"
102         android:text="Sports" />
103
104     <CheckBox
105         android:id="@+id/checkBoxYoga"
106         android:layout_width="wrap_content"
107         android:layout_height="wrap_content"
108         android:text="Yoga" />
109
110
111     <LinearLayout
112         android:layout_width="match_parent"
113         android:layout_height="wrap_content"
114         android:orientation="horizontal"
115         android:layout_marginTop="24dp">
116
117         <Button
118             android:id="@+id/submit"
119             android:layout_width="0dp"
120             android:layout_weight="1"
121             android:layout_height="wrap_content"
122             android:layout_marginEnd="8dp"
123             android:text="Submit" />
124
125         <Button
126             android:id="@+id/clear"
127             android:layout_width="0dp"
128             android:layout_weight="1"
129             android:layout_height="wrap_content"
130             android:text="Clear" />
131     </LinearLayout>
132
133
134     <TextView
135         android:id="@+id/output"
136         android:layout_width="match_parent"
137         android:layout_height="53dp"
138         android:layout_marginTop="20dp"
139         android:text="Output will appear here"
140         android:textAlignment="center"
141         android:visibility="invisible" />
142
143 </LinearLayout>

```

app/src/main/res/layout/activity_intent1.xml

LAYOUT XML • 43 LINES

```

1  <androidx.constraintlayout.widget.ConstraintLayout
2      xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".Intent1">
8
9      <RelativeLayout
10         android:id="@+id/container"
11         android:layout_width="match_parent"
12         android:layout_height="match_parent">
13
14         <ImageView
15             android:id="@+id/imageView"
16             android:layout_width="100dp"
17             android:layout_height="100dp"

```

```

18         android:layout_centerHorizontal="true"
19         android:layout_marginTop="40dp"
20         android:contentDescription="Profile image"
21         tools:srcCompat="@tools:sample/avatars" />
22
23     <TextView
24         android:id="@+id/output"
25         android:layout_width="wrap_content"
26         android:layout_height="wrap_content"
27         android:text="TextView"
28         android:layout_below="@id/imageView"
29         android:layout_centerHorizontal="true"
30         android:layout_marginTop="20dp" />
31
32     <Button
33         android:id="@+id/back"
34         android:layout_width="wrap_content"
35         android:layout_height="wrap_content"
36         android:text="Back"
37         android:layout_below="@id/output"
38         android:layout_centerHorizontal="true"
39         android:layout_marginTop="30dp" />
40
41 </RelativeLayout>
42
43 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```

1 <resources>
2     <string name="app_name">Intent</string>
3 </resources>

```

LAB1

AndroidStudioProjects • 4 files

120 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.lab1" >
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.LAB1"
15         tools:targetApi="31" >
16         <activity
17             android:name=".MainActivity"
18             android:exported="true" >
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/lab1/MainActivity.java

JAVA SOURCE • 18 LINES

```

1  package com.example.lab1;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5  import android.view.View;
6  import android.widget.TextView;
7
8  public class MainActivity extends AppCompatActivity {
9
10     @Override
11     protected void onCreate(Bundle savedInstanceState) {
12         super.onCreate(savedInstanceState);
13         setContentView(R.layout.activity_main);
14     }
15
16
17
18 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 70 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/

```

```

res/android"
3  xmlns:app="http://schemas.android.com/apk/res-auto"
4  xmlns:tools="http://schemas.android.com/tools"
5  android:layout_width="match_parent"
6  android:layout_height="match_parent"
7  android:background="@color/white"
8  tools:context=".MainActivity">
9
10 <TextView
11     android:layout_width="wrap_content"
12     android:layout_height="wrap_content"
13     android:text="@string/my_first_lab_program"
14     android:textColor="@color/purple_200"
15     app:layout_constraintBottom_toBottomOf="parent"
16     app:layout_constraintEnd_toEndOf="parent"
17     app:layout_constraintStart_toStartOf="parent"
18     app:layout_constraintTop_toTopOf="parent"
19     app:layout_constraintVertical_bias="0.119" />
20
21 <TextView
22
23     android:layout_width="400dp"
24     android:layout_height="100dp"
25     android:background="#00F3F3"
26     android:fontFamily="monospace"
27     android:text="NAME: NANDANRAJ"
28     android:textAllCaps="false"
29     android:textColor="@color/purple_200"
30     android:textSize="20sp"
31     android:textStyle="italic"
32     app:layout_constraintBottom_toBottomOf="parent"
33     app:layout_constraintEnd_toEndOf="parent"
34     app:layout_constraintHorizontal_bias="1.0"
35     app:layout_constraintStart_toStartOf="parent"
36     app:layout_constraintTop_toTopOf="parent"
37     app:layout_constraintVertical_bias="0.31"
38     tools:ignore="MissingConstraints" />
39
40 <TextView
41     android:layout_width="354dp"
42     android:layout_height="66dp"
43     android:text="College Name:Poornaprajna Institute of Management , Udupi "
44     android:textColor="@color/purple_200"
45     android:textSize="20sp"
46     android:textStyle="italic"
47     app:layout_constraintBottom_toBottomOf="parent"
48     app:layout_constraintEnd_toEndOf="parent"
49     app:layout_constraintHorizontal_bias="0.28"
50     app:layout_constraintStart_toStartOf="parent"
51     app:layout_constraintTop_toTopOf="parent"
52     app:layout_constraintVertical_bias="0.41"
53     tools:ignore="MissingConstraints" />
54
55 <Button
56     android:id="@+id/button"
57     android:layout_width="wrap_content"
58     android:layout_height="wrap_content"
59     android:onClick="Hello"
60     android:text="Hi"
61     app:layout_constraintBottom_toBottomOf="parent"
62     app:layout_constraintEnd_toEndOf="parent"
63     app:layout_constraintHorizontal_bias="0.445"
64     app:layout_constraintStart_toStartOf="parent"
65     app:layout_constraintTop_toTopOf="parent"
66     app:layout_constraintVertical_bias="0.55"
67     tools:ignore="OnClick" />
68
69 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 5 LINES

```
1 <resources>
2   <string name="app_name">LAB 1</string>
3   <string name="my_first_lab_program">MY FIRST LAB PROGRAM</string>
4   <string name="Hi">Hello</string>
5 </resources>
```

Lab3activitylifecycle

AndroidStudioProjects • 4 files

102 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.lab3activitylifecycle">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Lab3ActivityLifeCycle"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/lab3activitylifecycle/MainActivity.java

JAVA SOURCE • 54 LINES

```

1  package com.example.lab3activitylifecycle;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.util.Log;
7
8  public class MainActivity extends AppCompatActivity {
9
10     @Override
11     protected void onCreate(Bundle savedInstanceState) {
12         super.onCreate(savedInstanceState);
13         setContentView(R.layout.activity_main);
14         Log.d("Activity life cycle", "onCreateCalled");
15     }
16
17     @Override
18     protected void onStart(){
19         super.onStart();
20         Log.d("Activity life cycle", "onStartCalled");
21     }
22
23     @Override
24     protected void onResume(){
25         super.onResume();

```

```

26     Log.d("Activity life cycle","onResumeCalled");
27 }
28
29 @Override
30 protected void onPause(){
31     super.onPause();
32     Log.d("Activity life cycle","onPauseCalled");
33 }
34
35 @Override
36 protected void onRestart(){
37     super.onRestart();
38     Log.d("Activity life cycle","onRestartCalled");
39 }
40
41 @Override
42 protected void onStop(){
43     super.onStop();
44     Log.d("Activity life cycle","onStopCalled");
45 }
46
47 @Override
48 protected void onDestroy(){
49     super.onDestroy();
50     Log.d("Activity life cycle","onDestroyCalled");
51 }
52
53
54 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 18 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <TextView
10         android:layout_width="wrap_content"
11         android:layout_height="wrap_content"
12         android:text="Activity Life Cycle"
13         app:layout_constraintBottom_toBottomOf="parent"
14         app:layout_constraintEnd_toEndOf="parent"
15         app:layout_constraintStart_toStartOf="parent"
16         app:layout_constraintTop_toTopOf="parent" />
17
18  </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```

1  <resources>
2      <string name="app_name">Lab 3 activity life cycle</string>
3  </resources>

```

Lab4

AndroidStudioProjects • 4 files

136 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.lab4">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@android:drawable/bottom_bar"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.Lab4"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/lab4/MainActivity.java

JAVA SOURCE • 49 LINES

```

1  package com.example.lab4;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.text.TextUtils;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.EditText;
10 import android.widget.TextView;
11 import android.widget.Toast;
12
13
14 public class MainActivity extends AppCompatActivity {
15     EditText ET1,ET2;
16     TextView TV;
17     Button Btnsum,BtnReset;
18
19     @Override
20     protected void onCreate(Bundle savedInstanceState) {
21         super.onCreate(savedInstanceState);
22         setContentView(R.layout.activity_main);
23         ET1 = (EditText) findViewById(R.id.number1);
24         ET2 = (EditText) findViewById(R.id.number2);
25
26         TV = (TextView) findViewById(R.id.number1);
27         Btnsum = (Button) findViewById(R.id.calculTE);

```

```

28     BtnReset = (Button) findViewById(R.id.Reset);
29
30     Btnsum.setOnClickListener(new View.OnClickListener() {
31         @Override
32         public void onClick(View view) {
33             String val1 =getString().toString();
34             String val2 =getString().toString();
35
36             if (!val1)
37
38                 Integer sum=Integer.parseInt(val1)+Integer.parseInt(val2);
39             }
40             TV.setText(sum.toString());
41             Toast.makeText(getApplicationContext(),Toast.Length.short;
42         }
43         else{
44             Toast.makeText(getApplicationContext("Enter the number one").Toast.Length.short)))
45         }
46     }
47 }
48 BtnReset.setOnClickListener

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 52 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <Button
10         android:id="@+id/calculatE"
11         android:layout_width="169dp"
12         android:layout_height="64dp"
13         android:text="calculate"
14         tools:layout_editor_absoluteX="96dp"
15         tools:layout_editor_absoluteY="179dp" />
16
17      <TextView
18         android:id="@+id/textView"
19         android:layout_width="88dp"
20         android:layout_height="27dp"
21         android:text="Result=0"
22         tools:layout_editor_absoluteX="120dp"
23         tools:layout_editor_absoluteY="334dp" />
24
25      <Button
26         android:id="@+id/Reset"
27         android:layout_width="139dp"
28         android:layout_height="58dp"
29         android:text="reset"
30         tools:layout_editor_absoluteX="103dp"
31         tools:layout_editor_absoluteY="262dp" />
32
33      <EditText
34         android:layout_width="207dp"
35         android:layout_height="41dp"
36         android:hint="Enter number one"
37         android:id="@+id/number1"
38         android:inputType="number"
39         tools:layout_editor_absoluteX="72dp"
40         tools:layout_editor_absoluteY="16dp" />
41
42      <EditText
43         android:layout_width="200dp"
44         android:layout_height="49dp"
45         android:hint="Enter number two"

```

```
46         android:id="@+id/number2"  
47         android:inputType="number"  
48         tools:layout_editor_absoluteX="79dp"  
49         tools:layout_editor_absoluteY="85dp" />  
50  
51  
52     </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 8 LINES

```
1     <resources>  
2         <string name="app_name">Lab4</string>  
3         <string name="input_hint1">Enter first number</string>  
4         <string name="input_hint2">Enter second number</string>  
5         <string name="sum_button">calculate sum string</string>  
6         <string name="reset_button">reset string close</string>  
7         <string name="result_text">result will appear here</string>  
8     </resources>
```

MasterApp

AndroidStudioProjects • 8 files

247 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 33 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.masterapp">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.MasterApp"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity3"
18           android:exported="false" />
19       <activity
20           android:name=".MainActivity2"
21           android:exported="false" />
22       <activity
23           android:name=".MainActivity"
24           android:exported="true">
25           <intent-filter>
26               <action android:name="android.intent.action.MAIN" />
27
28               <category android:name="android.intent.category.LAUNCHER" />
29           </intent-filter>
30       </activity>
31   </application>
32
33 </manifest>

```

app/src/main/java/com/example/masterapp/MainActivity.java

JAVA SOURCE • 26 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.content.Intent;
5  import android.os.Bundle;
6  import android.view.View;
7
8  public class MainActivity extends AppCompatActivity {
9
10     @Override
11     protected void onCreate(Bundle savedInstanceState) {
12         super.onCreate(savedInstanceState);
13         setContentView(R.layout.activity_main);
14     }
15
16
17     public void hello(View view) {
18         Intent intent = new Intent(MainActivity.this, MainActivity2.class);
19         startActivity(intent);
20     }
21

```

```

22     public void toast(View view) {
23         Intent intent1 = new Intent(MainActivity.this, MainActivity3.class);
24         startActivity(intent1);
25     }
26 }

```

app/src/main/java/com/example/masterapp/MainActivity2.java

JAVA SOURCE • 13 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5
6  public class MainActivity2 extends AppCompatActivity {
7
8      @Override
9      protected void onCreate(Bundle savedInstanceState) {
10         super.onCreate(savedInstanceState);
11         setContentView(R.layout.activity_main2);
12     }
13 }

```

app/src/main/java/com/example/masterapp/MainActivity3.java

JAVA SOURCE • 21 LINES

```

1  package com.example.masterapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.Toast;
8
9  public class MainActivity3 extends AppCompatActivity {
10
11      @Override
12      protected void onCreate(Bundle savedInstanceState) {
13         super.onCreate(savedInstanceState);
14         setContentView(R.layout.activity_main);
15
16         Toast.makeText(getApplicationContext(), "Default Toast Message", Toast.LENGTH_LONG).show
O;
17     }
18     public void showtoast(View view){
19         Toast.makeText(getApplicationContext(), "onClickToastMessage", Toast.LENGTH_SHORT).show();
20     }
21 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 101 LINES

```

1  <androidx.constraintlayout.widget.ConstraintLayout
2      xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <TextView
10         android:id="@+id/titleText"
11         android:layout_width="wrap_content"
12         android:layout_height="wrap_content"
13         android:text="Master App"
14         android:textColor="#1472FF"

```

```

15         android:textSize="20sp"
16         app:layout_constraintBottom_toTopOf="@+id/button"
17         app:layout_constraintEnd_toEndOf="parent"
18         app:layout_constraintHorizontal_bias="0.498"
19         app:layout_constraintStart_toStartOf="parent"
20         app:layout_constraintTop_toTopOf="parent" />
21
22     <Button
23         android:id="@+id/button"
24         android:layout_width="270dp"
25         android:layout_height="51dp"
26         android:layout_marginTop="28dp"
27         android:text="Hello World"
28         android:onClick="hello"
29         app:layout_constraintEnd_toEndOf="parent"
30         app:layout_constraintHorizontal_bias="0.496"
31         app:layout_constraintStart_toStartOf="parent"
32         app:layout_constraintTop_toBottomOf="@id/titleText" />
33
34     <Button
35         android:id="@+id/button2"
36         android:layout_width="272dp"
37         android:layout_height="48dp"
38         android:layout_marginTop="28dp"
39         android:onClick="toast"
40         android:text="Toast"
41         app:layout_constraintEnd_toEndOf="parent"
42         app:layout_constraintHorizontal_bias="0.496"
43         app:layout_constraintStart_toStartOf="parent"
44         app:layout_constraintTop_toBottomOf="@id/button" />
45
46     <Button
47         android:id="@+id/button3"
48         android:layout_width="262dp"
49         android:layout_height="50dp"
50         android:layout_marginTop="40dp"
51         android:text="Life Cycle"
52         app:layout_constraintEnd_toEndOf="parent"
53         app:layout_constraintHorizontal_bias="0.51"
54         app:layout_constraintStart_toStartOf="parent"
55         app:layout_constraintTop_toBottomOf="@id/button2" />
56
57     <Button
58         android:id="@+id/button4"
59         android:layout_width="263dp"
60         android:layout_height="47dp"
61         android:layout_marginTop="36dp"
62         android:text="Sum of 2 nums"
63         app:layout_constraintEnd_toEndOf="parent"
64         app:layout_constraintHorizontal_bias="0.486"
65         app:layout_constraintStart_toStartOf="parent"
66         app:layout_constraintTop_toBottomOf="@id/button3" />
67
68     <Button
69         android:id="@+id/button5"
70         android:layout_width="251dp"
71         android:layout_height="43dp"
72         android:layout_marginTop="48dp"
73         android:text="User Info"
74         app:layout_constraintEnd_toEndOf="parent"
75         app:layout_constraintHorizontal_bias="0.475"
76         app:layout_constraintStart_toStartOf="parent"
77         app:layout_constraintTop_toBottomOf="@id/button4" />
78
79     <Button
80         android:id="@+id/button6"
81         android:layout_width="256dp"
82         android:layout_height="54dp"
83         android:layout_marginTop="44dp"
84         android:text="User Info Intent"
85         app:layout_constraintEnd_toEndOf="parent"
86         app:layout_constraintHorizontal_bias="0.529"
87         app:layout_constraintStart_toStartOf="parent"

```

```

88         app:layout_constraintTop_toBottomOf="@id/button5" />
89
90     <Button
91         android:id="@+id/button7"
92         android:layout_width="259dp"
93         android:layout_height="50dp"
94         android:layout_marginTop="36dp"
95         android:text="Calculator"
96         app:layout_constraintEnd_toEndOf="parent"
97         app:layout_constraintHorizontal_bias="0.48"
98         app:layout_constraintStart_toStartOf="parent"
99         app:layout_constraintTop_toBottomOf="@id/button6" />
100
101 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main2.xml

LAYOUT XML • 21 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      tools:context=".MainActivity2">
9
10     <TextView
11         android:id="@+id/textView"
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="Hello World"
15         android:textColor="#AC1B4C"
16         android:textSize="48sp"
17         app:layout_constraintTop_toTopOf="parent"
18         app:layout_constraintBottom_toBottomOf="parent"
19         app:layout_constraintStart_toStartOf="parent"
20         app:layout_constraintEnd_toEndOf="parent" />
21 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_main3.xml

LAYOUT XML • 29 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity3">
8
9
10     <Button
11         android:id="@+id/button"
12         android:layout_width="293dp"
13         android:layout_height="159dp"
14         android:insetLeft="50dp"
15         android:insetTop="50dp"
16         android:onClick="showtoast"
17         android:text=" On Click Toast"
18         tools:ignore="MissingConstraints"
19         tools:layout_editor_absoluteX="47dp"
20         tools:layout_editor_absoluteY="243dp" />
21
22     <androidx.constraintlayout.widget.Barrier
23         android:id="@+id/barrier2"
24         android:layout_width="wrap_content"

```

```
25         android:layout_height="wrap_content"
26         app:barrierDirection="top"
27         app:constraint_referenced_ids="button" />
28
29     </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">Master App</string>
3 </resources>
```

Merge

AndroidStudioProjects • 4 files

141 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.merge">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.Merge"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/merge/MainActivity.java

JAVA SOURCE • 59 LINES

```

1  package com.example.merge;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.util.Log;
7  import android.view.View;
8  import android.widget.Toast;
9
10 public class MainActivity extends AppCompatActivity {
11
12     @Override
13     protected void onCreate(Bundle savedInstanceState) {
14         super.onCreate(savedInstanceState);
15         setContentView(R.layout.activity_main);
16         Log.d("Activity life cycle","onCreateCalled");
17         Toast.makeText(getApplicationContext(), "Default Toast Message", Toast.LENGTH_LONG).show
18 ();
19     }
20     public void toast(View view){
21         Toast.makeText(getApplicationContext(), "onClickToastMessage", Toast.LENGTH_SHORT).show();
22     }
23
24     @Override
25     protected void onStart(){
26         super.onStart();
27         Log.d("Activity life cycle","onStartCalled");
28     }
29 }

```

```

27     }
28
29
30     @Override
31     protected void onResume(){
32         super.onResume();
33         Log.d("Activity life cycle","onResumeCalled");
34     }
35
36     @Override
37     protected void onPause(){
38         super.onPause();
39         Log.d("Activity life cycle","onPauseCalled");
40     }
41
42     @Override
43     protected void onRestart(){
44         super.onRestart();
45         Log.d("Activity life cycle","onRestartCalled");
46     }
47
48     @Override
49     protected void onStop(){
50         super.onStop();
51         Log.d("Activity life cycle","onStopCalled");
52     }
53
54     @Override
55     protected void onDestroy(){
56         super.onDestroy();
57         Log.d("Activity life cycle","onDestroyCalled");
58     }
59 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 52 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <TextView
10         android:id="@+id/textView2"
11         android:layout_width="wrap_content"
12         android:layout_height="wrap_content"
13         android:fontFamily="serif-monospace"
14         android:text="Hello World!"
15         android:textColor="#0D41FA"
16         android:textSize="20sp"
17         app:layout_constraintBottom_toBottomOf="parent"
18         app:layout_constraintEnd_toEndOf="parent"
19         app:layout_constraintHorizontal_bias="0.461"
20         app:layout_constraintStart_toStartOf="parent"
21         app:layout_constraintTop_toTopOf="parent"
22         app:layout_constraintVertical_bias="0.186" />
23
24      <Button
25         android:id="@+id/button"
26         android:layout_width="186dp"
27         android:layout_height="87dp"
28         android:onClick="toast"
29         android:text="on Click Toast"
30         app:layout_constraintBottom_toBottomOf="parent"
31         app:layout_constraintEnd_toEndOf="parent"
32         app:layout_constraintHorizontal_bias="0.5"
33         app:layout_constraintStart_toStartOf="parent"

```

```
34         app:layout_constraintTop_toBottomOf="@+id/textView2" />
35
36     <TextView
37         android:id="@+id/textView3"
38         android:layout_width="174dp"
39         android:layout_height="40dp"
40         android:clickable="true"
41         android:text="Name:Nandan Raj"
42         android:textAlignment="center"
43         android:textColor="#1D46D8"
44         android:textSize="20sp"
45         app:layout_constraintBottom_toTopOf="@+id/button"
46         app:layout_constraintEnd_toEndOf="parent"
47         app:layout_constraintHorizontal_bias="0.476"
48         app:layout_constraintStart_toStartOf="parent"
49         app:layout_constraintTop_toBottomOf="@+id/textView2"
50         app:layout_constraintVertical_bias="0.543" />
51
52 </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">Merge</string>
3 </resources>
```

MyApplication

AndroidStudioProjects • 4 files

62 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.myapplication">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.MyApplication"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/myapplication/MainActivity.java

JAVA SOURCE • 14 LINES

```

1  package com.example.myapplication;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6
7  public class MainActivity extends AppCompatActivity {
8
9      @Override
10     protected void onCreate(Bundle savedInstanceState) {
11         super.onCreate(savedInstanceState);
12         setContentView(R.layout.activity_main);
13     }
14 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 18 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3          xmlns:app="http://schemas.android.com/apk/res-auto"
4          xmlns:tools="http://schemas.android.com/tools"
5          android:layout_width="match_parent"

```

```
6     android:layout_height="match_parent"
7     tools:context=".MainActivity">
8
9     <TextView
10         android:layout_width="wrap_content"
11         android:layout_height="wrap_content"
12         android:text="Hello World!"
13         app:layout_constraintBottom_toBottomOf="parent"
14         app:layout_constraintEnd_toEndOf="parent"
15         app:layout_constraintStart_toStartOf="parent"
16         app:layout_constraintTop_toTopOf="parent" />
17
18 </androidx.constraintlayout.widget.ConstraintLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">My Application</string>
3 </resources>
```

Notification

AndroidStudioProjects • 8 files

308 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 33 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.notification">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.Notification"
15       tools:targetApi="31">
16       <activity
17           android:name=".CardActivity"
18           android:exported="false" />
19       <activity
20           android:name=".ScrollingActivity"
21           android:exported="false" />
22       <activity
23           android:name=".MainActivity"
24           android:exported="true">
25           <intent-filter>
26               <action android:name="android.intent.action.MAIN" />
27
28               <category android:name="android.intent.category.LAUNCHER" />
29           </intent-filter>
30       </activity>
31   </application>
32
33 </manifest>

```

app/src/main/java/com/example/notification/MainActivity.java

JAVA SOURCE • 90 LINES

```

1  package com.example.notification;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5
6  import androidx.appcompat.app.AlertDialog;
7  import androidx.appcompat.app.AppCompatActivity;
8  import androidx.core.app.NotificationCompat;
9  import androidx.core.app.NotificationManagerCompat;
10
11 import android.app.NotificationChannel;
12 import android.app.NotificationManager;
13 import android.content.Intent;
14 import android.os.Build;
15 import android.os.Bundle;
16
17 import android.widget.Button;
18
19 public class MainActivity extends AppCompatActivity {
20     private static final String CHANNEL_ID = "notify_channel";
21

```

```

22  @Override
23  protected void onCreate(Bundle savedInstanceState) {
24      super.onCreate(savedInstanceState);
25      setContentView(R.layout.activity_main);
26      createNotificationChannel();
27
28      Button btnNotification = findViewById(R.id.btnNotification);
29      Button btnAlert = findViewById(R.id.btnAlert);
30      Button btnScroll = findViewById(R.id.btnScroll);
31      Button btnCard = findViewById(R.id.btnCard);
32      Button btnTheme = findViewById(R.id.btnTheme);
33
34      // Notification
35      btnNotification.setOnClickListener(v → showNotification());
36
37      // Alert Dialog
38      btnAlert.setOnClickListener(v → {
39          new AlertDialog.Builder(MainActivity.this)
40              .setTitle("Alert")
41              .setMessage("This is an alert dialog example.")
42              .setPositiveButton("OK", null)
43              .show();
44      });
45
46      // ScrollView Activity
47      btnScroll.setOnClickListener(v → {
48          startActivity(new Intent(MainActivity.this, ScrollingActivity.class));
49      });
50
51      // CardView Activity
52      btnCard.setOnClickListener(v → {
53          startActivity(new Intent(MainActivity.this, CardActivity.class));
54      });
55
56      // Change Theme (Toggle Dark/Light)
57      btnTheme.setOnClickListener(v → {
58          int current = AppCompatDelegate.getDefaultNightMode();
59          if (current == AppCompatDelegate.MODE_NIGHT_YES) {
60              AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_NO);
61          } else {
62              AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
63          }
64      });
65  }
66
67  private void showNotification() {
68      NotificationCompat.Builder builder = new NotificationCompat.Builder(this, CHANNEL_ID)
69          .setSmallIcon(R.drawable.ic_launcher_foreground)
70          .setContentTitle("Hello Notification")
71          .setContentText("This is a demo notification.")
72          .setPriority(NotificationCompat.PRIORITY_DEFAULT);
73
74      NotificationManagerCompat manager = NotificationManagerCompat.from(this);
75      manager.notify(1, builder.build());
76  }
77
78  private void createNotificationChannel() {
79      if (Build.VERSION.SDK_INT ≥ Build.VERSION_CODES.O) {
80          CharSequence name = "NotifyChannel";
81          String description = "Channel for demo notifications";
82          int importance = NotificationManager.IMPORTANCE_DEFAULT;
83          NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name, importance);
84          channel.setDescription(description);
85          NotificationManager notificationManager = getSystemService(NotificationManager.class);
86          notificationManager.createNotificationChannel(channel);
87      }
88  }
89  }

```

app/src/main/java/com/example/notification/CardActivity.java

JAVA SOURCE • 26 LINES

```

1  package com.example.notification;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.widget.ImageView;
7  import android.widget.TextView;
8
9
10 public class CardActivity extends AppCompatActivity {
11     ImageView cardImage;
12     TextView cardText;
13     @Override
14     protected void onCreate(Bundle savedInstanceState) {
15         super.onCreate(savedInstanceState);
16         setContentView(R.layout.activity_card);
17         cardImage = findViewById(R.id.card_image);
18         cardText = findViewById(R.id.card_text);
19
20         // Update text
21         cardText.setText("Hello! I am Android 17!");
22
23         // Update image (replace with your drawable)
24         cardImage.setImageResource(R.drawable.img);
25     }
26 }

```

app/src/main/java/com/example/notification/ScrollingActivity.java

JAVA SOURCE • 14 LINES

```

1  package com.example.notification;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6
7  public class ScrollingActivity extends AppCompatActivity {
8
9      @Override
10     protected void onCreate(Bundle savedInstanceState) {
11         super.onCreate(savedInstanceState);
12         setContentView(R.layout.activity_scrolling);
13     }
14 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 43 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2
3  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:gravity="center"
7      android:orientation="vertical"
8      android:padding="20dp">
9
10     <Button
11         android:id="@+id/btnNotification"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"
14         android:text="Show Notification" />
15
16     <Button
17         android:id="@+id/btnAlert"

```

```

18         android:layout_width="match_parent"
19         android:layout_height="wrap_content"
20         android:text="Show Alert Dialog"
21         android:layout_marginTop="10dp" />
22
23     <Button
24         android:id="@+id/btnScroll"
25         android:layout_width="match_parent"
26         android:layout_height="wrap_content"
27         android:text="Show Scrolling"
28         android:layout_marginTop="10dp" />
29
30     <Button
31         android:id="@+id/btnCard"
32         android:layout_width="match_parent"
33         android:layout_height="wrap_content"
34         android:text="Show CardView"
35         android:layout_marginTop="10dp" />
36
37     <Button
38         android:id="@+id/btnTheme"
39         android:layout_width="match_parent"
40         android:layout_height="wrap_content"
41         android:text="Change Theme"
42         android:layout_marginTop="10dp" />
43 </LinearLayout>

```

app/src/main/res/layout/activity_card.xml

LAYOUT XML • 41 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:card_view="http://schemas.android.com/apk/res-auto"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:gravity="center"
7      android:orientation="vertical">
8
9      <androidx.cardview.widget.CardView
10         android:id="@+id/myCard"
11         android:layout_width="match_parent"
12         android:layout_height="wrap_content"
13         card_view:cardCornerRadius="12dp"
14         card_view:cardElevation="8dp"
15         android:layout_margin="16dp">
16
17         <LinearLayout
18             android:layout_width="match_parent"
19             android:layout_height="wrap_content"
20             android:orientation="vertical"
21             android:padding="16dp">
22
23             <ImageView
24                 android:id="@+id/card_image"
25                 android:layout_width="match_parent"
26                 android:layout_height="200dp"
27                 android:src="@drawable/ic_launcher_foreground"
28                 android:scaleType="centerCrop" />
29
30             <TextView
31                 android:id="@+id/card_text"
32                 android:layout_width="wrap_content"
33                 android:layout_height="wrap_content"
34                 android:text="Click me!"
35                 android:textSize="20sp"
36                 android:layout_marginTop="8dp" />
37
38         </LinearLayout>
39     </androidx.cardview.widget.CardView>

```

```

40
41 </LinearLayout>

```

app/src/main/res/layout/activity_scrolling.xml

LAYOUT XML • 12 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3     android:layout_width="match_parent"
4     android:layout_height="match_parent"
5     android:padding="16dp">
6
7     <TextView
8         android:layout_width="wrap_content"
9         android:layout_height="wrap_content"
10        android:text="@string/long_text"
11        android:textSize="18sp"/>
12 </ScrollView>

```

app/src/main/res/values/strings.xml

STRINGS XML • 49 LINES

```

1 <resources>
2     <string name="app_name">NotifyApp</string>
3     <string name="notification_demo">Show Notification</string>
4     <string name="alert_demo">Show Alert</string>
5     <string name="scroll_demo">Scroll Demo</string>
6     <string name="card_demo">Card Demo</string>
7     <string name="theme_demo">Change Theme</string>
8
9     <!-- Long text for scrolling demo -->
10    <string name="long_text"> <![CDATA[
11        Android is an open-source operating system developed by Google for mobile devices
12        such as smartphones and tablets. It is based on the Linux kernel and provides a
13        rich application framework that allows developers to create innovative apps and games
14        for users across the globe. Android applications are written primarily in Java,
15        Kotlin, and sometimes C++, and they run inside a managed environment known as the
16        Android Runtime (ART).
17
18        One of Android's key strengths is its flexibility. It supports a wide range of
19        hardware devices from different manufacturers, which has helped it become the
20        world's most widely used mobile operating system. Google Play Store provides
21        millions of apps that extend the core functionalities of Android devices, covering
22        areas such as communication, productivity, entertainment, navigation, and gaming.
23
24        Android also supports features like notifications, widgets, multi-tasking,
25        background services, and deep integration with Google's ecosystem, including
26        Gmail, Maps, and Drive. With regular updates and strong developer community support,
27        Android continues to evolve and power innovative technologies such as foldable phones,
28        wearables, smart TVs, and connected IoT devices.
29
30        Android is an open-source operating system developed by Google for mobile devices
31        such as smartphones and tablets. It is based on the Linux kernel and provides a
32        rich application framework that allows developers to create innovative apps and games
33        for users across the globe. Android applications are written primarily in Java,
34        Kotlin, and sometimes C++, and they run inside a managed environment known as the
35        Android Runtime (ART).
36
37        One of Android's key strengths is its flexibility. It supports a wide range of
38        hardware devices from different manufacturers, which has helped it become the
39        world's most widely used mobile operating system. Google Play Store provides
40        millions of apps that extend the core functionalities of Android devices, covering
41        areas such as communication, productivity, entertainment, navigation, and gaming.
42
43        Android also supports features like notifications, widgets, multi-tasking,
44        background services, and deep integration with Google's ecosystem, including
45        Gmail, Maps, and Drive. With regular updates and strong developer community support,
46        Android continues to evolve and power innovative technologies such as foldable phones,

```

```
47         wearables, smart TVs, and connected IoT devices.  
48     ]]> </string>  
49 </resources>
```

Notificationapp

AndroidStudioProjects • 8 files

339 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 38 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.notificationapp">
5
6      <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />
7
8      <application
9          android:allowBackup="true"
10         android:dataExtractionRules="@xml/data_extraction_rules"
11         android:fullBackupContent="@xml/backup_rules"
12         android:icon="@mipmap/ic_launcher"
13         android:label="@string/app_name"
14         android:roundIcon="@mipmap/ic_launcher_round"
15         android:supportsRtl="true"
16         android:theme="@style/Theme.NotificationApp"
17         tools:targetApi="31">
18
19         <activity android:name=".CardActivity"
20                 android:exported="false" />
21
22         <activity android:name=".ScrollingActivity"
23                 android:exported="false" />
24
25
26
27         <activity
28             android:name=".MainActivity"
29             android:exported="true">
30             <intent-filter>
31                 <action android:name="android.intent.action.MAIN" />
32
33                 <category android:name="android.intent.category.LAUNCHER" />
34             </intent-filter>
35         </activity>
36     </application>
37
38 </manifest>

```

app/src/main/java/com/example/notificationapp/MainActivity.java

JAVA SOURCE • 114 LINES

```

1  package com.example.notificationapp;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import androidx.appcompat.app.AlertDialog;
5  import androidx.appcompat.app.AppCompatActivity;
6  import androidx.core.app.ActivityCompat;
7  import androidx.core.app.NotificationCompat;
8  import androidx.core.app.NotificationManagerCompat;
9  import androidx.core.content.ContextCompat;
10
11 import android.Manifest;
12 import android.app.NotificationChannel;
13 import android.app.NotificationManager;
14 import android.content.Intent;
15 import android.content.pm.PackageManager;
16 import android.os.Build;

```

```

17 import android.os.Bundle;
18
19 import android.widget.Button;
20
21
22 public class MainActivity extends AppCompatActivity {
23
24     private static final String CHANNEL_ID = "notify_channel";
25
26     @Override
27     protected void onCreate(Bundle savedInstanceState) {
28         super.onCreate(savedInstanceState);
29         setContentView(R.layout.activity_main);
30
31         createNotificationChannel();
32         requestNotificationPermission();
33
34         Button btnNotification = findViewById(R.id.btnNotification);
35         Button btnAlert = findViewById(R.id.btnAlert);
36         Button btnScroll = findViewById(R.id.btnScroll);
37         Button btnCard = findViewById(R.id.btnCard);
38         Button btnTheme = findViewById(R.id.btnTheme);
39
40         // Notification
41         btnNotification.setOnClickListener(v → showNotification());
42
43         // Alert Dialog
44         btnAlert.setOnClickListener(v → {
45             new AlertDialog.Builder(MainActivity.this)
46                 .setTitle("Alert")
47                 .setMessage("This is an alert dialog example.")
48                 .setPositiveButton("OK", null)
49                 .show();
50         });
51
52         // ScrollView Activity
53         btnScroll.setOnClickListener(v → {
54             startActivity(new Intent(MainActivity.this, ScrollingActivity.class));
55         });
56
57         // CardView Activity
58         btnCard.setOnClickListener(v → {
59             startActivity(new Intent(MainActivity.this, CardActivity.class));
60         });
61
62         // Change Theme (Toggle Dark/Light)
63         btnTheme.setOnClickListener(v → {
64             int current = AppCompatDelegate.getDefaultNightMode();
65             if (current == AppCompatDelegate.MODE_NIGHT_YES) {
66                 AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_NO);
67             } else {
68                 AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
69             }
70         });
71     }
72
73     private void requestNotificationPermission() {
74         if (Build.VERSION.SDK_INT ≥ Build.VERSION_CODES.TIRAMISU) {
75             if (ContextCompat.checkSelfPermission(this, Manifest.permission.POST_NOTIFICATIONS)
76                 ≠ PackageManager.PERMISSION_GRANTED) {
77                 ActivityCompat.requestPermissions(this,
78                     new String[]{Manifest.permission.POST_NOTIFICATIONS}, 101);
79             }
80         }
81     }
82
83     private void showNotification() {
84         NotificationManagerCompat manager = NotificationManagerCompat.from(this);
85         if (!manager.areNotificationsEnabled()) {
86             new AlertDialog.Builder(this)
87                 .setTitle("Notifications Blocked")
88                 .setMessage("Please enable notifications for this app in Settings.")
89                 .setPositiveButton("OK", null)

```

```

90         .show();
91         return;
92     }
93
94     NotificationCompat.Builder builder = new NotificationCompat.Builder(this, CHANNEL_ID)
95         .setSmallIcon(R.drawable.ic_notification)
96         .setContentTitle("Hello Notification")
97         .setContentText("This is a demo notification.")
98         .setPriority(NotificationCompat.PRIORITY_DEFAULT);
99
100     manager.notify(1, builder.build());
101 }
102
103 private void createNotificationChannel() {
104     if (Build.VERSION.SDK_INT ≥ Build.VERSION_CODES.O) {
105         CharSequence name = "NotifyChannel";
106         String description = "Channel for demo notifications";
107         int importance = NotificationManager.IMPORTANCE_DEFAULT;
108         NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name, importance);
109         channel.setDescription(description);
110         NotificationManager notificationManager = getSystemService(NotificationManager.class);
111         notificationManager.createNotificationChannel(channel);
112     }
113 }
114 }

```

app/src/main/java/com/example/notificationapp/CardActivity.java

JAVA SOURCE • 29 LINES

```

1  package com.example.notificationapp;
2
3  import android.os.Bundle;
4  import android.widget.ImageView;
5  import android.widget.TextView;
6
7
8  import androidx.appcompat.app.AppCompatActivity;
9
10 public class CardActivity extends AppCompatActivity {
11
12     ImageView cardImage;
13     TextView cardText;
14
15     @Override
16     protected void onCreate(Bundle savedInstanceState) {
17         super.onCreate(savedInstanceState);
18         setContentView(R.layout.activity_card); // Make sure activity_main includes card_item.xml
19         or has it inside
20         cardImage = findViewById(R.id.card_image);
21         cardText = findViewById(R.id.card_text);
22
23         // Update text
24         cardText.setText("Hello! I am Android 17!");
25
26         // Update image (replace with your drawable)
27         cardImage.setImageResource(R.drawable.img);
28     }
29 }

```

app/src/main/java/com/example/notificationapp/ScrollingActivity.java

JAVA SOURCE • 13 LINES

```

1  package com.example.notificationapp;
2
3
4  import android.os.Bundle;

```

```

5  import androidx.appcompat.app.AppCompatActivity;
6
7  public class ScrollingActivity extends AppCompatActivity {
8      @Override
9      protected void onCreate(Bundle savedInstanceState) {
10         super.onCreate(savedInstanceState);
11         setContentView(R.layout.activity_scrolling);
12     }
13 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 43 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2
3  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:gravity="center"
7      android:orientation="vertical"
8      android:padding="20dp">
9
10     <Button
11         android:id="@+id/btnNotification"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"
14         android:text="Show Notification" />
15
16     <Button
17         android:id="@+id/btnAlert"
18         android:layout_width="match_parent"
19         android:layout_height="wrap_content"
20         android:text="Show Alert Dialog"
21         android:layout_marginTop="10dp" />
22
23     <Button
24         android:id="@+id/btnScroll"
25         android:layout_width="match_parent"
26         android:layout_height="wrap_content"
27         android:text="Show Scrolling"
28         android:layout_marginTop="10dp" />
29
30     <Button
31         android:id="@+id/btnCard"
32         android:layout_width="match_parent"
33         android:layout_height="wrap_content"
34         android:text="Show CardView"
35         android:layout_marginTop="10dp" />
36
37     <Button
38         android:id="@+id/btnTheme"
39         android:layout_width="match_parent"
40         android:layout_height="wrap_content"
41         android:text="Change Theme"
42         android:layout_marginTop="10dp" />
43 </LinearLayout>

```

app/src/main/res/layout/activity_card.xml

LAYOUT XML • 41 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:card_view="http://schemas.android.com/apk/res-auto"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:gravity="center"
7      android:orientation="vertical">
8

```

```

9      <androidx.cardview.widget.CardView
10          android:id="@+id/myCard"
11          android:layout_width="match_parent"
12          android:layout_height="wrap_content"
13          card_view:cardCornerRadius="12dp"
14          card_view:cardElevation="8dp"
15          android:layout_margin="16dp">
16
17          <LinearLayout
18              android:layout_width="match_parent"
19              android:layout_height="wrap_content"
20              android:orientation="vertical"
21              android:padding="16dp">
22
23              <ImageView
24                  android:id="@+id/card_image"
25                  android:layout_width="match_parent"
26                  android:layout_height="200dp"
27                  android:src="@drawable/ic_launcher_foreground"
28                  android:scaleType="centerCrop" />
29
30              <TextView
31                  android:id="@+id/card_text"
32                  android:layout_width="wrap_content"
33                  android:layout_height="wrap_content"
34                  android:text="Click me!"
35                  android:textSize="20sp"
36                  android:layout_marginTop="8dp" />
37
38          </LinearLayout>
39      </androidx.cardview.widget.CardView>
40
41 </LinearLayout>

```

app/src/main/res/layout/activity_scrolling.xml

LAYOUT XML • 12 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:padding="16dp">
6
7      <TextView
8          android:layout_width="wrap_content"
9          android:layout_height="wrap_content"
10         android:text="@string/long_text"
11         android:textSize="18sp" />
12 </ScrollView>

```

app/src/main/res/values/strings.xml

STRINGS XML • 49 LINES

```

1  <resources>
2      <string name="app_name">NotifyApp</string>
3      <string name="notification_demo">Show Notification</string>
4      <string name="alert_demo">Show Alert</string>
5      <string name="scroll_demo">Scroll Demo</string>
6      <string name="card_demo">Card Demo</string>
7      <string name="theme_demo">Change Theme</string>
8
9      <!-- Long text for scrolling demo -->
10     <string name="long_text"> <![CDATA[
11         Android is an open-source operating system developed by Google for mobile devices
12         such as smartphones and tablets. It is based on the Linux kernel and provides a
13         rich application framework that allows developers to create innovative apps and games
14         for users across the globe. Android applications are written primarily in Java,
15         Kotlin, and sometimes C++, and they run inside a managed environment known as the

```

Android Runtime (ART).

One of Android's key strengths is its flexibility. It supports a wide range of hardware devices from different manufacturers, which has helped it become the world's most widely used mobile operating system. Google Play Store provides millions of apps that extend the core functionalities of Android devices, covering areas such as communication, productivity, entertainment, navigation, and gaming.

Android also supports features like notifications, widgets, multi-tasking, background services, and deep integration with Google's ecosystem, including Gmail, Maps, and Drive. With regular updates and strong developer community support, Android continues to evolve and power innovative technologies such as foldable phones, wearables, smart TVs, and connected IoT devices.

Android is an open-source operating system developed by Google for mobile devices such as smartphones and tablets. It is based on the Linux kernel and provides a rich application framework that allows developers to create innovative apps and games for users across the globe. Android applications are written primarily in Java, Kotlin, and sometimes C++, and they run inside a managed environment known as the Android Runtime (ART).

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```
]]> </string>  
</resources>
```

President

AndroidStudioProjects • 12 files

313 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3      package="com.example.president">
4
5      <application
6          android:allowBackup="true"
7          android:icon="@mipmap/ic_launcher"
8          android:label="@string/app_name"
9          android:roundIcon="@mipmap/ic_launcher_round"
10         android:supportsRtl="true"
11         android:theme="@style/Theme.President">
12
13         <activity
14             android:name=".PresidentActivity" />
15
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19
20             <intent-filter>
21                 <action android:name="android.intent.action.MAIN" />
22
23                 <category android:name="android.intent.category.LAUNCHER" />
24             </intent-filter>
25
26         </activity>
27
28     </application>
29
30 </manifest>

```

app/src/main/java/com/example/president/MainActivity.java

JAVA SOURCE • 46 LINES

```

1  package com.example.president;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.Menu;
8  import android.view.MenuItem;
9  import android.widget.Toast;
10
11 public class MainActivity extends AppCompatActivity {
12
13     @Override
14     protected void onCreate(Bundle savedInstanceState) {
15         super.onCreate(savedInstanceState);
16         setContentView(R.layout.activity_main);
17     }
18
19     @Override
20     public boolean onCreateOptionsMenu(Menu menu) {
21         getMenuInflater().inflate(R.menu.menu, menu);
22         return true;
23     }
24

```

```

25     @Override
26     public boolean onOptionsItemSelected(MenuItem item) {
27
28         switch (item.getItemId()) {
29
30             case R.id.action_toast:
31                 Toast.makeText(this, "This is a toast message", Toast.LENGTH_SHORT).show();
32                 return true;
33
34             case R.id.action_president:
35                 startActivity(new Intent(this, PresidentActivity.class));
36                 return true;
37
38             case R.id.action_exit:
39                 finish();
40                 return true;
41
42             default:
43                 return super.onOptionsItemSelected(item);
44         }
45     }
46 }

```

app/src/main/java/com/example/president/FragmentDetail.java

JAVA SOURCE • 36 LINES

```

1  package com.example.president;
2
3  import android.os.Bundle;
4
5  import androidx.fragment.app.Fragment;
6
7  import android.view.LayoutInflater;
8  import android.view.View;
9  import android.view.ViewGroup;
10 import android.widget.ImageView;
11 import android.widget.TextView;
12 //import androidx.fragment.app.Fragment;
13
14 public class FragmentDetail extends Fragment {
15
16     private TextView detailsView;
17     private ImageView imageView;
18
19     @Override
20     public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
21         View view = inflater.inflate(R.layout.fragment_detail, container, false);
22         detailsView = view.findViewById(R.id.presidentDetails);
23         imageView = view.findViewById(R.id.presidentImage);
24         return view;
25     }
26
27     public void updateDetails(President p) {
28         String detail = "Name: " + p.name + "\n"
29             + "Period: " + p.period + "\n"
30             + "Qualification: " + p.qualification + "\n"
31             + "Achievements: " + p.achievements + "\n"
32             + "Experience: " + p.experience;
33         detailsView.setText(detail);
34         imageView.setImageResource(p.imageResId);
35     }
36 }

```

app/src/main/java/com/example/president/ListFragment.java

JAVA SOURCE • 67 LINES

```

1  package com.example.president;
2
3  import android.os.Bundle;
4
5  import androidx.fragment.app.Fragment;
6
7  import android.view.LayoutInflater;
8  import android.view.View;
9  import android.view.ViewGroup;
10
11
12  import android.widget.ArrayAdapter;
13  import android.widget.ListView;
14
15
16
17  public class ListFragment extends Fragment {
18
19      String[] names = {
20          "Dr. Rajendra Prasad",
21          "Dr. S. Radhakrishnan",
22          "Dr. A.P.J. Abdul Kalam",
23          "Shri Ram Nath Kovind",
24          "Smt. Droupadi Murmu",
25          "Giani Zail Singh",
26          "Pranab Mukherjee",
27          "Pratibha Patil",
28          "Ramaswamy Venkataraman",
29          "Shankar Dayal Sharma"
30      };
31      President[] presidents;
32
33      @Override
34      public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
35          View view = inflater.inflate(R.layout.fragment_list, container, false);
36          ListView listView = view.findViewById(R.id.presidentListView);
37
38
39          presidents=new President[] {
40              new President("Dr. Rajendra Prasad", "1950-1962", "Lawyer", "Indian Independence
41              Movement", "Lifetime service to the nation", R.drawable.img),
42              new President("Dr. S. Radhakrishnan", "1962-1967", "Philosopher", "Education & Cu
43              lture", "Promoted Indian philosophy", R.drawable.img_1),
44              new President("Dr. A.P.J. Abdul Kalam", "2002-2007", "Scientist", "Missile Man of
45              India", "ISRO, DRDO, and education reform", R.drawable.img_2),
46              new President("Shri Ramnath Kovind", "2017-2022", "Lawyer", "Dalit rights advocac
47              y", "Legal and social justice", R.drawable.img_3),
48              new President("Smt. Droupadi Murmu", "2022-present", "Teacher", "Tribal upliftmen
49              t", "Governance and tribal education", R.drawable.img_4),
50              new President("Giani Zail Singh", "1982-1987", "Political Leader", "Punjab & Nati
51              onal Unity", "Ministerial roles and unity efforts", R.drawable.img_5),
52              new President("Pranab Mukherjee", "2012-2017", "Economist", "Finance & Governanc
53              e", "Decades of political leadership", R.drawable.img_6),
54              new President("Pratibha Patil", "2007-2012", "Lawyer", "Women's Rights", "First w
55              oman president", R.drawable.img_7),
56              new President("Ramaswamy Venkataraman", "1987-1992", "Lawyer", "Indian law & inde
57              pendence", "Indian Constituent Assembly", R.drawable.img_8),
58              new President("Shankar Dayal Sharma", "1992-1997", "Lawyer", "Governance", "Forme
59              r VP and Governor", R.drawable.img_9)
60          };
61
62          ArrayAdapter<String> adapter = new ArrayAdapter<>(getActivity(),
63              android.R.layout.simple_list_item_1, names);
64          listView.setAdapter(adapter);
65
66          listView.setOnItemClickListener((parent, v, position, id) -> {
67
68              PresidentActivity activity =

```

```

59         (PresidentActivity) requireActivity();
60
61         activity.showPresidentDetails(presidents[position]);
62
63     });
64
65     return view;
66 }
67 }

```

app/src/main/java/com/example/president/President.java

JAVA SOURCE • 21 LINES

```

1  package com.example.president;
2
3  public class President {
4
5      String name;
6      String period;
7      String qualification;
8      String achievements;
9      String experience;
10     int imageResId;
11
12     public President(String name, String period, String qualification,
13                     String achievements, String experience, int imageResId) {
14         this.name = name;
15         this.period = period;
16         this.qualification = qualification;
17         this.achievements = achievements;
18         this.experience = experience;
19         this.imageResId = imageResId;
20     }
21 }

```

app/src/main/java/com/example/president/PresidentActivity.java

JAVA SOURCE • 23 LINES

```

1  package com.example.president;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5
6  public class PresidentActivity extends AppCompatActivity {
7
8      @Override
9      protected void onCreate(Bundle savedInstanceState) {
10         super.onCreate(savedInstanceState);
11         setContentView(R.layout.activity_president);
12     }
13
14     public void showPresidentDetails(President president) {
15         FragmentDetail detailFragment =
16             (FragmentDetail) getSupportFragmentManager()
17                 .findFragmentById(R.id.fragment_detail);
18
19         if (detailFragment != null) {
20             detailFragment.updateDetails(president);
21         }
22     }
23 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 13 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent">
5
6      <TextView
7          android:id="@+id/mainText"
8          android:layout_width="wrap_content"
9          android:layout_height="wrap_content"
10         android:text="Welcome to the President App"
11         android:layout_centerInParent="true"
12         android:textSize="18sp"/>
13 </RelativeLayout>

```

app/src/main/res/layout/activity_president.xml

LAYOUT XML • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="horizontal">
6
7      <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
8          android:layout_width="match_parent"
9          android:layout_height="match_parent"
10         android:orientation="horizontal">
11
12         <fragment
13             android:id="@+id/fragment_list"
14             android:name="com.example.president.ListFragment"
15             android:layout_width="0dp"
16             android:layout_weight="1"
17             android:layout_height="match_parent"/>
18
19         <fragment
20             android:id="@+id/fragment_detail"
21             android:name="com.example.president.FragmentDetail"
22             android:layout_width="0dp"
23             android:layout_weight="2"
24             android:layout_height="match_parent"/>
25     </LinearLayout>
26
27 </LinearLayout>

```

app/src/main/res/layout/fragment_detail.xml

LAYOUT XML • 28 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent">
5
6      <LinearLayout
7          android:orientation="vertical"
8          android:padding="16dp"
9          android:layout_width="match_parent"
10         android:layout_height="wrap_content">
11
12         <ImageView
13             android:id="@+id/presidentImage"
14             android:layout_width="200dp"
15             android:layout_height="200dp"
16             android:layout_gravity="center"
17             android:scaleType="centerCrop"

```

```

18         android:src="@drawable/ic_launcher_background"
19         android:contentDescription="TODO" />
20
21     <TextView
22         android:id="@+id/presidentDetails"
23         android:layout_width="wrap_content"
24         android:layout_height="wrap_content"
25         android:textSize="16sp"
26         android:layout_marginTop="16dp" />
27     </LinearLayout>
28 </ScrollView>

```

app/src/main/res/layout/fragment_list.xml

LAYOUT XML • 5 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <ListView xmlns:android="http://schemas.android.com/apk/res/android"
3     android:id="@+id/presidentListView"
4     android:layout_width="match_parent"
5     android:layout_height="match_parent" />

```

app/src/main/res/menu/menu.xml

MENU XML • 12 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <menu xmlns:android="http://schemas.android.com/apk/res/android">
3     <item
4         android:id="@+id/action_toast"
5         android:title="Toast" />
6     <item
7         android:id="@+id/action_exit"
8         android:title="Exit" />
9     <item
10        android:id="@+id/action_president"
11        android:title="President" />
12 </menu>

```

app/src/main/res/values/strings.xml

STRINGS XML • 5 LINES

```

1 <resources>
2     <string name="app_name">President</string>
3     <!-- TODO: Remove or change this placeholder text -->
4     <string name="hello_blank_fragment">Hello blank fragment</string>
5 </resources>

```

SharedReference

AndroidStudioProjects • 6 files

243 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 31 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.sharedreference">
5    <application
6      android:allowBackup="true"
7      android:dataExtractionRules="@xml/data_extraction_rules"
8      android:fullBackupContent="@xml/backup_rules"
9      android:icon="@mipmap/ic_launcher"
10     android:label="@string/app_name"
11     android:roundIcon="@mipmap/ic_launcher_round"
12     android:supportsRtl="true"
13     android:theme="@style/Theme.SharedReference"
14     tools:targetApi="31">
15
16     <activity
17       android:name=".Home"
18       android:exported="false" />
19
20     <activity
21       android:name=".MainActivity"
22       android:exported="true">
23       <intent-filter>
24         <action android:name="android.intent.action.MAIN" />
25         <category android:name="android.intent.category.LAUNCHER" />
26       </intent-filter>
27     </activity>
28
29   </application>
30
31 </manifest>

```

app/src/main/java/com/example/sharedreference/MainActivity.java

JAVA SOURCE • 65 LINES

```

1  package com.example.sharedreference;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.content.SharedPreferences;
8  import android.view.View;
9  import android.widget.Button;
10 import android.widget.EditText;
11 import android.widget.Toast;
12
13 public class MainActivity extends AppCompatActivity {
14
15     EditText et1,et2;
16     Button login;
17
18     public static final String SHARED_PREFS = "loginPrefs";
19     public static final String KEY_USERNAME = "username";
20     public static final String KEY_LOGGED_IN = "loggedin";
21
22
23     @Override

```

```

24     protected void onCreate(Bundle savedInstanceState) {
25         super.onCreate(savedInstanceState);
26
27         SharedPreferences prefs=getSharedPreferences(SHARED_PREFS,MODE_PRIVATE);
28
29         if (prefs.getBoolean(KEY_LOGGED_IN,false))
30         {
31             startActivity(new Intent(MainActivity.this,Home.class));
32             finish();
33             return;
34         }
35
36         setContentView(R.layout.activity_main);
37
38         et1=findViewById(R.id.et1);
39         et2=findViewById(R.id.et2);
40         login=findViewById(R.id.loginbtn);
41
42         login.setOnClickListener(new View.OnClickListener() {
43             @Override
44             public void onClick(View view) {
45                 String uname=et1.getText().toString();
46                 String pass=et2.getText().toString();
47
48                 if(uname.equals("admin") && pass.equals("1234"))
49                 {
50                     SharedPreferences.Editor editor=getSharedPreferences(SHARED_PREFS,MODE_PRIVAT
E).edit();
51                     editor.putString(KEY_USERNAME,uname);
52                     editor.putBoolean(KEY_LOGGED_IN,true);
53                     editor.apply();
54                     startActivity(new Intent(MainActivity.this,Home.class));
55                     finish();
56                     return;
57                 }
58                 else
59                 {
60                     Toast.makeText(MainActivity.this, "Invalid Credentials", Toast.LENGTH_SHORT).
show();
61                 }
62             }
63         });
64     }
65 }

```

app/src/main/java/com/example/sharedreference/Home.java

JAVA SOURCE • 45 LINES

```

1  package com.example.sharedreference;
2  import androidx.appcompat.app.AppCompatActivity;
3
4  import android.content.Intent;
5  import android.content.SharedPreferences;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.TextView;
10
11
12  public class Home extends AppCompatActivity {
13      TextView welcometext;
14      Button btn1;
15
16      public static final String SHARED_PREFS = "loginPrefs";
17      public static final String KEY_USERNAME = "username";
18      public static final String KEY_LOGGED_IN = "loggedin";
19
20      @Override
21      protected void onCreate(Bundle savedInstanceState) {
22          super.onCreate(savedInstanceState);
23          setContentView(R.layout.activity_home);

```

```

24
25     welcometext=findViewById(R.id.tv1);
26     btn1=findViewById(R.id.btn1);
27
28     SharedPreferences prefs=getSharedPreferences(SHARED_PREFS,MODE_PRIVATE);
29     String name =prefs.getString(KEY_USERNAME,"user");
30     welcometext.setText("Welcome" + name + "!");
31
32     btn1.setOnClickListener(new View.OnClickListener() {
33         @Override
34         public void onClick(View view) {
35             SharedPreferences.Editor editor=getSharedPreferences(SHARED_PREFS,MODE_PRIVATE).e
dit();
36
37             editor.clear();
38             editor.apply();
39             startActivity(new Intent(Home.this,MainActivity.class));
40             finish();
41
42         }
43     });
44 }
45 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 66 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      android:background="@color/purple_200"
9      tools:context=".MainActivity">
10
11      <TextView
12          android:id="@+id/textView"
13          android:layout_width="174dp"
14          android:layout_height="73dp"
15          android:text="Login Here "
16          android:textAlignment="center"
17          android:textSize="30dp"
18          app:layout_constraintBottom_toBottomOf="parent"
19          app:layout_constraintEnd_toEndOf="parent"
20          app:layout_constraintHorizontal_bias="0.469"
21          app:layout_constraintStart_toStartOf="parent"
22          app:layout_constraintTop_toTopOf="parent"
23          app:layout_constraintVertical_bias="0.054" />
24
25      <EditText
26          android:id="@+id/et1"
27          android:layout_width="232dp"
28          android:layout_height="54dp"
29          android:layout_marginTop="184dp"
30          android:ems="10"
31          android:inputType="textPersonName"
32          android:hint="Enter Username"
33          app:layout_constraintEnd_toEndOf="parent"
34          app:layout_constraintHorizontal_bias="0.441"
35          app:layout_constraintStart_toStartOf="parent"
36          app:layout_constraintTop_toTopOf="parent" />
37
38      <EditText
39          android:id="@+id/et2"
40          android:layout_width="238dp"
41          android:layout_height="54dp"
42          android:ems="10"
43          android:hint="Enter Password"
44          android:inputType="textPersonName"

```

```

45     app:layout_constraintBottom_toBottomOf="parent"
46     app:layout_constraintEnd_toEndOf="parent"
47     app:layout_constraintHorizontal_bias="0.421"
48     app:layout_constraintStart_toStartOf="parent"
49     app:layout_constraintTop_toTopOf="parent"
50     app:layout_constraintVertical_bias="0.48" />
51
52     <Button
53         android:id="@+id/loginbtn"
54         android:layout_width="109dp"
55         android:layout_height="49dp"
56         android:backgroundTint="#FFFF0000"
57         android:text="Login"
58         app:layout_constraintBottom_toBottomOf="parent"
59         app:layout_constraintEnd_toEndOf="parent"
60         app:layout_constraintHorizontal_bias="0.47"
61         app:layout_constraintStart_toStartOf="parent"
62         app:layout_constraintTop_toTopOf="parent"
63         app:layout_constraintVertical_bias="0.695" />
64
65
66 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/activity_home.xml

LAYOUT XML • 33 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent"
8      android:background="@color/purple_200"
9      tools:context=".Home">
10
11     <TextView
12         android:id="@+id/tv1"
13         android:layout_width="243dp"
14         android:layout_height="83dp"
15         android:text="Welcome Username"
16         android:textSize="30dp"
17         app:layout_constraintBottom_toBottomOf="parent"
18         app:layout_constraintEnd_toEndOf="parent"
19         app:layout_constraintStart_toStartOf="parent"
20         app:layout_constraintTop_toTopOf="parent"
21         app:layout_constraintVertical_bias="0.082" />
22
23     <Button
24         android:id="@+id/btn1"
25         android:layout_width="132dp"
26         android:layout_height="54dp"
27         android:text="Logout"
28         android:backgroundTint="#FFFF0000"
29         app:layout_constraintBottom_toBottomOf="parent"
30         app:layout_constraintEnd_toEndOf="parent"
31         app:layout_constraintStart_toStartOf="parent"
32         app:layout_constraintTop_toTopOf="parent" />
33 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```

1  <resources>
2      <string name="app_name">Shared Reference</string>
3  </resources>

```

Toast

81 Lines

AndroidStudioProjects • 4 files

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.toast">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.Toast"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/toast/MainActivity.java

JAVA SOURCE • 22 LINES

```

1  package com.example.toast;
2
3  import androidx.appcompat.app.AppCompatActivity;
4  import android.os.Bundle;
5  import android.view.View;
6  import android.widget.Toast;
7
8  public class MainActivity extends AppCompatActivity {
9
10     @Override
11     protected void onCreate(Bundle savedInstanceState) {
12         super.onCreate(savedInstanceState);
13         setContentView(R.layout.activity_main);
14
15         Toast.makeText(getApplicationContext(), "Default Toast Message", Toast.LENGTH_LONG).show
16     };
17     public void showtoast(View view){
18         Toast.makeText(getApplicationContext(), "onClickToastMessage", Toast.LENGTH_SHORT).show();
19     }
20
21 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 29 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9
10     <Button
11         android:id="@+id/button"
12         android:layout_width="293dp"
13         android:layout_height="159dp"
14         android:insetLeft="50dp"
15         android:insetTop="50dp"
16         android:onClick="showtoast"
17         android:text=" On Click Toast"
18         tools:ignore="MissingConstraints"
19         tools:layout_editor_absoluteX="47dp"
20         tools:layout_editor_absoluteY="243dp" />
21
22     <androidx.constraintlayout.widget.Barrier
23         android:id="@+id/barrier2"
24         android:layout_width="wrap_content"
25         android:layout_height="wrap_content"
26         app:barrierDirection="top"
27         app:constraint_referenced_ids="button" />
28
29 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```

1  <resources>
2      <string name="app_name">Toast</string>
3  </resources>

```

linearlayoutcalculator

178 Lines

AndroidStudioProjects • 4 files

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.linearlayoutcalculator">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="@string/app_name"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.LinearLayoutCalculator"
15       tools:targetApi="31">
16       <activity
17           android:name=".MainActivity"
18           android:exported="true">
19           <intent-filter>
20               <action android:name="android.intent.action.MAIN" />
21
22               <category android:name="android.intent.category.LAUNCHER" />
23           </intent-filter>
24       </activity>
25   </application>
26
27 </manifest>

```

app/src/main/java/com/example/linearlayoutcalculator/MainActivity.java

JAVA SOURCE • 72 LINES

```

1  package com.example.linearlayoutcalculator;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.content.Intent;
6  import android.os.Bundle;
7  import android.view.View;
8  import android.widget.TextView;
9
10
11 public class MainActivity extends AppCompatActivity {
12
13     TextView txtDisplay;
14     String input="";
15     double num1=0;
16     String operator="";
17
18     @Override
19     protected void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_main);
22         txtDisplay=findViewById(R.id.txtDisplay);
23     }
24
25     public void numberClick(View view){

```

```

26     String value=((android.widget.Button)view).getText().toString();
27     input+=value;
28     txtDisplay.setText(input);
29 }
30
31 public void operatorClick(View view){
32     num1=Double.parseDouble(input);
33     operator=((android.widget.Button)view).getText().toString();
34     input="";
35 }
36
37 public void equalClick(View view){
38     double num2=Double.parseDouble(input);
39     double result=0;
40     switch (operator){
41         case "+":
42             result=num1+num2;
43             break;
44
45         case "-":
46             result=num1-num2;
47             break;
48
49         case "*":
50             result=num1*num2;
51             break;
52
53         case "/":
54             result=num1/num2;
55             break;
56     }
57     txtDisplay.setText(String.valueOf(result));
58     input=String.valueOf(result);
59 }
60
61 public void clearClick(View view){
62     input="";
63     num1=0;
64     operator="";
65
66     txtDisplay.setText("0");
67 }
68
69 }
70
71
72 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 76 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp">
8
9
10     <TextView
11         android:id="@+id/txtDisplay"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"
14         android:text="0"
15         android:textSize="32sp"
16         android:gravity="end"
17         android:padding="12dp"
18         android:background="#EEEEEE" />
19
20

```

```

21     <LinearLayout
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
24         android:orientation="horizontal">
25
26         <Button android:text="7" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
27         <Button android:text="8" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
28         <Button android:text="9" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
29         <Button android:text="/" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="operatorClick" />
30     </LinearLayout>
31
32
33     <LinearLayout
34         android:layout_width="match_parent"
35         android:layout_height="wrap_content"
36         android:orientation="horizontal">
37
38         <Button android:text="4" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
39         <Button android:text="5" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
40         <Button android:text="6" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
41         <Button android:text="*" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="operatorClick" />
42     </LinearLayout>
43
44
45     <LinearLayout
46         android:layout_width="match_parent"
47         android:layout_height="wrap_content"
48         android:orientation="horizontal">
49
50         <Button android:text="1" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
51         <Button android:text="2" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
52         <Button android:text="3" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
53         <Button android:text="-" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="operatorClick" />
54     </LinearLayout>
55
56     <LinearLayout
57         android:layout_width="match_parent"
58         android:layout_height="wrap_content"
59         android:orientation="horizontal">
60
61         <Button android:text="0" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="2" android:onClick="numberClick" />
62         <Button android:text="." android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="numberClick" />
63         <Button android:text="+" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="operatorClick" />
64     </LinearLayout>
65
66
67     <LinearLayout
68         android:layout_width="match_parent"
69         android:layout_height="wrap_content"
70         android:orientation="horizontal">
71
72         <Button android:text="C" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="1" android:onClick="clearClick" />
73         <Button android:text="=" android:layout_width="0dp" android:layout_height="wrap_content"
android:layout_weight="3" android:onClick="equalClick" />
74     </LinearLayout>

```

```
75  
76 </LinearLayout>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>  
2   <string name="app_name">linear layout calculator</string>  
3 </resources>
```

presidentnew - gallery

AndroidStudioProjects • 16 files

405 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 30 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3      package="com.example.presidentnew">
4
5      <application
6          android:allowBackup="true"
7          android:icon="@mipmap/ic_launcher"
8          android:label="@string/app_name"
9          android:roundIcon="@mipmap/ic_launcher_round"
10         android:supportsRtl="true"
11         android:theme="@style/Theme.Presidentnew">
12         <activity
13             android:name=".FullImageActivity"
14             android:exported="false" />
15         <activity
16             android:name=".gallery_activity"
17             android:exported="false" />
18         <activity android:name=".PresidentActivity" />
19         <activity
20             android:name=".MainActivity"
21             android:exported="true">
22             <intent-filter>
23                 <action android:name="android.intent.action.MAIN" />
24
25                 <category android:name="android.intent.category.LAUNCHER" />
26             </intent-filter>
27         </activity>
28     </application>
29
30 </manifest>

```

app/src/main/java/com/example/presidentnew/MainActivity.java

JAVA SOURCE • 52 LINES

```

1  package com.example.presidentnew;
2
3  import
4      androidx.appcompat.app.AppCompatActivity;
5
6  import android.content.Intent;
7  import android.os.Bundle;
8  import android.view.Menu;
9  import android.view.MenuItem;
10 import android.widget.Toast;
11
12 public class MainActivity extends AppCompatActivity {
13
14     @Override
15     protected void onCreate(Bundle savedInstanceState) {
16         super.onCreate(savedInstanceState);
17         setContentView(R.layout.activity_main);
18     }
19
20     @Override
21     public boolean onCreateOptionsMenu(Menu menu) {
22         getMenuInflater().inflate(R.menu.menu, menu);
23         return true;
24     }

```

```

25
26 @Override
27 public boolean onOptionsItemSelected(MenuItem item) {
28
29     switch (item.getItemId()) {
30
31         case R.id.action_toast:
32             Toast.makeText(this, "This is a toast message", Toast.LENGTH_SHORT).show();
33             return true;
34
35         case R.id.action_president:
36             startActivity(new Intent(this, PresidentActivity.class));
37             return true;
38
39         case R.id.action_gallery:
40             startActivity(new Intent(this, gallery_activity.class));
41             return true;
42
43
44         case R.id.action_exit:
45             finish();
46             return true;
47
48         default:
49             return super.onOptionsItemSelected(item);
50     }
51 }
52 }

```

app/src/main/java/com/example/presidentnew/FullImageActivity.java

JAVA SOURCE • 20 LINES

```

1 package com.example.presidentnew;
2
3 import androidx.appcompat.app.AppCompatActivity;
4
5 import android.os.Bundle;
6 import android.widget.ImageView;
7 public class FullImageActivity extends AppCompatActivity {
8     ImageView imageView;
9
10    @Override
11    protected void onCreate(Bundle savedInstanceState) {
12        super.onCreate(savedInstanceState);
13        setContentView(R.layout.activity_full_image);
14
15        imageView = findViewById(R.id.fullImage);
16
17        int image = getIntent().getIntExtra("image", 0);
18        imageView.setImageResource(image);
19    }
20 }

```

app/src/main/java/com/example/presidentnew/gallery_activity.java

JAVA SOURCE • 37 LINES

```

1 package com.example.presidentnew;
2
3 import androidx.appcompat.app.AppCompatActivity;
4
5 import android.content.Intent;
6 import android.os.Bundle;
7 import android.util.Log;
8 import android.widget.ImageView;
9
10
11 public class gallery_activity extends AppCompatActivity {
12

```

```

13     private ImageView img_1, img_2, img_3;
14
15     @Override
16     protected void onCreate(Bundle savedInstanceState) {
17         super.onCreate(savedInstanceState);
18         setContentView(R.layout.activity_gallary);
19
20         img_1 = findViewById(R.id.img_1);
21         img_2 = findViewById(R.id.img_2);
22         img_3 = findViewById(R.id.img_3);
23
24         img_1.setOnClickListener(v → openImage(R.drawable.img_1));
25         img_2.setOnClickListener(v → openImage(R.drawable.img_2));
26         img_3.setOnClickListener(v → openImage(R.drawable.img_3));
27     }
28
29     private void openImage(int imageResId) {
30
31         Log.d("GallaryActivity", "Sending Image ID = " + imageResId);
32
33         Intent intent = new Intent(gallary_activity.this, FullImageActivity.class);
34         intent.putExtra("image", imageResId);
35         startActivity(intent);
36     }
37 }

```

app/src/main/java/com/example/presidentnew/President.java

JAVA SOURCE • 21 LINES

```

1  package com.example.presidentnew;
2
3
4
5  public class President {
6      String name;
7      String period;
8      String qualification;
9      String achievements;
10     String experience;
11     int imageResId;
12
13     public President(String name, String period, String qualification, String achievements, String
14 experience, int imageResId) {
15         this.name = name;
16         this.period = period;
17         this.qualification = qualification;
18         this.achievements = achievements;
19         this.experience = experience;
20         this.imageResId = imageResId;
21     }
22 }

```

app/src/main/java/com/example/presidentnew/PresidentActivity.java

JAVA SOURCE • 22 LINES

```

1  package com.example.presidentnew;
2  import androidx.appcompat.app.AppCompatActivity;
3  import android.os.Bundle;
4
5  public class PresidentActivity extends AppCompatActivity {
6
7      @Override
8      protected void onCreate(Bundle savedInstanceState) {
9          super.onCreate(savedInstanceState);
10         setContentView(R.layout.activity_president);
11     }
12
13     public void showPresidentDetails(President president) {

```

```

14     PresidentDetailFragment detailFragment =
15         (PresidentDetailFragment) getSupportFragmentManager()
16             .findFragmentById(R.id.fragment_detail);
17
18     if (detailFragment != null) {
19         detailFragment.updateDetails(president);
20     }
21 }
22 }

```

app/src/main/java/com/example/presidentnew/PresidentDetailFragment.java

JAVA SOURCE • 40
LINES

```

1  package com.example.presidentnew;
2
3  import android.os.Bundle;
4  import android.view.LayoutInflater;
5  import android.view.View;
6  import android.view.ViewGroup;
7  import android.widget.ImageView;
8  import android.widget.TextView;
9
10 import androidx.fragment.app.Fragment;
11
12 public class PresidentDetailFragment extends Fragment {
13
14     private TextView detailsView;
15     private ImageView imageView;
16
17     @Override
18     public View onCreateView(LayoutInflater inflater, ViewGroup container,
19                             Bundle savedInstanceState) {
20
21         View view = inflater.inflate(R.layout.fragment_president_detail, container, false);
22
23         detailsView = view.findViewById(R.id.presidentDetails);
24         imageView = view.findViewById(R.id.presidentImage);
25
26         return view;
27     }
28
29     public void updateDetails(President p) {
30
31         String detail = "Name: " + p.name + "\n"
32             + "Period: " + p.period + "\n"
33             + "Qualification: " + p.qualification + "\n"
34             + "Achievements: " + p.achievements + "\n"
35             + "Experience: " + p.experience;
36
37         detailsView.setText(detail);
38         imageView.setImageResource(p.imageResId);
39     }
40 }

```

app/src/main/java/com/example/presidentnew/PresidentListFragment.java

JAVA SOURCE • 61
LINES

```

1  package com.example.presidentnew;
2
3  import androidx.fragment.app.Fragment;
4  import android.os.Bundle;
5  import android.view.LayoutInflater;
6  import android.view.View;
7  import android.view.ViewGroup;
8  import android.widget.AdapterView;
9  import android.widget.AdapterView;

```

```

10 import android.widget.ListView;
11
12 public class PresidentListFragment extends Fragment {
13
14     String[] names = {
15         "Dr. Rajendra Prasad",
16         "Dr. S. Radhakrishnan",
17         "Dr. A.P.J. Abdul Kalam",
18         "Shri Ramnath Kovind",
19         "Smt. Droupadi Murmu",
20         "Giani Zail Singh",
21         "Pranab Mukherjee",
22         "Pratibha Patil",
23         "Ramaswamy Venkataraman",
24         "Shankar Dayal Sharma"
25     };
26     President[] presidents;
27
28     @Override
29     public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
30         View view = inflater.inflate(R.layout.fragment_president_list, container, false);
31         ListView listView = view.findViewById(R.id.presidentListView);
32
33
34         presidents = new President[] {
35             new President("Dr. Rajendra Prasad", "1950-1962", "Lawyer", "Indian Independence Movement", "Lifetime service to the nation", R.drawable.img),
36             new President("Dr. S. Radhakrishnan", "1962-1967", "Philosopher", "Education & Culture", "Promoted Indian philosophy", R.drawable.img_1),
37             new President("Dr. A.P.J. Abdul Kalam", "2002-2007", "Scientist", "Missile Man of India", "ISRO, DRDO, and education reform", R.drawable.img_2),
38             new President("Shri Ramnath Kovind", "2017-2022", "Lawyer", "Dalit rights advocacy", "Legal and social justice", R.drawable.img_3),
39             new President("Smt. Droupadi Murmu", "2022-present", "Teacher", "Tribal upliftment", "Governance and tribal education", R.drawable.img_4),
40             new President("Giani Zail Singh", "1982-1987", "Political Leader", "Punjab & National Unity", "Ministerial roles and unity efforts", R.drawable.img_5),
41             new President("Pranab Mukherjee", "2012-2017", "Economist", "Finance & Governance", "Decades of political leadership", R.drawable.img_6),
42             new President("Pratibha Patil", "2007-2012", "Lawyer", "Women's Rights", "First woman president", R.drawable.img_7),
43             new President("Ramaswamy Venkataraman", "1987-1992", "Lawyer", "Indian law & independence", "Indian Constituent Assembly", R.drawable.img_8),
44             new President("Shankar Dayal Sharma", "1992-1997", "Lawyer", "Governance", "Former VP and Governor", R.drawable.img_9)
45         };
46
47         ArrayAdapter<String> adapter = new ArrayAdapter<>(getActivity(),
48             android.R.layout.simple_list_item_1, names);
49         listView.setAdapter(adapter);
50
51         listView.setOnItemClickListener((parent, v, position, id) → {
52
53             PresidentActivity activity =
54                 (PresidentActivity) requireActivity();
55
56             activity.showPresidentDetails(presidents[position]);
57
58         });
59         return view;
60     }
61 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 13 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:layout_width="match_parent"
4     android:layout_height="match_parent">

```

```

5
6      <TextView
7          android:id="@+id/mainText"
8          android:layout_width="wrap_content"
9          android:layout_height="wrap_content"
10         android:text="Welcome to the President App"
11         android:layout_centerInParent="true"
12         android:textSize="18sp" />
13 </RelativeLayout>

```

app/src/main/res/layout/activity_full_image.xml

LAYOUT XML • 8 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2
3 <ImageView xmlns:android="http://schemas.android.com/apk/res/android"
4     android:id="@+id/fullImage"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent"
7     android:scaleType="fitCenter"
8     android:background="#000000" />

```

app/src/main/res/layout/activity_gallery.xml

LAYOUT XML • 30 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2
3 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
4     android:layout_width="match_parent"
5     android:layout_height="match_parent"
6     android:gravity="center"
7     android:orientation="vertical">
8
9     <ImageView
10         android:id="@+id/img_1"
11         android:layout_width="150dp"
12         android:layout_height="150dp"
13         android:src="@drawable/img_1"
14         android:layout_margin="10dp" />
15
16     <ImageView
17         android:id="@+id/img_2"
18         android:layout_width="150dp"
19         android:layout_height="150dp"
20         android:src="@drawable/img_2"
21         android:layout_margin="10dp" />
22
23     <ImageView
24         android:id="@+id/img_3"
25         android:layout_width="150dp"
26         android:layout_height="150dp"
27         android:src="@drawable/img_3"
28         android:layout_margin="10dp" />
29
30 </LinearLayout>

```

app/src/main/res/layout/activity_president.xml

LAYOUT XML • 27 LINES

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:layout_width="match_parent"
4     android:layout_height="match_parent"
5     android:orientation="horizontal">
6
7     <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

```

```

8         android:layout_width="match_parent"
9         android:layout_height="match_parent"
10        android:orientation="horizontal">
11
12        <fragment
13            android:id="@+id/fragment_list"
14            android:name="com.example.presidentnew.PresidentListFragment"
15            android:layout_width="0dp"
16            android:layout_weight="1"
17            android:layout_height="match_parent" />
18
19        <fragment
20            android:id="@+id/fragment_detail"
21            android:name="com.example.presidentnew.PresidentDetailFragment"
22            android:layout_width="0dp"
23            android:layout_weight="2"
24            android:layout_height="match_parent" />
25    </LinearLayout>
26
27 </LinearLayout>

```

app/src/main/res/layout/fragment_president_detail.xml

LAYOUT XML • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent">
5
6      <LinearLayout
7          android:orientation="vertical"
8          android:padding="16dp"
9          android:layout_width="match_parent"
10         android:layout_height="wrap_content">
11
12         <ImageView
13             android:id="@+id/presidentImage"
14             android:layout_width="200dp"
15             android:layout_height="200dp"
16             android:layout_gravity="center"
17             android:scaleType="centerCrop"
18             android:src="@drawable/ic_launcher_background" />
19
20         <TextView
21             android:id="@+id/presidentDetails"
22             android:layout_width="wrap_content"
23             android:layout_height="wrap_content"
24             android:textSize="16sp"
25             android:layout_marginTop="16dp" />
26     </LinearLayout>
27 </ScrollView>

```

app/src/main/res/layout/fragment_president_list.xml

LAYOUT XML • 5 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ListView xmlns:android="http://schemas.android.com/apk/res/android"
3      android:id="@+id/presidentListView"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent" />

```

app/src/main/res/menu/menu.xml

MENU XML • 9 LINES

```

1  <menu xmlns:android="http://schemas.android.com/apk/res/android">
2      <item android:id="@+id/action_toast" android:title="Toast" />

```

```
3     <item android:id="@+id/action_president" android:title="President" />
4     <item android:id="@+id/action_gallery" android:title="Gallery" />
5     <item android:id="@+id/action_exit" android:title="Exit" />
6
7
8
9 </menu>
```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2     <string name="app_name">presidentnew</string>
3 </resources>
```

sum of 2 num

AndroidStudioProjects • 6 files

253 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.layout">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@android:drawable/ic_menu_compass"
11         android:label="@string/Ashok"
12         android:roundIcon="@android:drawable/ic_menu_compass"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Layout"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/layout/MainActivity.java

JAVA SOURCE • 53 LINES

```

1  package com.example.layout;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.text.TextUtils;
7  import android.view.View;
8  import android.widget.Button;
9  import android.widget.EditText;
10 import android.widget.TextView;
11 import android.widget.Toast;
12
13 public class MainActivity extends AppCompatActivity {
14     EditText et1, et2;
15     TextView tv;
16     Button btn_sum, btn_reset;
17
18     @Override
19     protected void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_main);
22
23         et1 = (EditText) findViewById(R.id.number1);
24         et2 = (EditText) findViewById(R.id.number2);
25         tv = (TextView) findViewById(R.id.result_Text);
26         btn_sum = (Button) findViewById(R.id.calculate_button);
27         btn_reset = (Button) findViewById(R.id.reset_button);

```

```

28
29     btn_reset.setOnClickListener(new View.OnClickListener() {
30         @Override
31         public void onClick(View view) {
32             et1.setText("");
33             et2.setText("");
34         }
35     });
36
37     btn_sum.setOnClickListener(new View.OnClickListener() {
38         @Override
39         public void onClick(View view) {
40             String val1 = et1.getText().toString();
41             String val2 = et2.getText().toString();
42
43             if (!TextUtils.isEmpty(val1) && !TextUtils.isEmpty(val2)) {
44                 Integer sum = Integer.parseInt(val1) + Integer.parseInt(val2);
45                 tv.setText(sum.toString());
46                 Toast.makeText(getApplicationContext(), "sum is " + sum, Toast.LENGTH_SHORT).
show();
47             } else {
48                 Toast.makeText(getApplicationContext(), "enter the num first", Toast.LENGTH_S
HORT).show();
49             }
50         }
51     });
52 }
53 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 58 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/
res/android"
3      xmlns:app="http://schemas.android.com/apk/res-auto"
4      xmlns:tools="http://schemas.android.com/tools"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      tools:context=".MainActivity">
8
9      <Button
10         android:id="@+id/calculate_button"
11         android:layout_width="wrap_content"
12         android:layout_height="wrap_content"
13         android:text="Calculate"
14         android:textColor="#000000"
15         tools:layout_editor_absoluteX="141dp"
16         tools:layout_editor_absoluteY="297dp" />
17
18      <TextView
19         android:id="@+id/result_Text"
20         android:layout_width="128dp"
21         android:layout_height="46dp"
22         android:autoSizeTextType="uniform"
23         android:text="Result=0"
24         tools:layout_editor_absoluteX="141dp"
25         tools:layout_editor_absoluteY="460dp" />
26
27      <EditText
28         android:id="@+id/number1"
29         android:layout_width="136dp"
30         android:layout_height="48dp"
31         android:layout_marginTop="109dp"
32         android:hint="Enter number 1"
33         android:inputType="number"
34         android:textColor="@color/black"
35         app:layout_constraintTop_toTopOf="parent"
36         tools:layout_editor_absoluteX="141dp" />
37
38      <EditText

```

```

39         android:id="@+id/number2"
40         android:layout_width="133dp"
41         android:layout_height="48dp"
42         android:layout_marginTop="193dp"
43         android:hint="Enter number 2"
44         android:inputType="number"
45         android:textColor="@color/black"
46         app:layout_constraintTop_toTopOf="parent"
47         tools:layout_editor_absoluteX="141dp" />
48
49     <Button
50         android:id="@+id/reset_button"
51         android:layout_width="wrap_content"
52         android:layout_height="wrap_content"
53         android:text="Reset"
54         android:textColor="@color/black"
55         tools:layout_editor_absoluteX="161dp"
56         tools:layout_editor_absoluteY="378dp" />
57
58 </androidx.constraintlayout.widget.ConstraintLayout>

```

app/src/main/res/layout/layout.xml

LAYOUT XML • 57 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent">
5
6      <EditText
7          android:id="@+id/number1"
8          android:layout_width="136dp"
9          android:layout_height="48dp"
10         android:layout_centerHorizontal="true"
11         android:layout_marginTop="40dp"
12         android:hint="Enter number 1"
13         android:inputType="number"
14         android:textColor="@color/black" />
15
16     <EditText
17         android:id="@+id/number2"
18         android:layout_width="136dp"
19         android:layout_height="48dp"
20         android:layout_below="@id/number1"
21         android:layout_centerHorizontal="true"
22         android:layout_marginTop="16dp"
23         android:hint="Enter number 2"
24         android:inputType="number"
25         android:textColor="@color/black" />
26
27     <Button
28         android:id="@+id/calculate_button"
29         android:layout_width="wrap_content"
30         android:layout_height="wrap_content"
31         android:layout_below="@id/number2"
32         android:layout_centerHorizontal="true"
33         android:layout_marginTop="16dp"
34         android:text="Calculate"
35         android:textColor="#000000" />
36
37     <TextView
38         android:id="@+id/result_Text"
39         android:layout_width="128dp"
40         android:layout_height="46dp"
41         android:layout_below="@id/calculate_button"
42         android:layout_centerHorizontal="true"
43         android:layout_marginTop="16dp"
44         android:autoSizeTextType="uniform"
45         android:text="Result=0" />
46
47     <Button

```

```

48         android:id="@+id/reset_button"
49         android:layout_width="wrap_content"
50         android:layout_height="wrap_content"
51         android:layout_below="@id/result_Text"
52         android:layout_centerHorizontal="true"
53         android:layout_marginTop="16dp"
54         android:text="Reset"
55         android:textColor="@color/black" />
56
57 </RelativeLayout>

```

app/src/main/res/layout/layout2.xml

LAYOUT XML • 50 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="vertical"
6      android:gravity="center_horizontal">
7
8      <EditText
9          android:id="@+id/number1"
10         android:layout_width="136dp"
11         android:layout_height="48dp"
12         android:layout_marginTop="50dp"
13         android:hint="Enter number 1"
14         android:inputType="number"
15         android:textColor="@color/black" />
16
17     <EditText
18         android:id="@+id/number2"
19         android:layout_width="136dp"
20         android:layout_height="48dp"
21         android:layout_marginTop="16dp"
22         android:hint="Enter number 2"
23         android:inputType="number"
24         android:textColor="@color/black" />
25
26     <Button
27         android:id="@+id/calculate_button"
28         android:layout_width="wrap_content"
29         android:layout_height="wrap_content"
30         android:layout_marginTop="16dp"
31         android:text="Calculate"
32         android:textColor="#000000" />
33
34     <TextView
35         android:id="@+id/result_Text"
36         android:layout_width="128dp"
37         android:layout_height="46dp"
38         android:layout_marginTop="16dp"
39         android:autoSizeTextType="uniform"
40         android:text="Result=0" />
41
42     <Button
43         android:id="@+id/reset_button"
44         android:layout_width="wrap_content"
45         android:layout_height="wrap_content"
46         android:layout_marginTop="16dp"
47         android:text="Reset"
48         android:textColor="@color/black" />
49
50 </LinearLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 8 LINES

```
1 <resources>
2   <string name="Ashok">layout</string>
3   <string name="input_hint1">Enter first number</string>
4   <string name="input_hint2">Enter second number</string>
5   <string name="Result_text">Result will appear here</string>
6   <string name="sum_button">Calculate sum</string>
7   <string name="reset_button">Reset</string>
8 </resources>
```

userinfo

AndroidStudioProjects • 4 files

262 Lines

app/src/main/AndroidManifest.xml

MANIFEST • 27 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.userinfo">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="@string/app_name"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.Userinfo"
15         tools:targetApi="31">
16         <activity
17             android:name=".MainActivity"
18             android:exported="true">
19             <intent-filter>
20                 <action android:name="android.intent.action.MAIN" />
21
22                 <category android:name="android.intent.category.LAUNCHER" />
23             </intent-filter>
24         </activity>
25     </application>
26
27 </manifest>

```

app/src/main/java/com/example/userinfo/MainActivity.java

JAVA SOURCE • 92 LINES

```

1  package com.example.userinfo;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.view.View;
7  import android.widget.Button;
8  import android.widget.CheckBox;
9  import android.widget.RadioButton;
10 import android.widget.EditText;
11 import android.widget.RadioGroup;
12 import android.widget.TextView;
13 import android.widget.Toast;
14
15 public class MainActivity extends AppCompatActivity {
16     EditText editText;
17     RadioGroup radioGender;
18     RadioButton radioMale,radioFemale;
19     CheckBox checkBoxReading,checkBoxMusic,checkBoxSports,checkBoxYoga;
20     Button submit;
21     Button clear;
22     TextView output;
23
24     @Override
25     protected void onCreate(Bundle savedInstanceState) {
26         super.onCreate(savedInstanceState);
27         setContentView(R.layout.activity_main);

```

```

28
29     editText = findViewById(R.id.editText);
30     radioGender = findViewById(R.id.genderGroup);
31     radioMale = findViewById(R.id.radioMale);
32     radioFemale = findViewById(R.id.radioFemale);
33     checkBoxReading = findViewById(R.id.checkBoxReading);
34     checkBoxMusic = findViewById(R.id.checkBoxMusic);
35     checkBoxSports = findViewById(R.id.checkBoxSports);
36     checkBoxYoga = findViewById(R.id.checkBoxYoga);
37     submit = findViewById(R.id.submit);
38     clear = findViewById(R.id.clear);
39     output = findViewById(R.id.output);
40
41     View.OnClickListener radioListener = View → {
42         RadioButton selected = (RadioButton) View;
43         Toast.makeText(MainActivity.this, selected.getText() + "selected", Toast.LENGTH_SHOR
T).show();
44
45     };
46     radioMale.setOnClickListener(radioListener);
47     radioFemale.setOnClickListener(radioListener);
48
49     View.OnClickListener checkListener = View → {
50         CheckBox box = (CheckBox) View;
51         if (box.isChecked()) {
52             Toast.makeText(MainActivity.this, box.getText() + "selected", Toast.LENGTH_SHOR
T).show();
53         }
54     };
55
56     checkBoxReading.setOnClickListener(checkListener);
57     checkBoxMusic.setOnClickListener(checkListener);
58     checkBoxSports.setOnClickListener(checkListener);
59     checkBoxYoga.setOnClickListener(checkListener);
60
61
62     submit.setOnClickListener(v → {
63         String name = editText.getText().toString();
64         int selectedId = radioGender.getCheckedRadioButtonId();
65         String gender = "";
66         if (selectedId ≠ -1) {
67             RadioButton selected = findViewById(selectedId);
68             gender = selected.getText().toString();
69         }
70         StringBuilder hobbies = new StringBuilder();
71         if (checkBoxReading.isChecked()) hobbies.append("Reading");
72         if (checkBoxMusic.isChecked()) hobbies.append("Music");
73         if (checkBoxSports.isChecked()) hobbies.append("Sports");
74         if (checkBoxYoga.isChecked()) hobbies.append("Yoga");
75
76
77         String message = "Name :" + name + "\nGender :" + gender + "\nHobbies :" + hobbies;
78         output.setText(message);
79
80         clear.setOnClickListener(clear → {
81             editText.setText("");
82             radioGender.clearCheck();
83             checkBoxReading.setChecked(false);
84             checkBoxMusic.setChecked(false);
85             checkBoxSports.setChecked(false);
86             checkBoxYoga.setChecked(false);
87             output.setText("Output will appear here");
88         });
89
90     });
91 }
92 }

```

app/src/main/res/layout/activity_main.xml

LAYOUT XML • 140 LINES

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:tools="http://schemas.android.com/tools"
4      android:layout_width="match_parent"
5      android:layout_height="match_parent"
6      android:orientation="vertical"
7      android:padding="16dp">
8
9
10     <ImageButton
11         android:id="@+id/imageButton2"
12         android:layout_width="185dp"
13         android:layout_height="120dp"
14         android:layout_gravity="center_horizontal"
15         android:layout_marginBottom="16dp"
16         android:scaleType="centerCrop"
17         tools:srcCompat="@drawable/ic_launcher_foreground" />
18
19     <LinearLayout
20         android:layout_width="match_parent"
21         android:layout_height="wrap_content"
22         android:orientation="horizontal"
23         android:gravity="center_vertical"
24         android:layout_marginBottom="12dp">
25
26         <TextView
27             android:id="@+id/textView2"
28             android:layout_width="80dp"
29             android:layout_height="wrap_content"
30             android:text="Name:" />
31
32         <EditText
33             android:id="@+id/editText"
34             android:layout_width="match_parent"
35             android:layout_height="wrap_content"
36             android:ems="10"
37             android:hint="Enter name"
38             android:inputType="textPersonName" />
39     </LinearLayout>
40
41
42     <LinearLayout
43         android:layout_width="match_parent"
44         android:layout_height="wrap_content"
45         android:orientation="horizontal"
46         android:gravity="center_vertical"
47         android:layout_marginBottom="12dp">
48
49         <TextView
50             android:id="@+id/radioGenderLabel"
51             android:layout_width="80dp"
52             android:layout_height="wrap_content"
53             android:text="Gender:" />
54
55         <RadioGroup
56             android:id="@+id/genderGroup"
57             android:layout_width="match_parent"
58             android:layout_height="wrap_content"
59             android:orientation="horizontal">
60
61             <RadioButton
62                 android:id="@+id/radioMale"
63                 android:layout_width="wrap_content"
64                 android:layout_height="wrap_content"
65                 android:text="Male" />
66
67             <RadioButton
68                 android:id="@+id/radioFemale"
69                 android:layout_width="wrap_content"

```

```

70         android:layout_height="wrap_content"
71         android:text="Female" />
72     </RadioGroup>
73 </LinearLayout>
74
75
76 <TextView
77     android:id="@+id/textView5"
78     android:layout_width="wrap_content"
79     android:layout_height="wrap_content"
80     android:text="Hobbies: "
81     android:textStyle="bold"
82     android:layout_marginBottom="8dp" />
83
84 <CheckBox
85     android:id="@+id/checkBoxReading"
86     android:layout_width="wrap_content"
87     android:layout_height="wrap_content"
88     android:text="Reading" />
89
90 <CheckBox
91     android:id="@+id/checkBoxMusic"
92     android:layout_width="wrap_content"
93     android:layout_height="wrap_content"
94     android:text="Music" />
95
96 <CheckBox
97     android:id="@+id/checkBoxSports"
98     android:layout_width="wrap_content"
99     android:layout_height="wrap_content"
100    android:text="Sports" />
101
102 <CheckBox
103     android:id="@+id/checkBoxYoga"
104     android:layout_width="wrap_content"
105     android:layout_height="wrap_content"
106     android:text="Yoga" />
107
108
109 <LinearLayout
110     android:layout_width="match_parent"
111     android:layout_height="wrap_content"
112     android:orientation="horizontal"
113     android:layout_marginTop="24dp">
114
115     <Button
116         android:id="@+id/submit"
117         android:layout_width="0dp"
118         android:layout_weight="1"
119         android:layout_height="wrap_content"
120         android:layout_marginEnd="8dp"
121         android:text="Submit" />
122
123     <Button
124         android:id="@+id/clear"
125         android:layout_width="0dp"
126         android:layout_weight="1"
127         android:layout_height="wrap_content"
128         android:text="Clear" />
129 </LinearLayout>
130
131
132 <TextView
133     android:id="@+id/output"
134     android:layout_width="match_parent"
135     android:layout_height="53dp"
136     android:layout_marginTop="20dp"
137     android:text="Output will appear here"
138     android:textAlignment="center" />
139
140 </LinearLayout>

```

app/src/main/res/values/strings.xml

STRINGS XML • 3 LINES

```
1 <resources>
2   <string name="app_name">userinfo</string>
3 </resources>
```